Coarse-Graining and the Closure Problem in Plasma Turbulence

KAAI, November 2064

Muni Zhou (Dartmouth College)

Two main problems

- **The coarse graining problem:** How to include the microscopic physics in macroscopic dynamics? *(E.g., Plasma physics for the magnetogenesis problem.)*

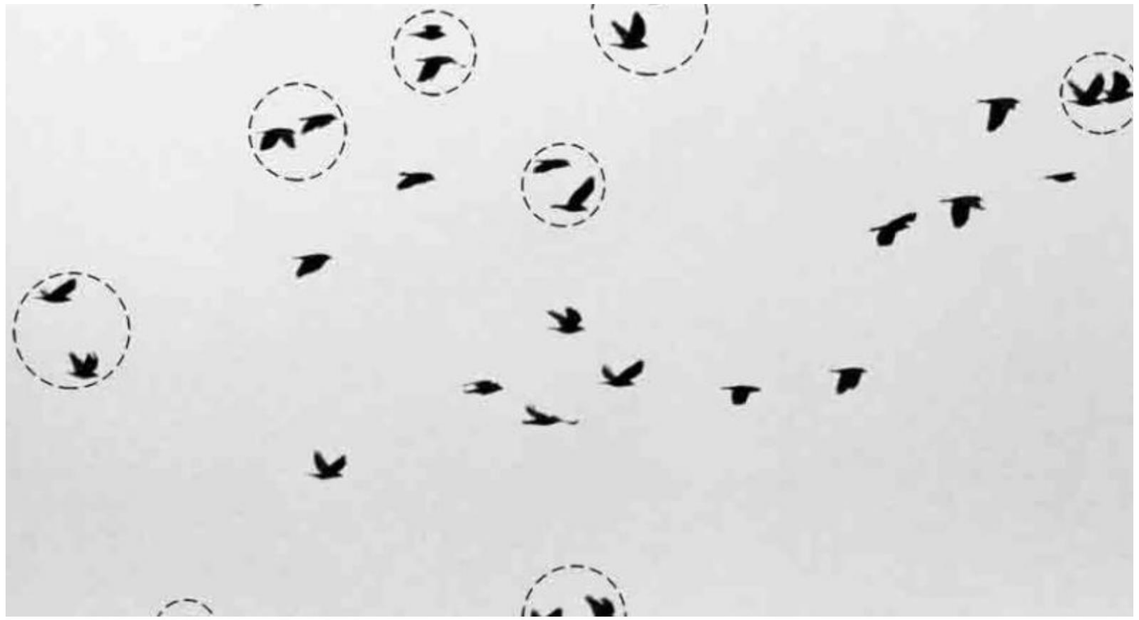
- **The closure problem**: How to use a set of fluid-type equations to describe a kinetic system? (*E.g., Phase-Space Dynamics of Kinetic Turbulence)*

A simple analogy



[credit: *National Geographic*]





Macroscopic scales turbulent fluid

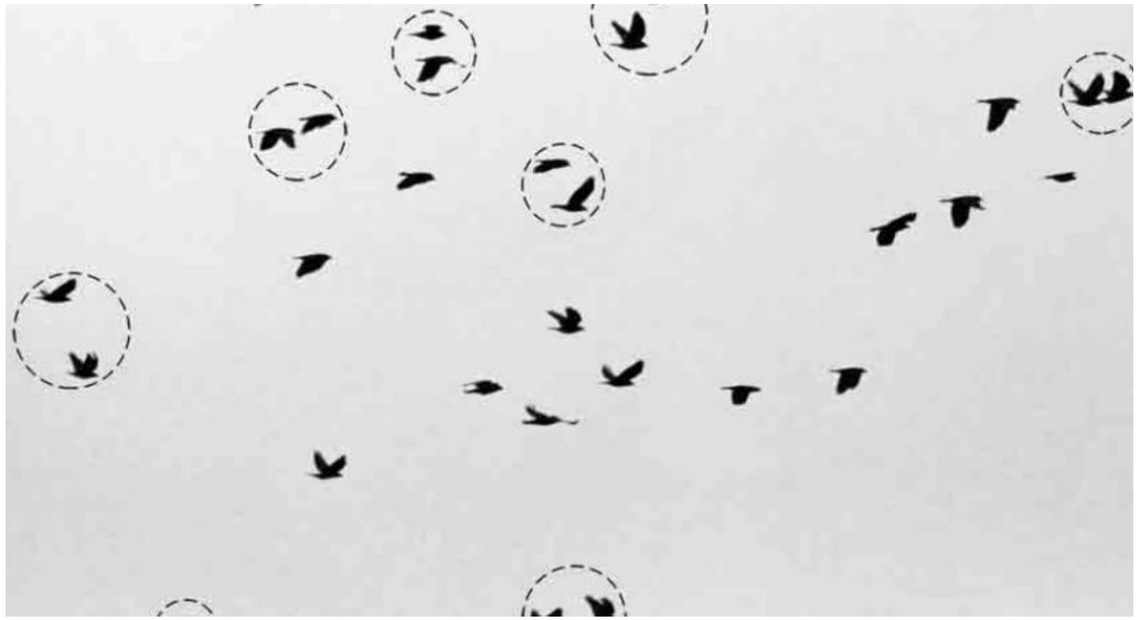


Integrating micro-physics to macro-phenomena is critical

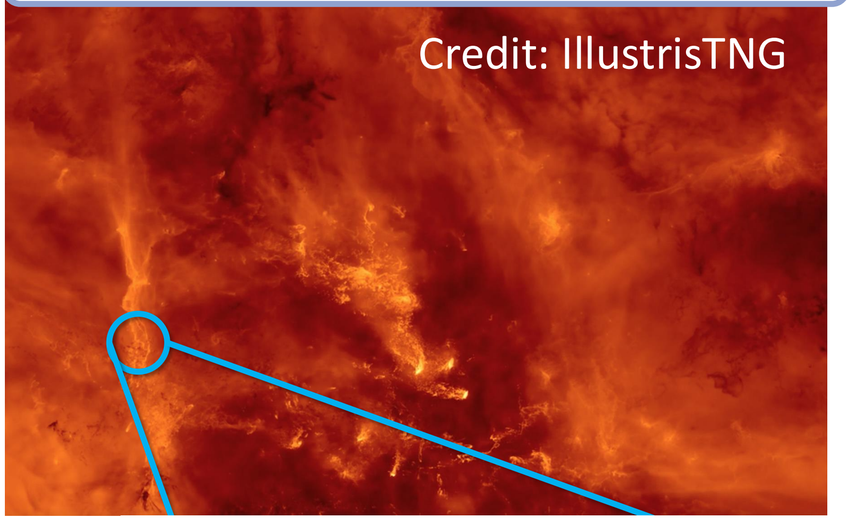
~100 m

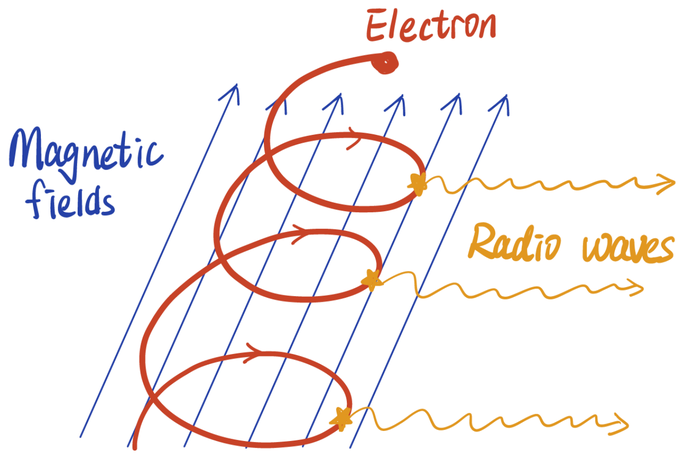


~10 cm



Macroscopic scales turbulent fluid





Microscopic scales of kinetic plasmas

~10^{17} km

~10^{3} km

Fluid v.s. kinetic description of plasma

Magnetohydrodynamics

(magnetized fluid description)

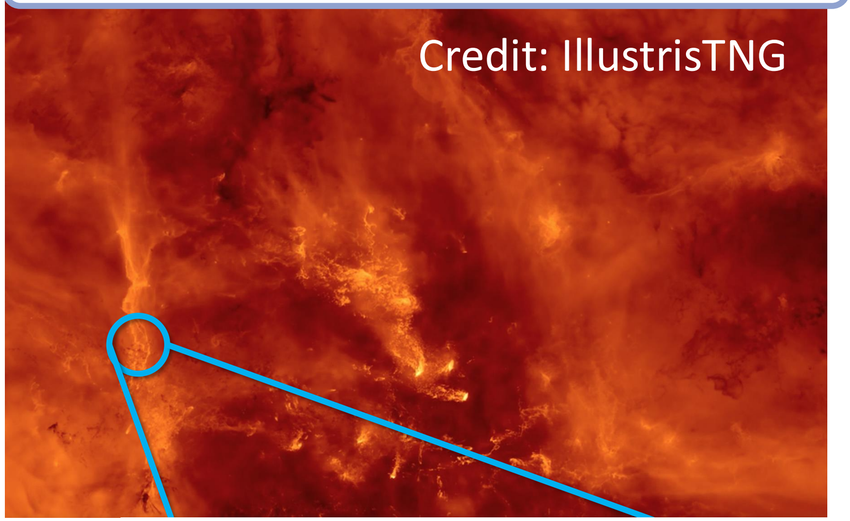
Fully kinetic description

Boltzmann-like equation without collision

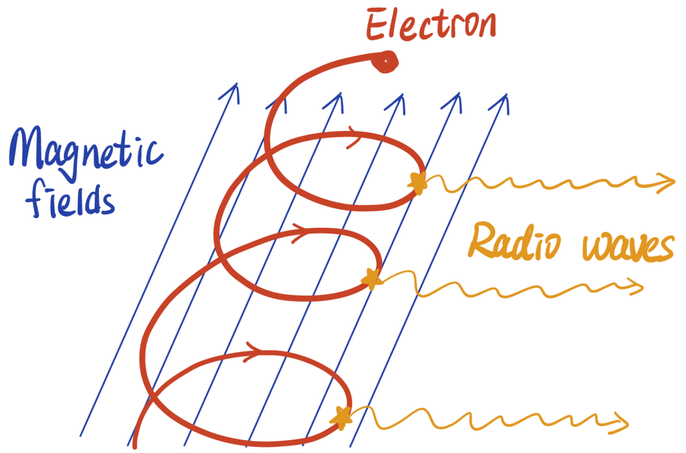
$$\frac{\partial f}{\partial t} + \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \cdot \frac{\partial f}{\partial \mathbf{r}} + \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} \cdot \frac{\partial f}{\partial \mathbf{p}} = 0$$

$f(\boldsymbol{x}, \boldsymbol{v}, t)$ is the distribution function of plasma in the 6-dimensional $(\boldsymbol{x} - \boldsymbol{v})$ phase space

Macroscopic scales turbulent fluid



$\sim 10^{17}$ km



$\sim 10^{3}$ km

Microscopic scales of kinetic plasmas

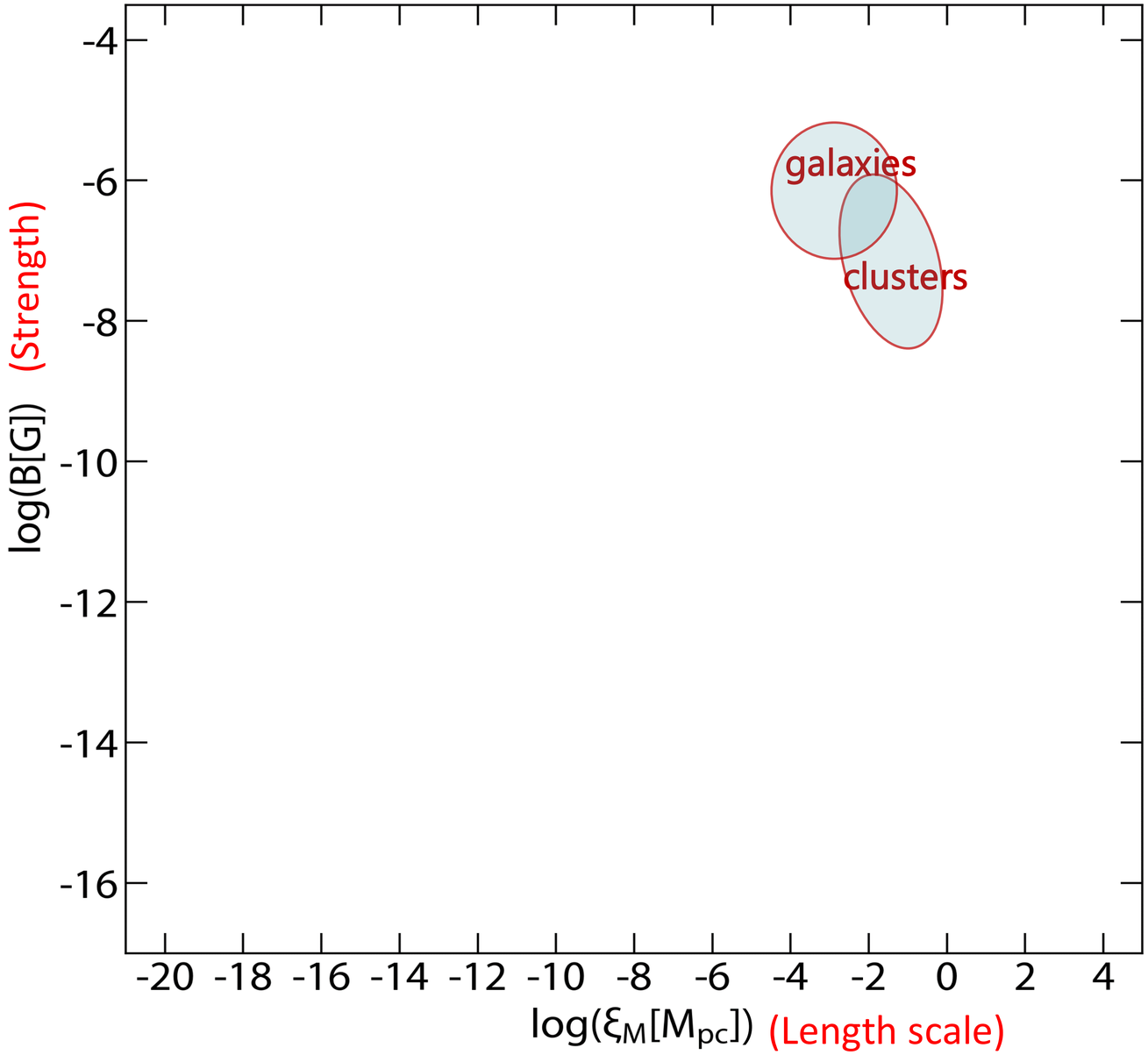
Velocity moments of Vlasov equation

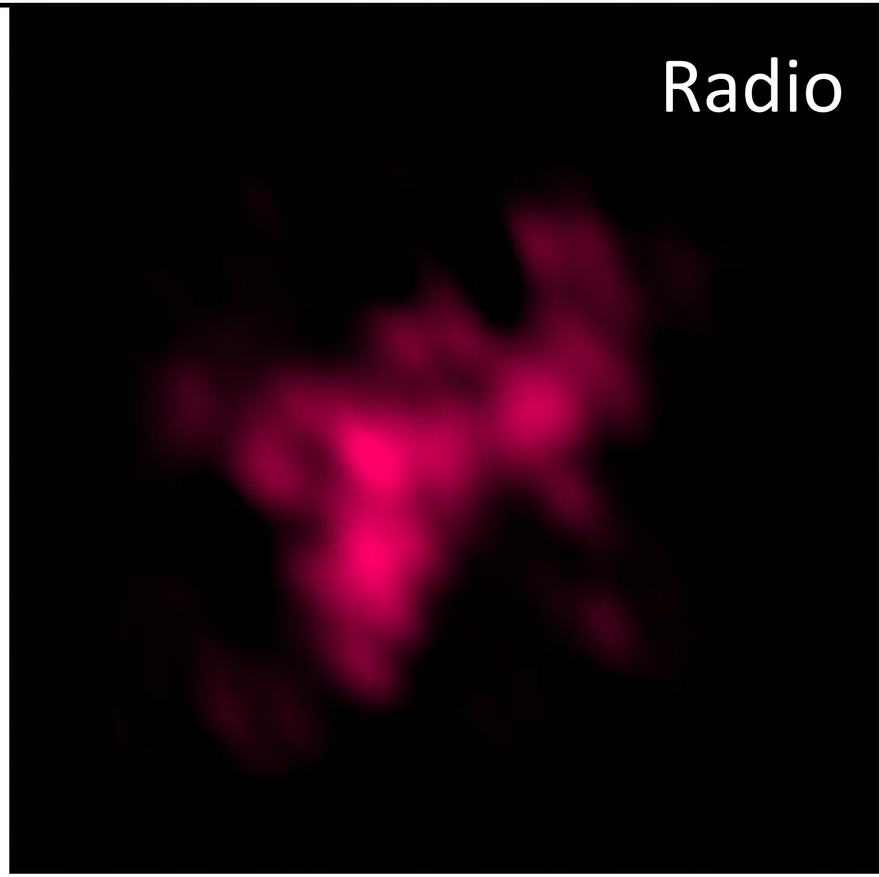
For species s: $\dfrac{\partial f_s}{\partial t} + \mathbf{v}\cdot\nabla f_s + \dfrac{q_s}{m_s}(\mathbf{E}+\mathbf{v}\times\mathbf{B})\cdot\nabla_v f_s = 0$

Moment of f_s	Equation obtained
$m=0:\quad n_s = \int f_s\, d^3v$	**Continuity equation** $\dfrac{\partial n_s}{\partial t} + \nabla\cdot(n_s\mathbf{u}_s) = 0$
$m=1:\quad n_s\mathbf{u}_s = \int \mathbf{v} f_s\, d^3v$	**Momentum equation** $m_s n_s \dfrac{D_s\mathbf{u}_s}{Dt} = q_s n_s(\mathbf{E}+\mathbf{u}_s\times\mathbf{B}) - \nabla\cdot\mathsf{P}_s$
$m=2:\quad P_{ij} = m_s\int c_i c_j f_s\, d^3v$ Scalar pressure: $p_s = \mathrm{Tr}(\mathsf{P}_s)/3$	**Pressure-tensor equation** $\dfrac{D_s\mathsf{P}_s}{Dt} + \mathsf{P}_s(\nabla\cdot\mathbf{u}_s) + \mathsf{P}_s\cdot\nabla\mathbf{u}_s + (\mathsf{P}_s\cdot\nabla\mathbf{u}_s)^T = -\nabla\cdot\mathsf{Q}_s + \dfrac{q_s}{m_s}\mathcal{R}_B(\mathsf{P}_s)$
$m=3:\quad Q_{ijk} = m_s\int c_i c_j c_k f_s\, d^3v$	**Third-moment equation** Its evolution depends on the fourth moment ...

Cosmic magnetic fields

Large coherent structure and strong fields are observed



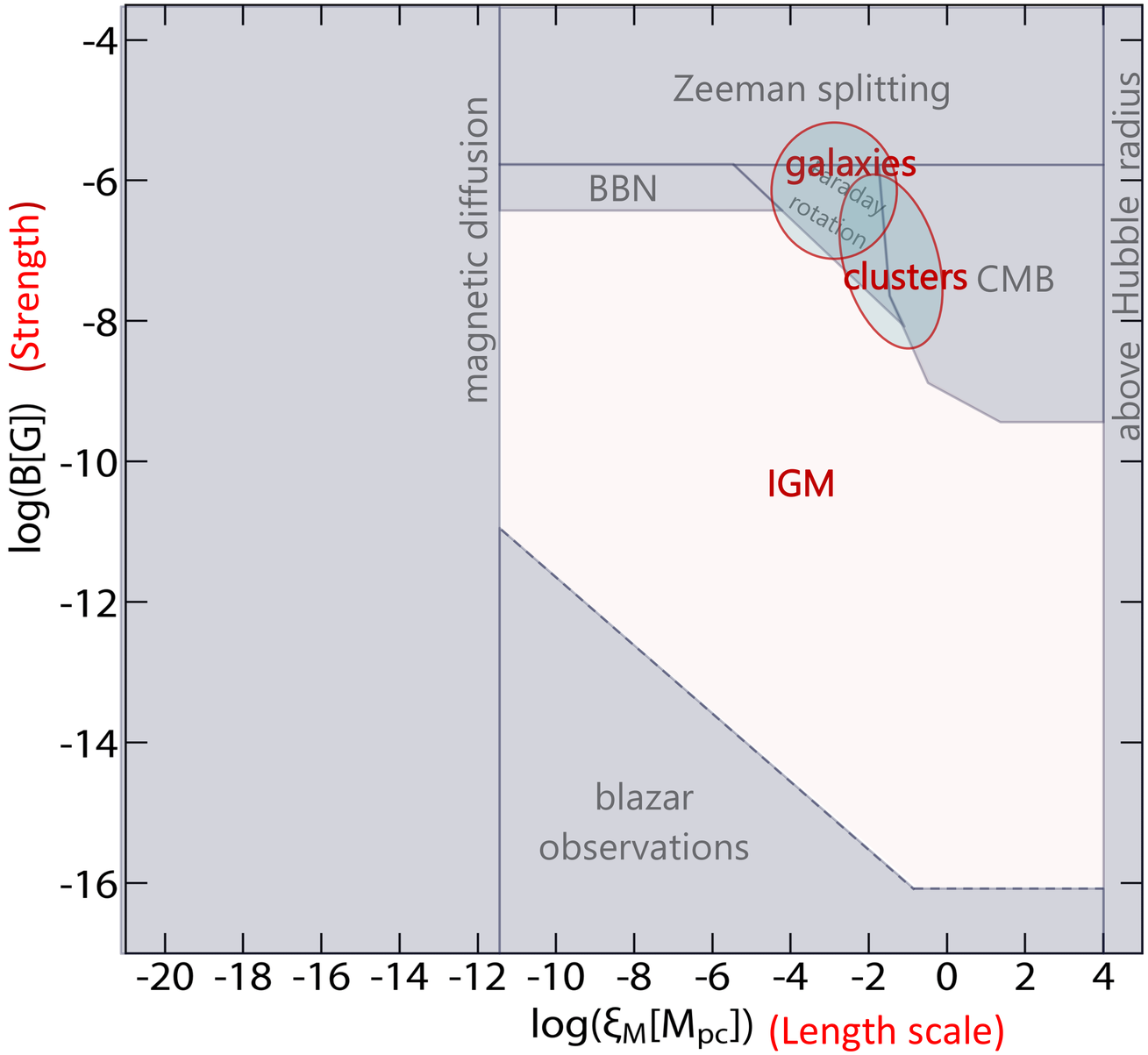


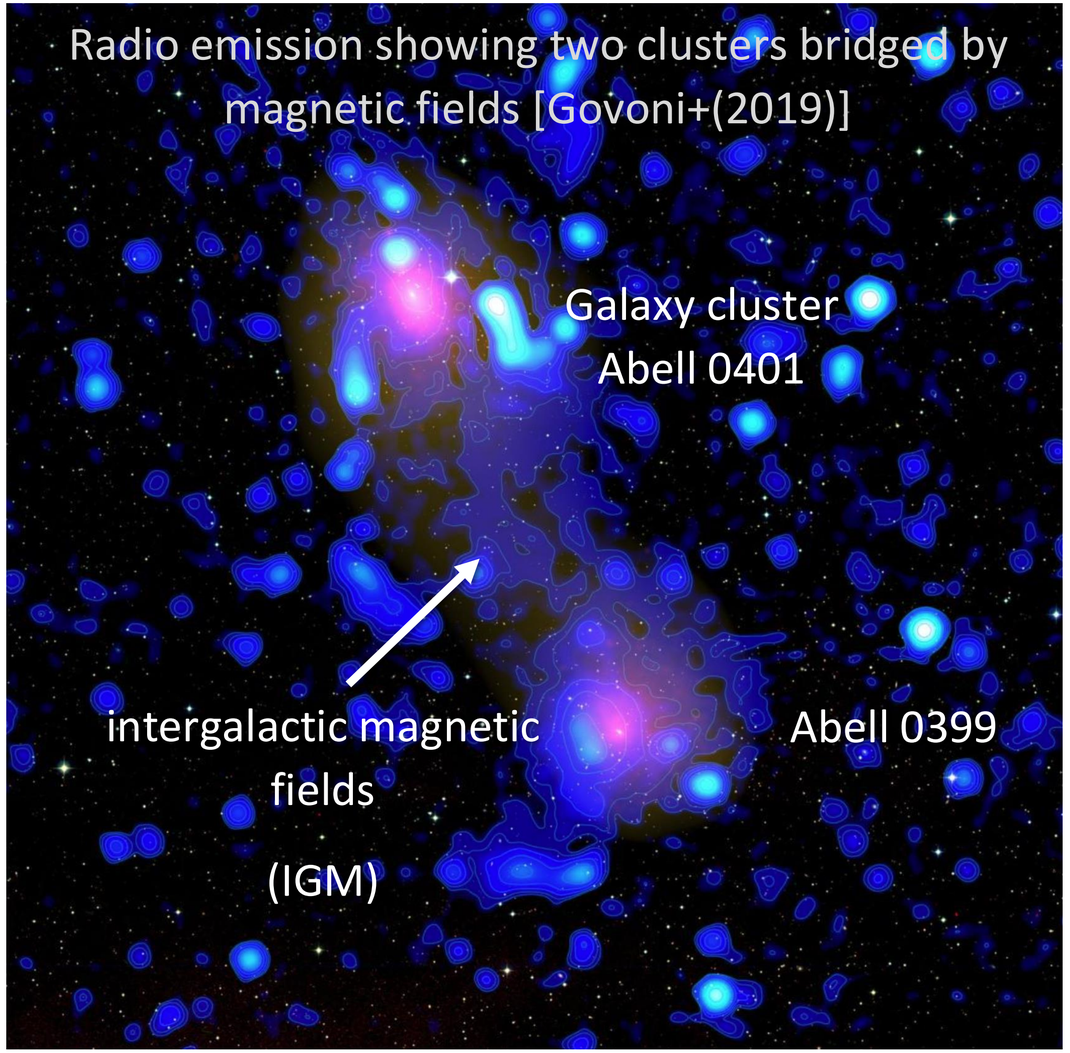
Galaxy cluster Abell 1758

[Giant Metrewave Radio Telescope]

Cosmic magnetic fields

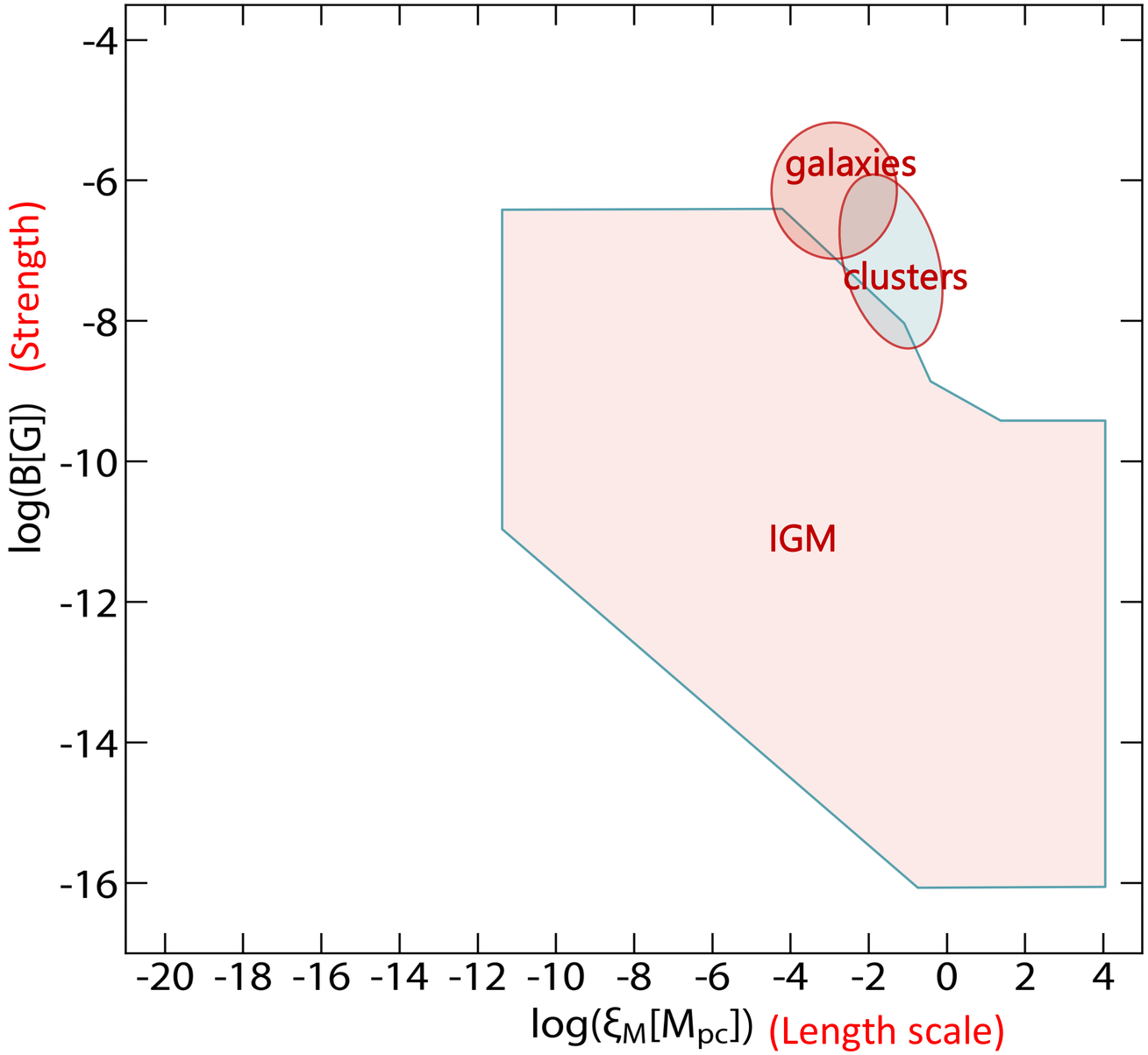
Large coherent structure and strong fields are observed





[Observational constraints are adapted from Neronov *et al.* 2010]

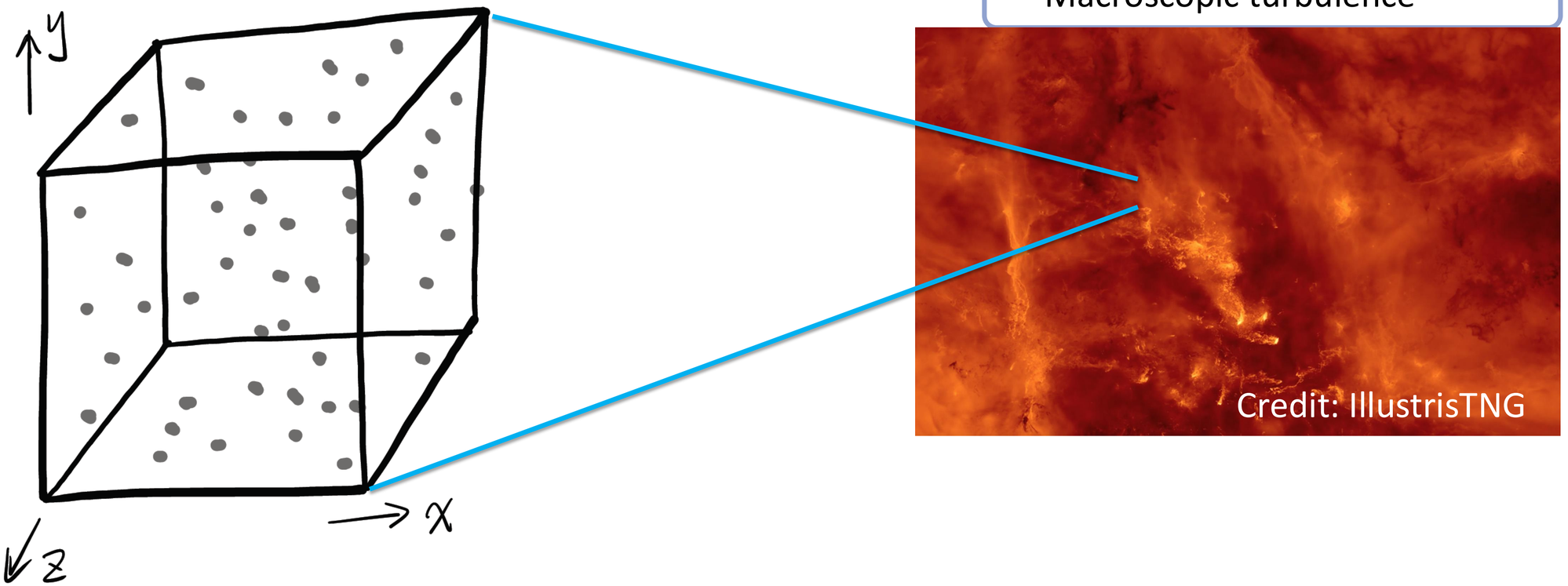
Main Questions



- **Magneto-genesis**: Where are the first magnetic fields from? How do cosmic plasmas get magnetized in the first place?
- **Dynamo**: How do microscopic turbulent flows efficiently amplify the seed magnetic fields? How does a collisionless dynamo work?
- **Relaxation**: How fast do dynamo-generated magnetic fields decay if no longer stirred? What equilibrium do they relax to?

Origin of cosmic magnetic fields? --Magnetogenesis

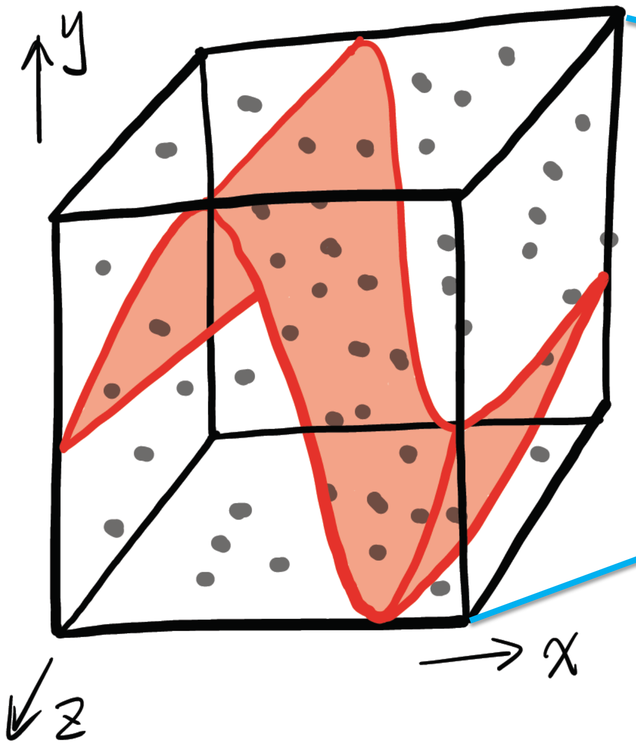
Initial condition: uniform Maxwellian plasma, no magnetic fields.



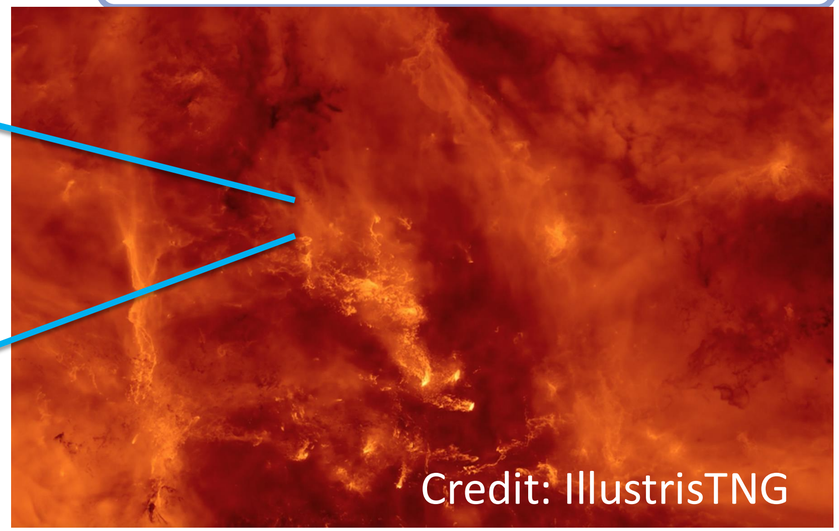
How are microscopic seed fields generated by macroscopic flows?

Initial condition: uniform Maxwellian plasma, no magnetic fields.

Applying constant system-scale shear force $\mathbf{F}_{\mathrm{ext}}$



Macroscopic turbulence

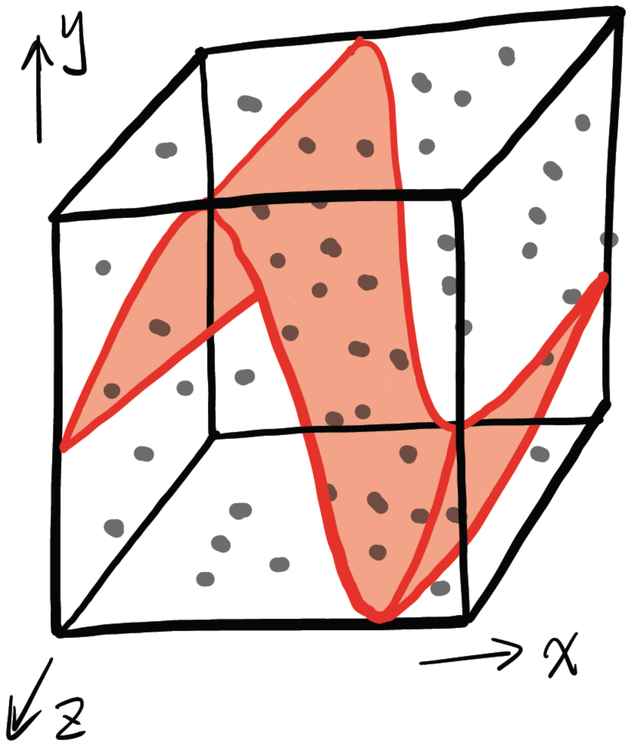


Consider a system driven by a shear force

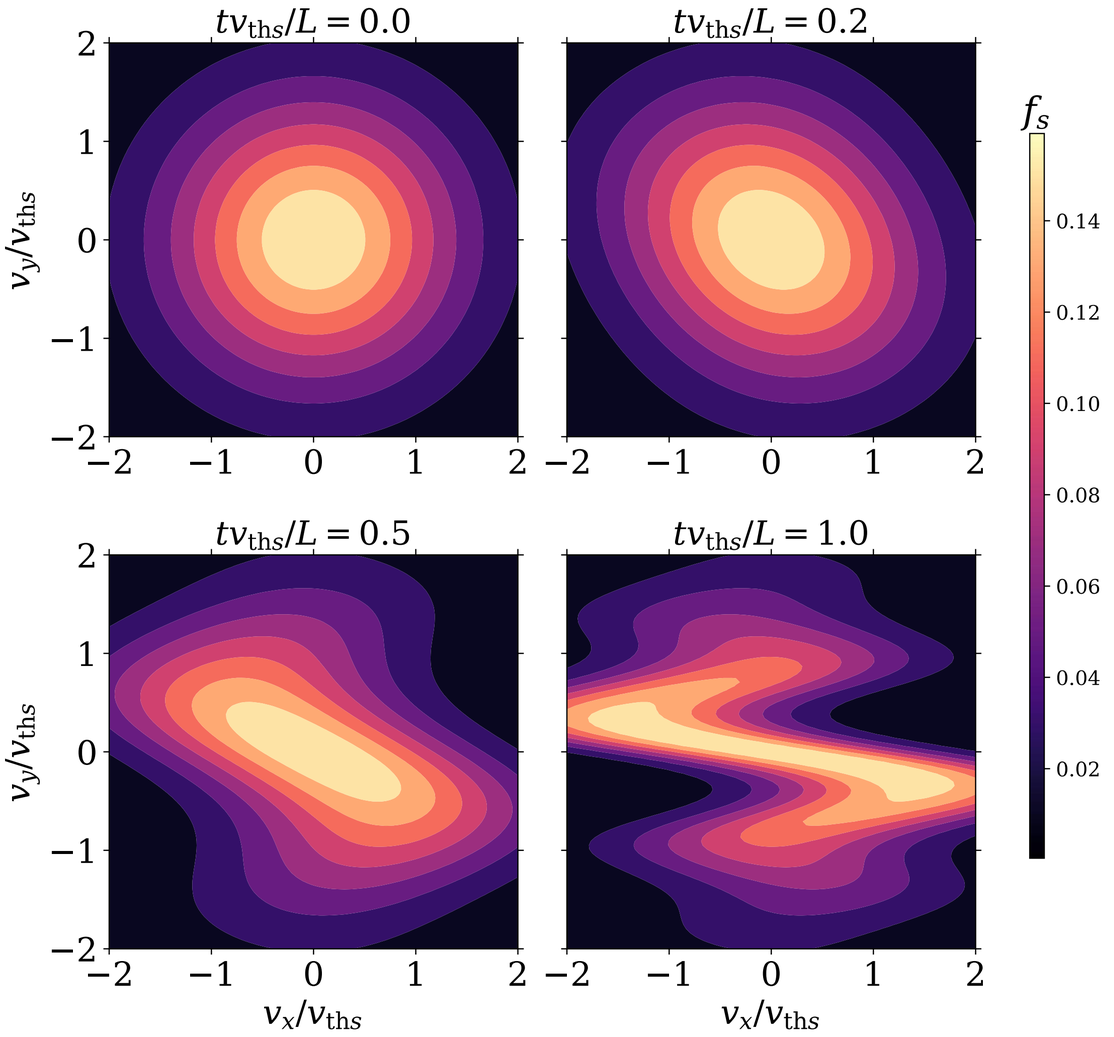
Initial condition: uniform Maxwellian plasma, no magnetic fields.

Applying constant system-scale shear force $\mathbf{F}_{\text{ext}}$

$$\frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + \frac{\mathbf{F}_{\text{ext}}}{m} \frac{\partial f}{\partial v_y} = 0$$



Analytical solution of distribution function – deviating from Maxwellian



[Zhou+(2022) *PNAS*]

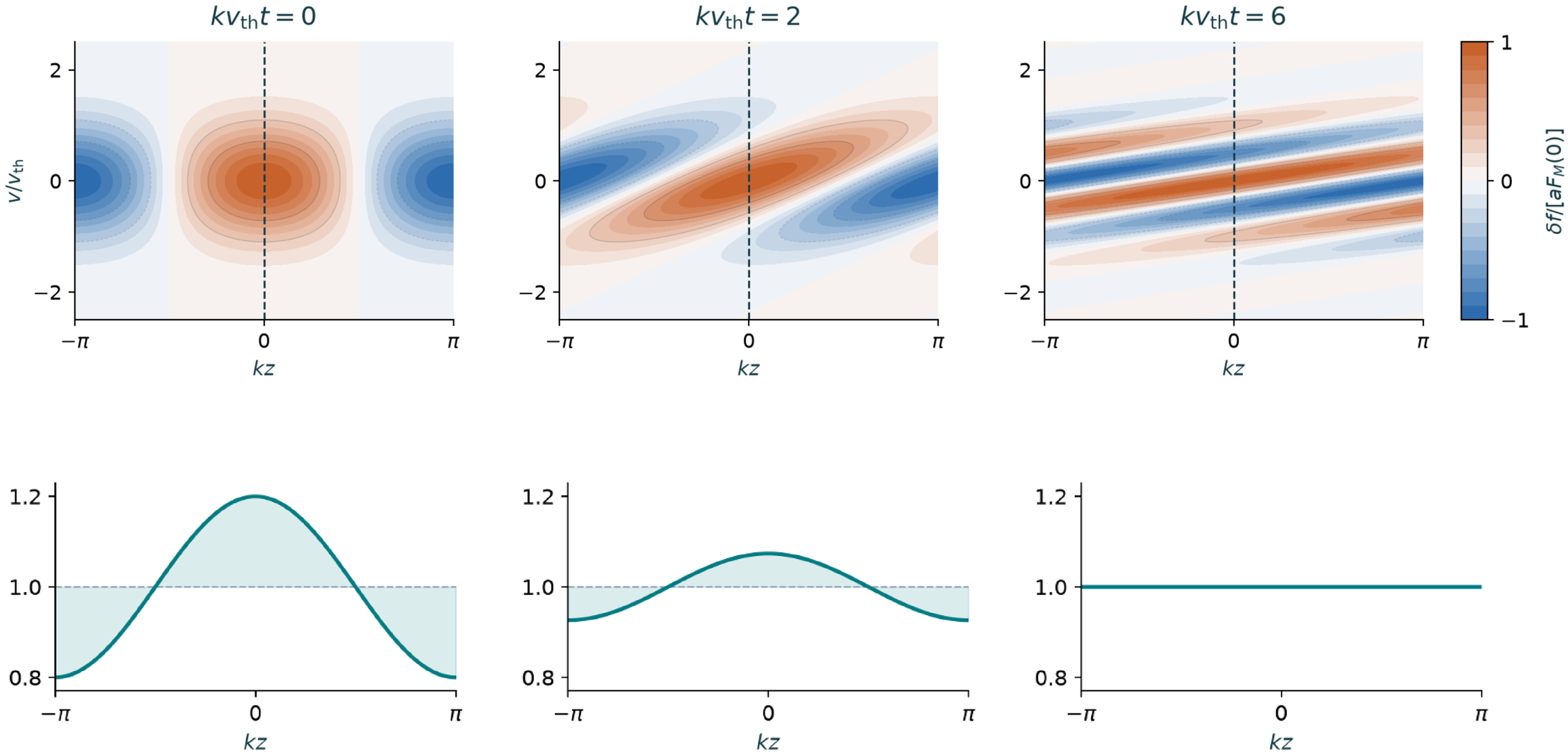
$$\frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + \frac{\mathbf{F}_{\rm ext}}{m} \frac{\partial f}{\partial v_y} = 0$$

Exact solution:

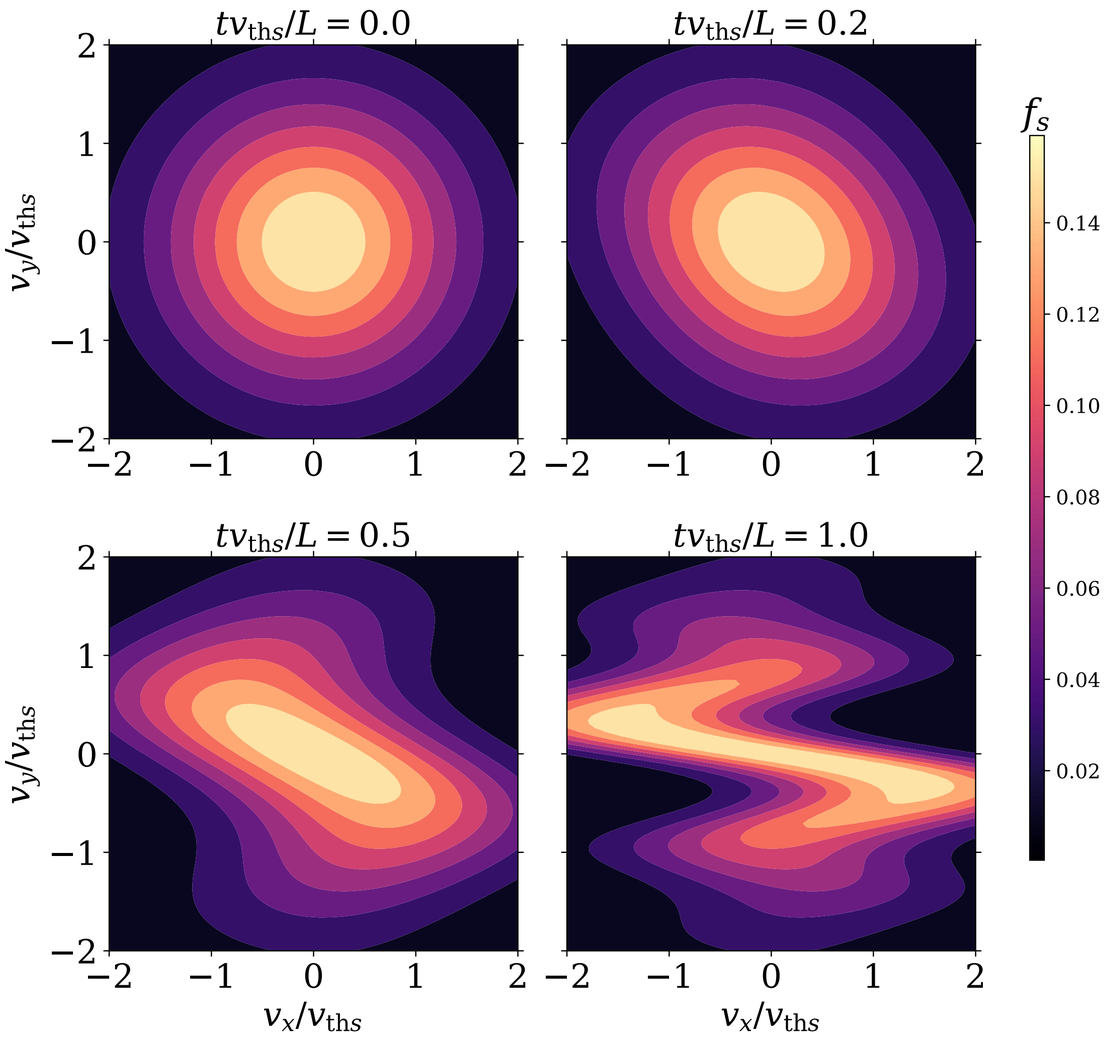
$$f_s(t, x, \boldsymbol{v}) = f_{{\rm M},s}\left(\sqrt{v_x^2 + v_z^2 + \widetilde{v_y}^2}\right),$$

$$\widetilde{v_y} \equiv v_y + \frac{La_0}{2\pi v_x}\left[\cos\left(\frac{2\pi}{L}x\right) - \cos\left(\frac{2\pi}{L}(x - v_x t)\right)\right]$$

Phase mixing



Free energy can be released through instabilities



[Zhou+(2022) *PNAS*]

Phase mixing of the large-scale inhomogeneity leads to the complex structure in the velocity space.

Thermal pressure anisotropy develops --- free energy build up in the plasma!

Plasma has the intendency to release the free energy and relax back to equilibrium

Seed field generation by Weibel instability

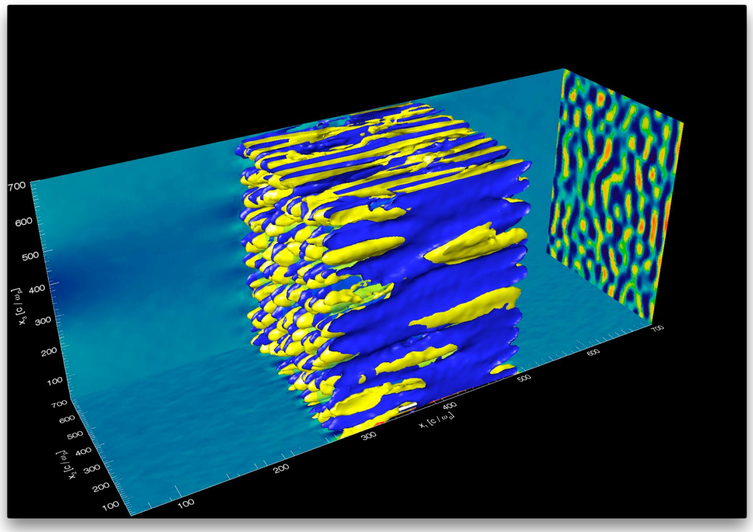
Weibel Instability

Generate seed magnetic fields from unmagnetized plasmas.

Generate strong seed fields, but only at length scales of plasma skin depth d_e.

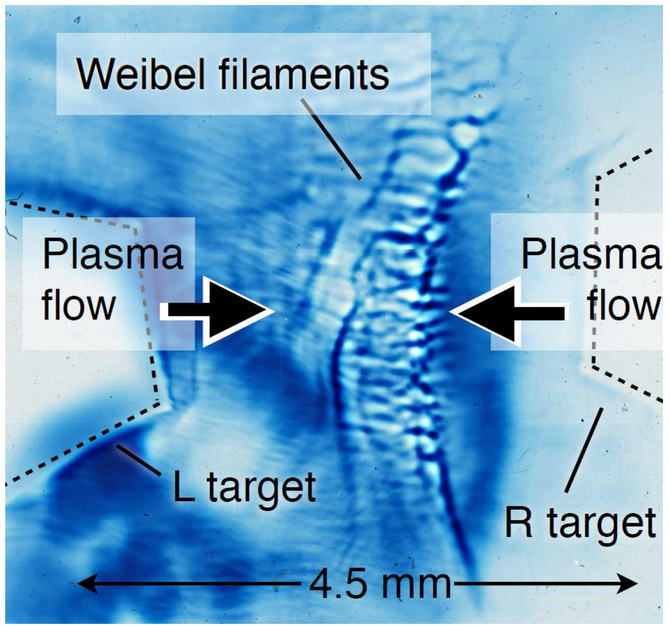
Only necessary condition: anisotropy in plasma distributions.

[Erich Weibel 1959; Burton Fried 1959]



kinetic simulations

[Silva *et al*. 2003]



laser experiments

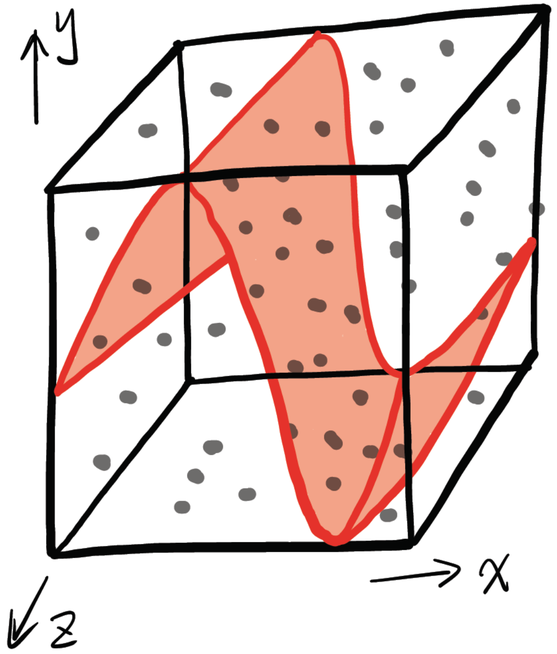
[Huntington *et al.* 2015]

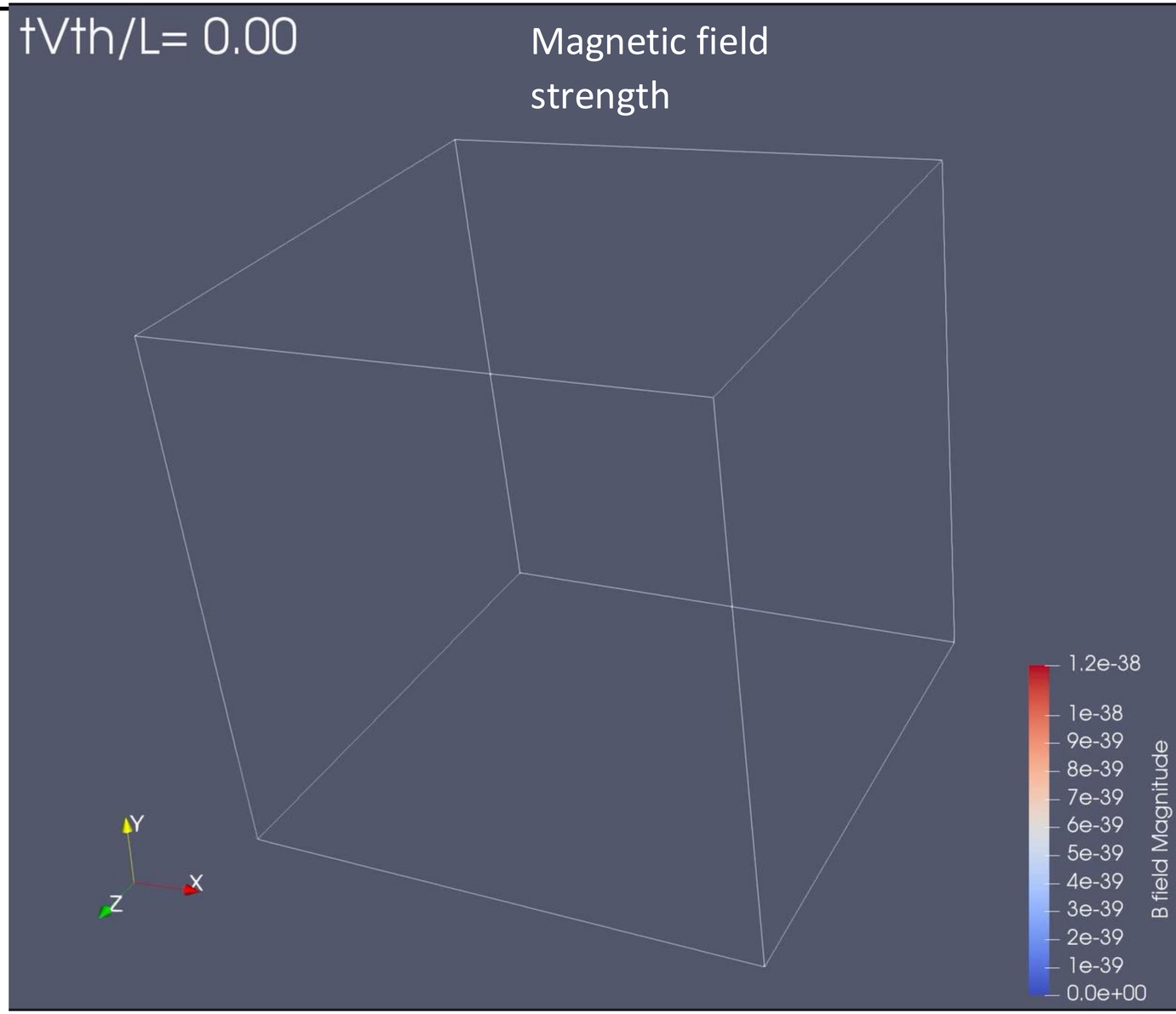
A numerical experiment: Let's shear an unmagnetized plasma!

3D particle-in-cell (PIC) simulation

--- numerically solving the first-principles Vlasov equation

For simplicity, using electron-positron plasmas

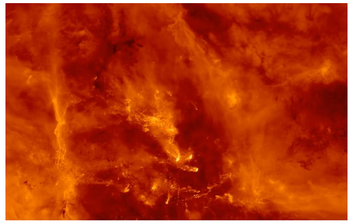




[Zhou+(2022) *PNAS*]

Fast, micro. Weibel instability grows on macro. slowly evolving background

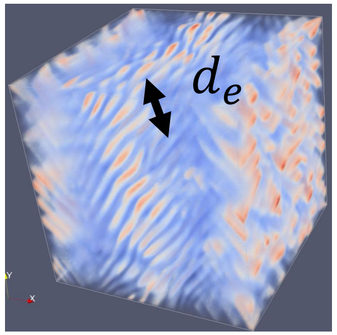
Macroscopic



L

Shear flow

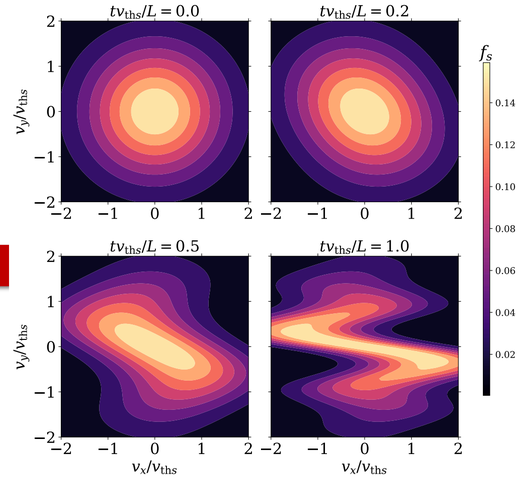
L/U~ Gyr



Microscopic

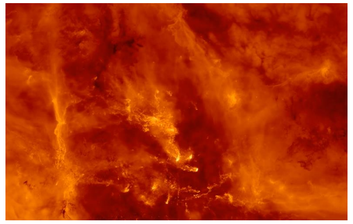
Weibel

$d_e/c \sim 10^{-3}$s



Massive scale separation allows perturbation theory for Weibel seed fields

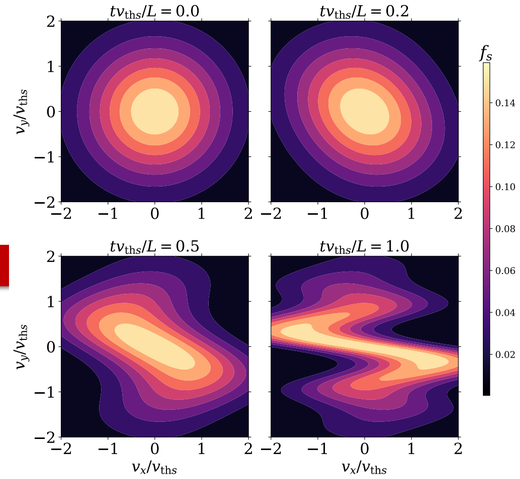
Macroscopic

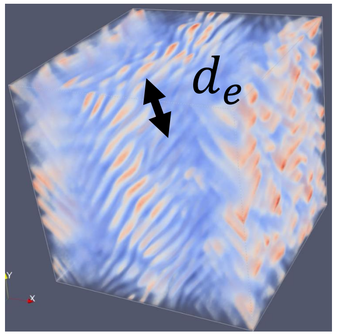


L

Shear flow

$L/U \sim$ Gyr





Microscopic

Weibel

$d_e/c \sim 10^{-3}$s

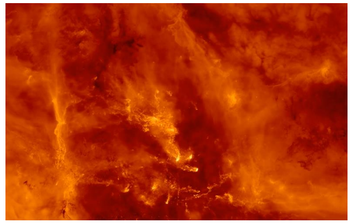
Perturbative calculation + nonlinear argument for Weibel saturation

Estimation for the Weibel seed fields in an ICM-like system

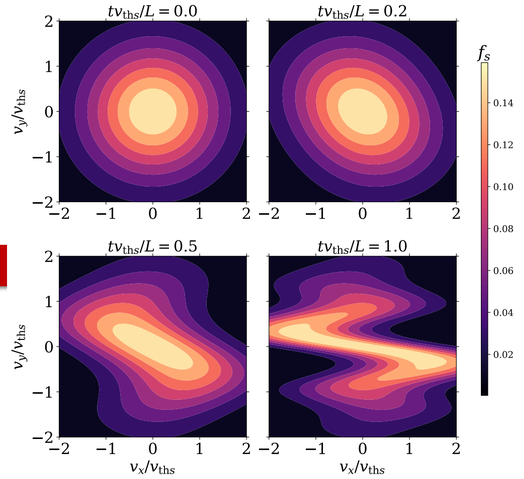
Macroscopic

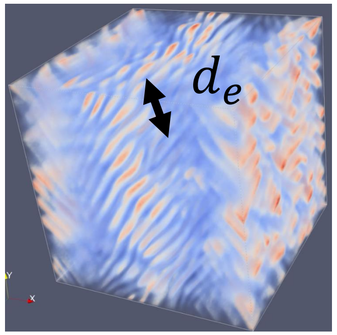
Shear flow

L/U~ Gyr



L





d_e

Microscopic

Weibel

d_e/c ~ 10^{-3}s

Perturbative calculation + nonlinear argument for Weibel saturation

Magnetic energy $\beta_{\rm w}^{-1} \sim (L/d_e)^{-1/2}$

Length scale $l_{\rm w}/d_e \sim (L/d_e)^{1/4}$

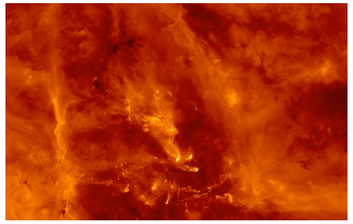
Scale separation

Estimation for the Weibel seed fields in an ICM-like system

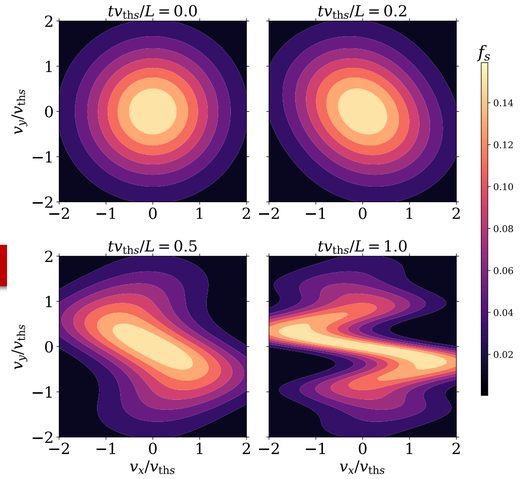
Macroscopic

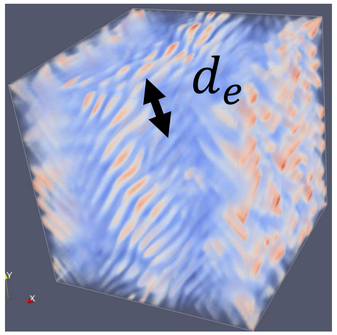
Shear flow

L/U~ Gyr



L





d_e

Microscopic

Weibel

d_e/c ~ 10^{-3}s

Perturbative calculation + nonlinear argument for Weibel saturation

Magnetic energy $\beta_{\rm w}^{-1} \sim (L/d_e)^{-1/2}$

Length scale $l_{\rm w}/d_e \sim (L/d_e)^{1/4}$

Scale separation

In the hot and dilute ICM: $L \gtrsim 10$ kpc $\quad d_e \sim 10^{-12}$ pc

Leading to: B fields ~ 0.1 nG $\gg 10^{-20}$G

Primordial fields

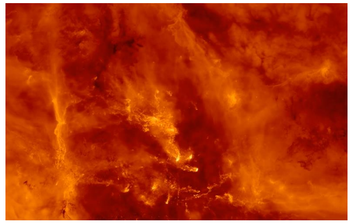
on the scale $\sim 10^4 d_e$

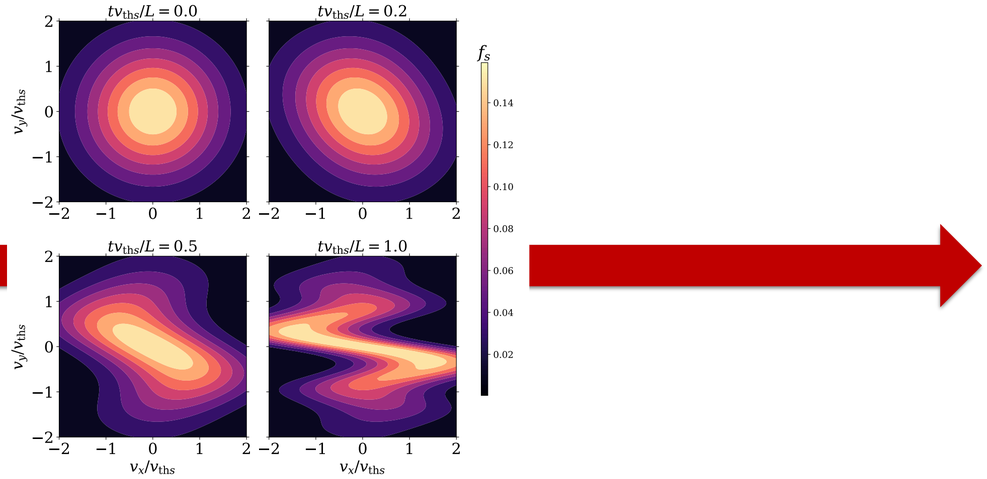
Plasmas are spontaneously magnetized under macro. flows

Macroscopic

Shear flow

L/U~ Gyr

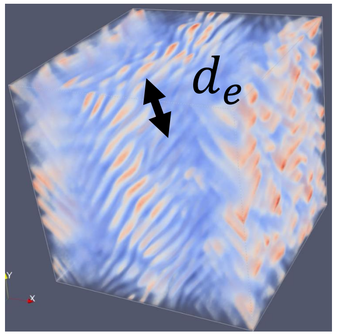




Microscopic

Weibel

d_e/c ~ 10^{-3}s



Perturbative calculation + nonlinear argument for Weibel saturation

Magnetic energy $\beta_{\rm w}^{-1} \sim (L/d_e)^{-1/2}$

Length scale $l_{\rm w}/d_e \sim (L/d_e)^{1/4}$

Scale separation

In the hot and dilute ICM: $L \gtrsim 10$ kpc $\quad d_e \sim 10^{-12}$ pc

Leading to: B fields ~ 0.1 nG $\gg 10^{-20}$G

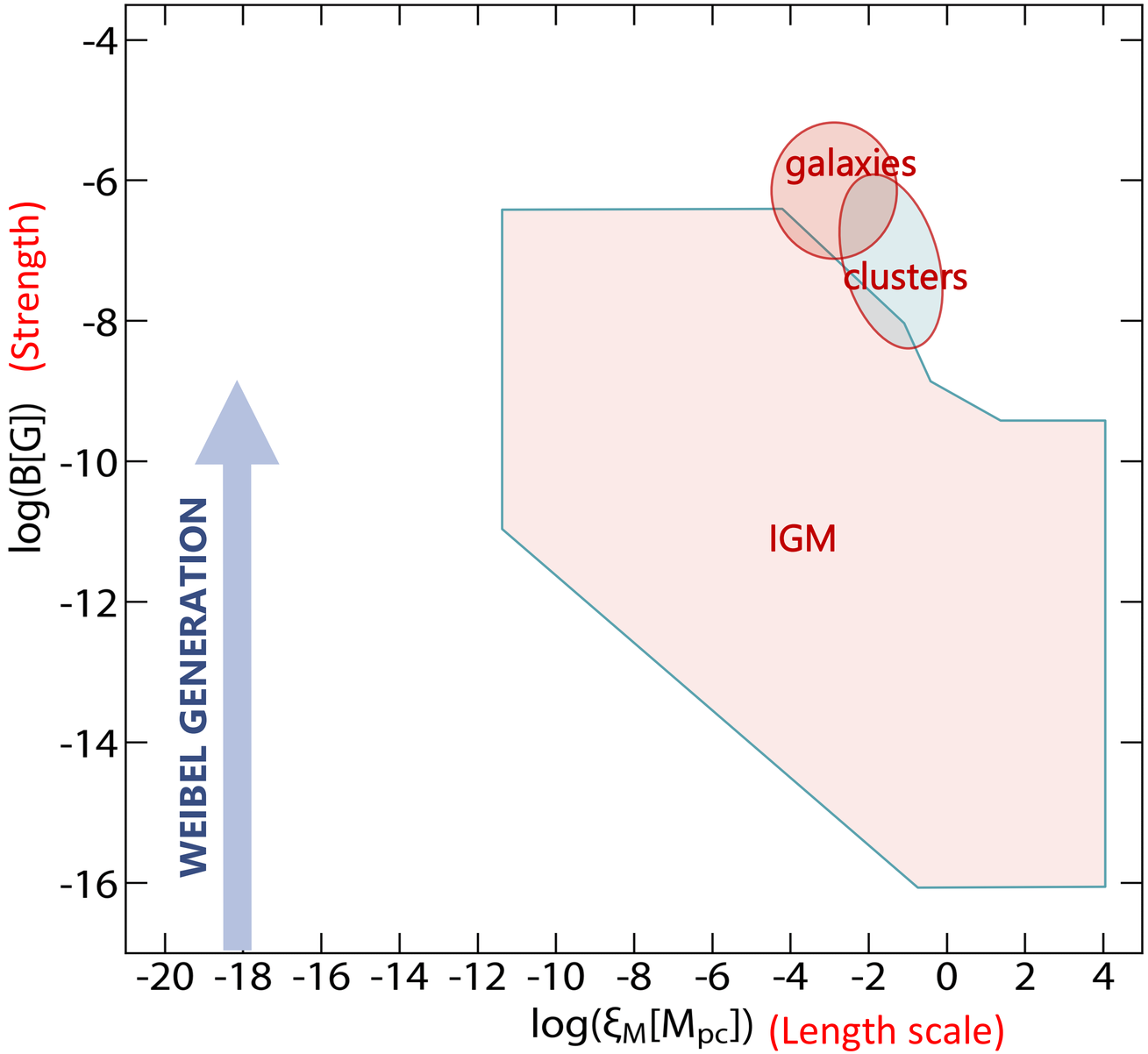
Primordial fields

on the scale $\sim 10^4 d_e$

Collisionless plasma can be spontaneously magnetized by large-scale motions as simple as a shear flow through the Weibel instability.

[Zhou+(2022) *PNAS*]

Generation of Weibel seed fields; plasma become magnetized

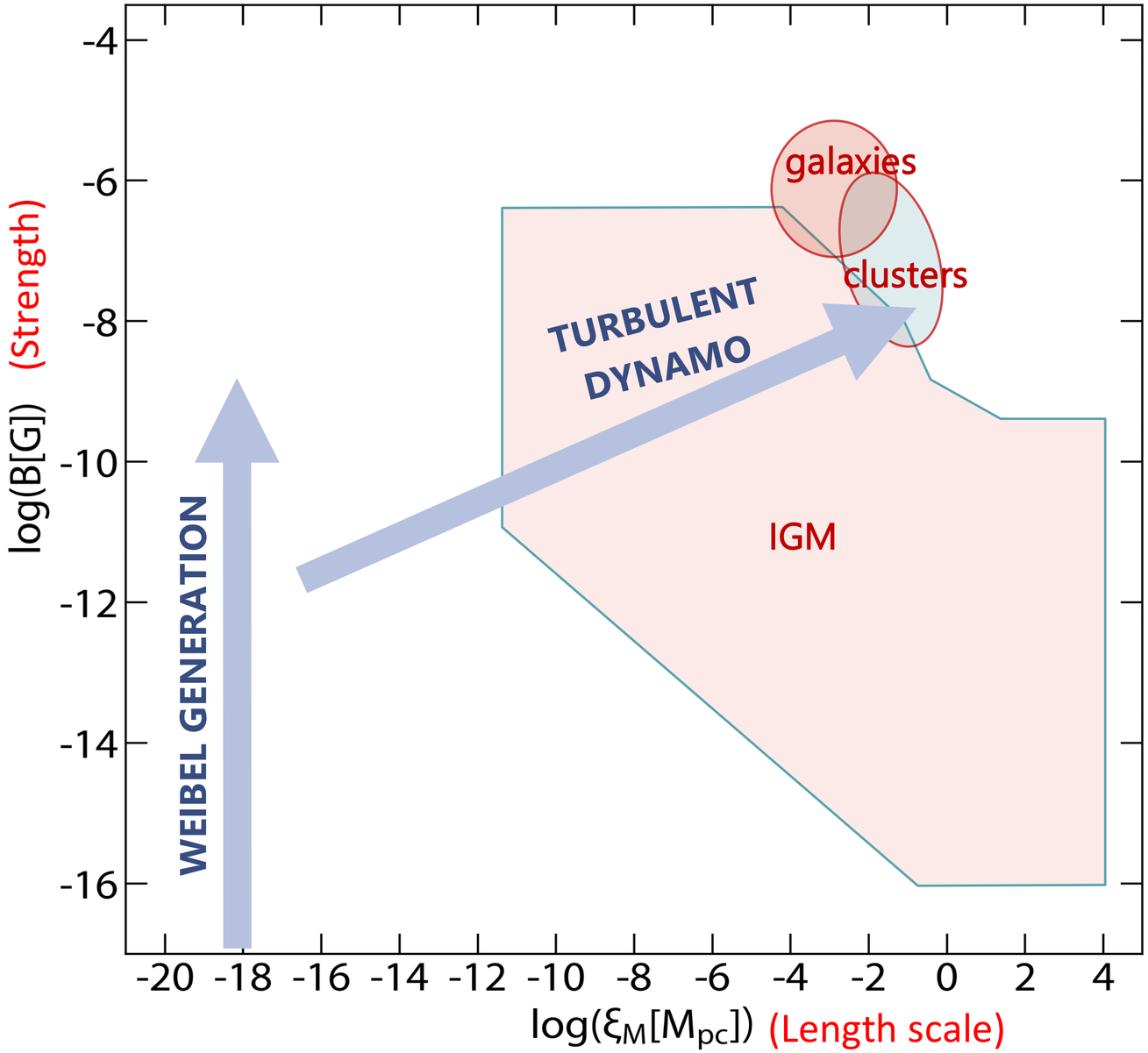


At $t = 0 + \epsilon$ *(almost instantaneously)*

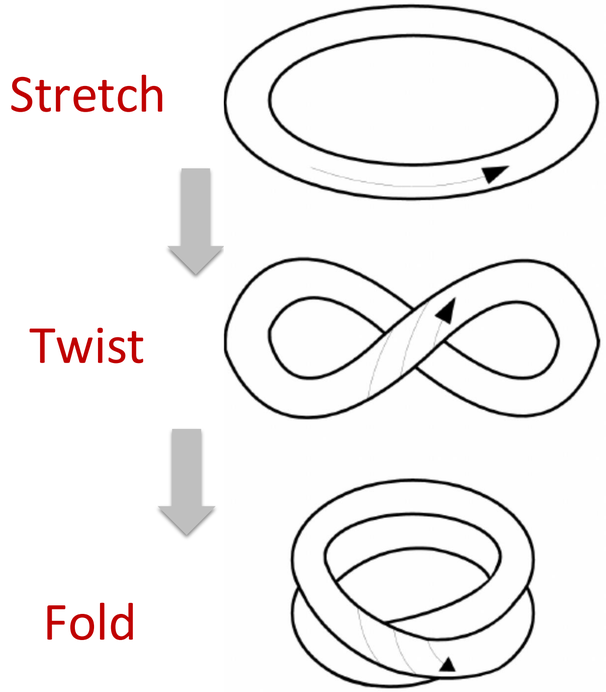
Weibel instability magnetizes the plasma

Seed fields ~ 0.1 nG to 1 nG in ICM

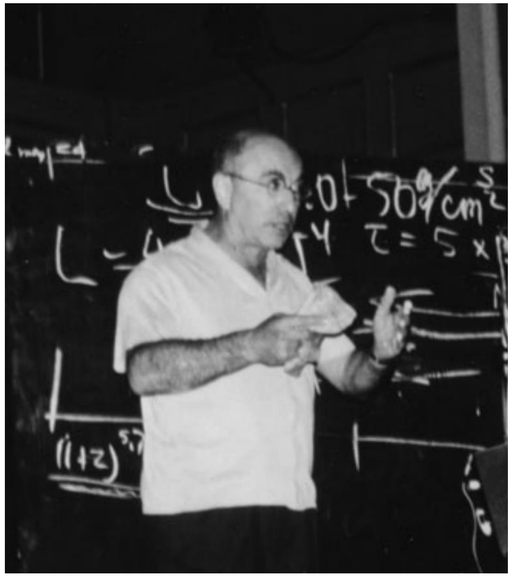
How can turbulent flow amply seed fields?



Fast plasma dynamo via “stretch” -- “twist” -- “fold”

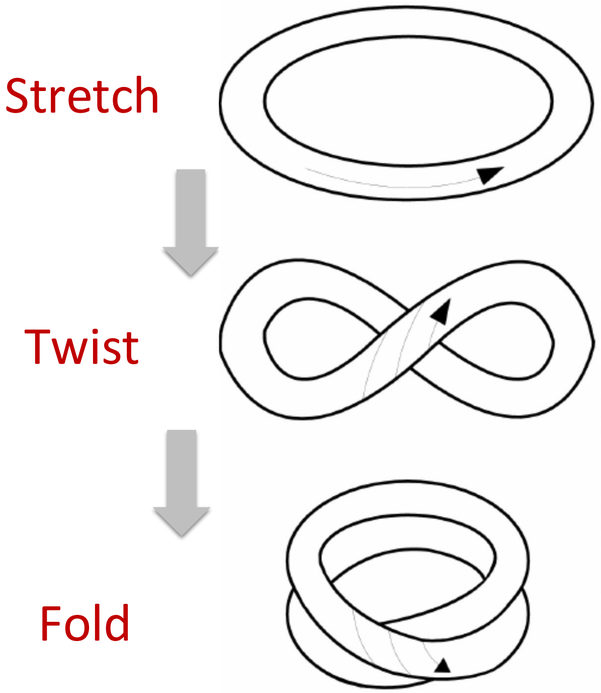


Ya. B. **Zeldovich**, 1967, at international astronomical union (IAU) meeting, Prague

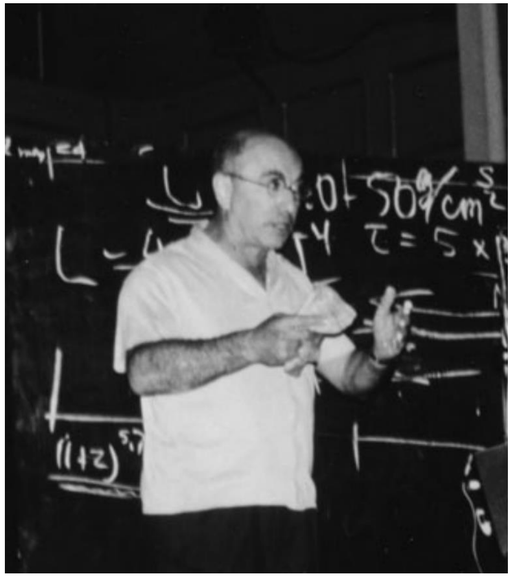


Fast plasma dynamo via “stretch” -- “twist” -- “fold”



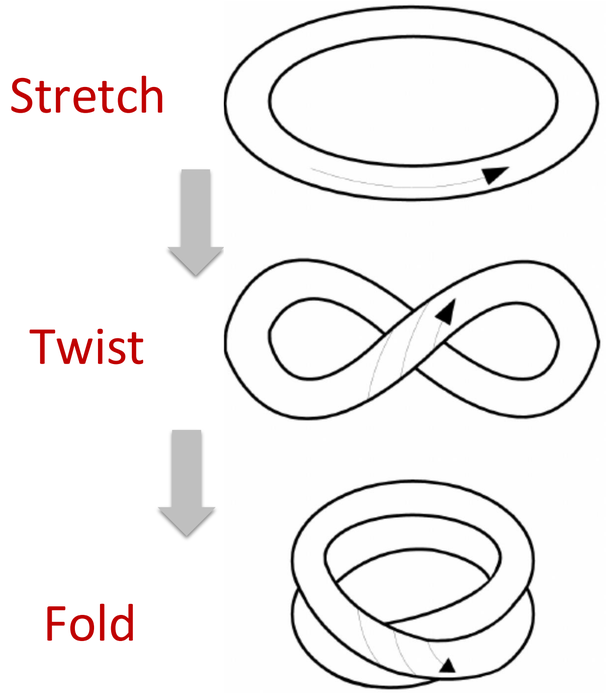


Ya. B. **Zeldovich**, 1967, at international astronomical union (IAU) meeting, Prague



Exponential growth of noodles/magnetic fields



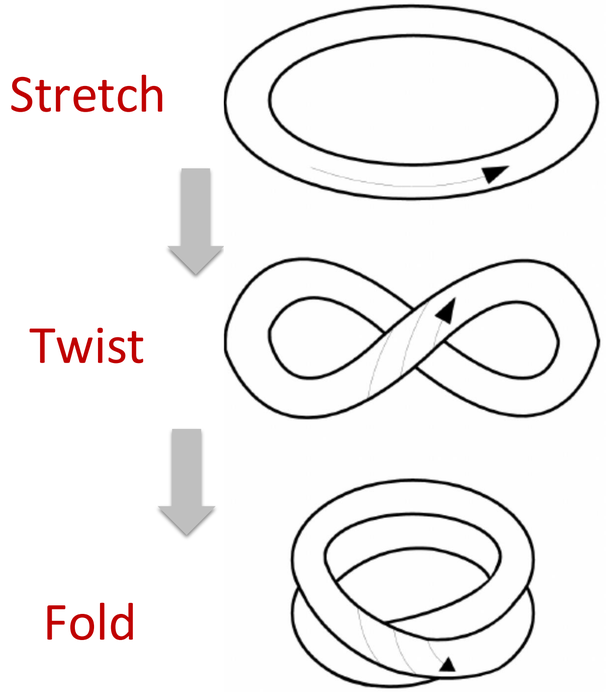


Number of noodles $= 2^n$

$$n = t/\tau_{\text{turnover}}$$

$$B = B_0 e^{t/\tau_{\text{turnover}}}$$

Faster the turnover time, faster the amplification



Number of noodles $= 2^n$

$$n = t/\tau_{\text{turnover}}$$

$$B = B_0 e^{t/\tau_{\text{turnover}}}$$

Small τ_{turnover} (faster turnover) → faster amplification

How to get a faster turnover?

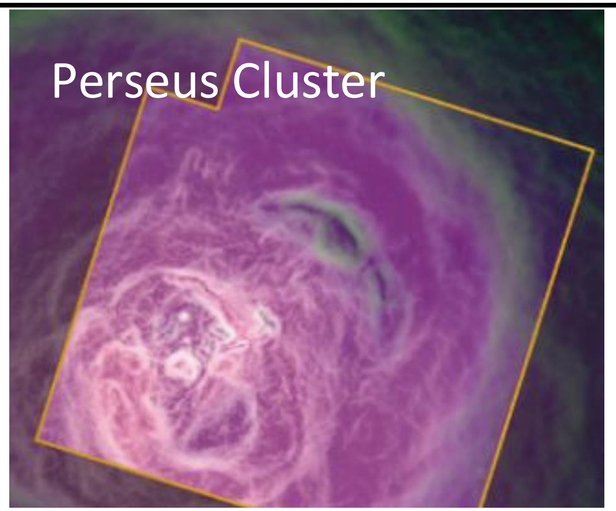
Turbulent cascade to small scales

At galaxy cluster scale: $\tau_{\text{turnover}} = \frac{\text{L}}{\text{U}} = \frac{10^{18}\text{km}}{150\text{km/s}} \sim \text{Gyr}$

Smaller viscosity → more turbulent

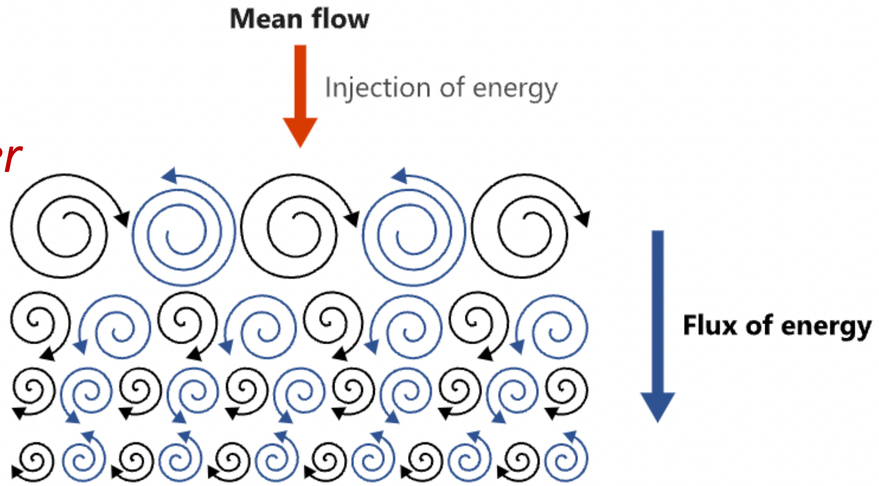
Turbulence is disordered/chaotic

It spans a large range of length scales and time scales.





Cascade of energy from large to small scales



Dissipate via viscosity

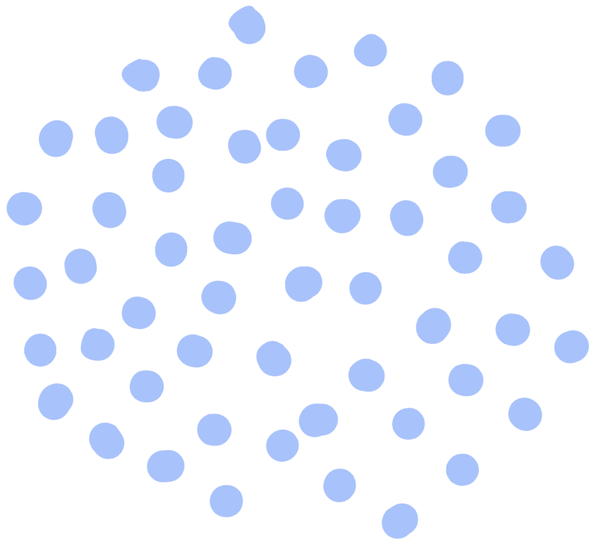


Small viscosity → turbulent

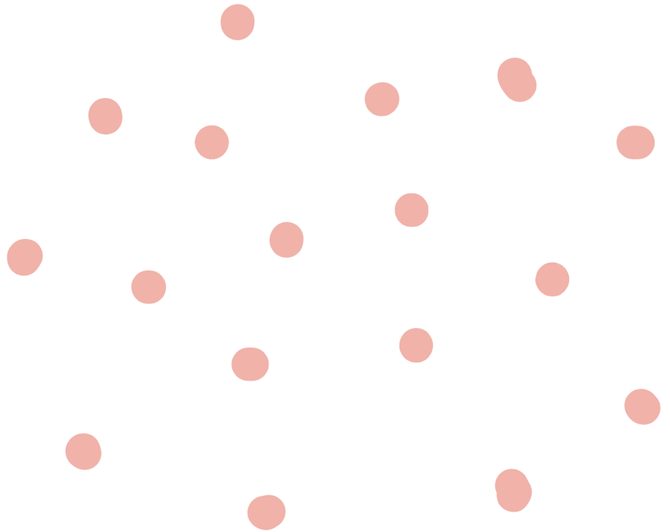
Large viscosity → laminar

What about viscosity for (ionized) gas?

High density, low temperature



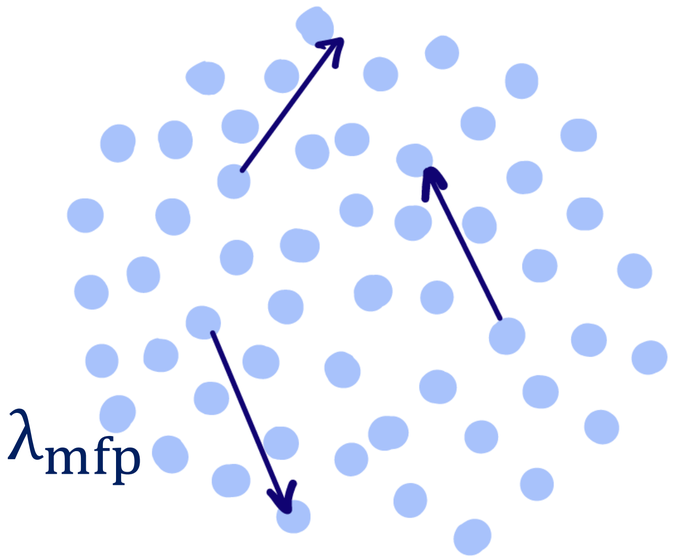
Low density, high temperature



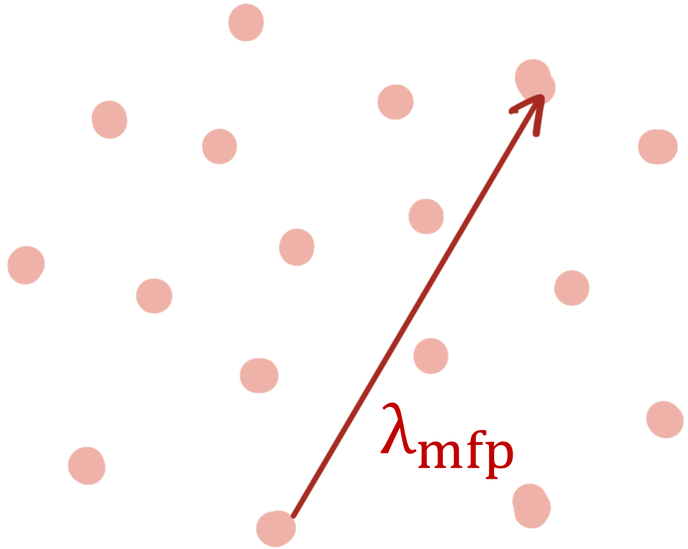
Which system has a higher viscosity?

Less collision, larger mean free path, larger viscosity

High density, low temperature



Low density, high temperature



Viscosity measures a material's resistance to deformation

This resistance is realized by momentum transfer

In gas or plasma, momentum transfer is done by (Coulomb) collisions

Coulomb collisions result in a large viscosity in ICM

Plasmas in intracluster/intergalactic (ICM/IGM) are weakly collisional.

When unmagnetized, considering only Coulomb collisions:

$$\text{viscosity} \quad \nu = v_{\mathrm{th}} \lambda_{\mathrm{mfp}} \rightarrow \text{large number}$$

How does turbulence develop in weakly collisional ICM/IGM?

Plasmas in intracluster/intergalactic (ICM/IGM) are weakly collisional.

When unmagnetized, considering only Coulomb collision:

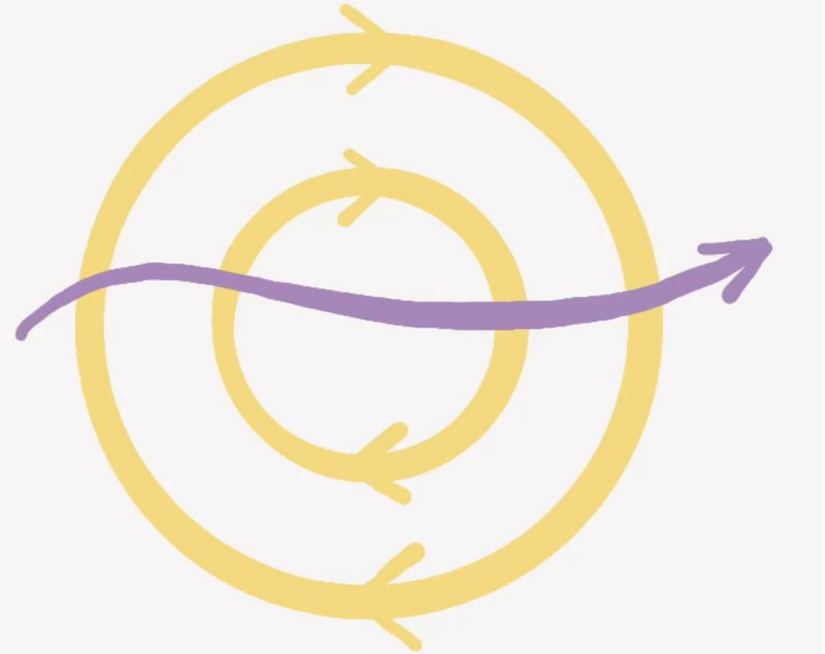
$$\text{viscosity} \quad \nu = v_{\mathrm{th}} \lambda_{\mathrm{mfp}} \rightarrow \text{large number}$$

Q: How does turbulence develop in ICM/IGM medium?

(i.e. how to attain a low viscosity, fast flow turnover?)

Include the *microscopic kinetic physics* of plasmas into *macroscopic astrophysical processes.*

“Stretching and folding” from a kinetic perspective

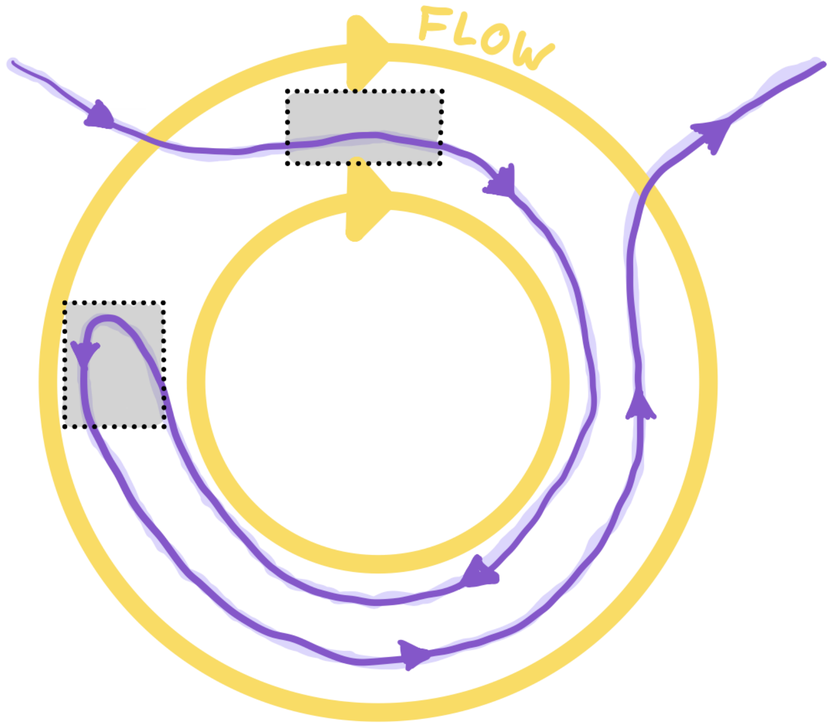


How to achieve a low viscosity, and thus a fast turnover in this system?

Local change of B (via stretching and folding) leads to local change of $\boldsymbol{P}$

Adiabatic invariance of magnetic moment of a particle: $\mu \equiv m\, P_{\perp}\, /2B$

($\perp$ and $\parallel$ with respective to B)

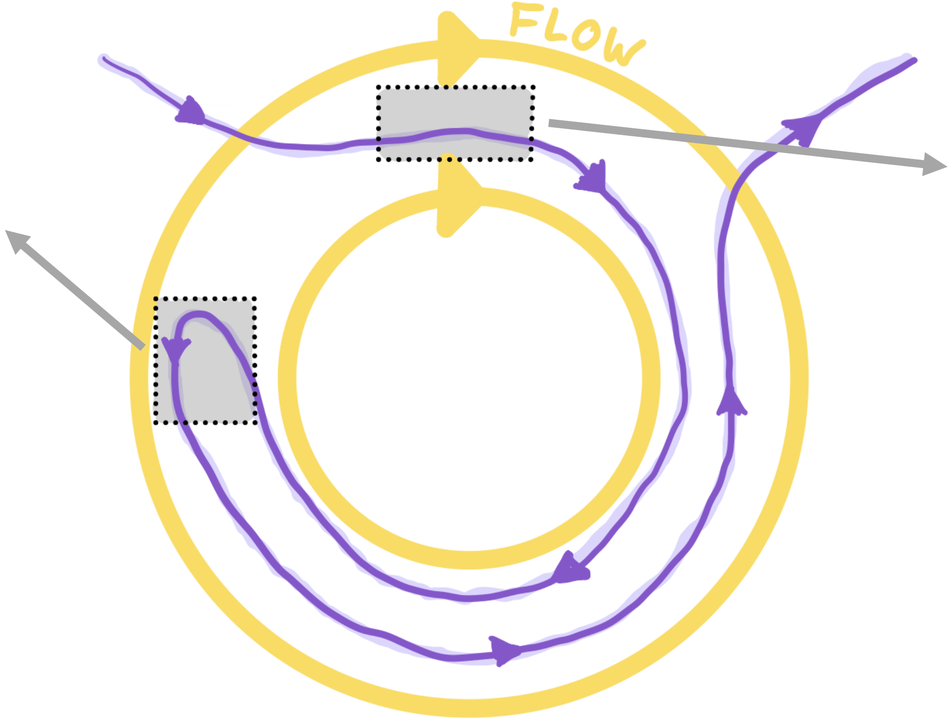


Adiabatic invariance of particles yields pressure anisotropy

Adiabatic invariance of magnetic moment of a particle: $\mu \equiv m\, P_\perp\, /2B$

($\perp$ and $\parallel$ with respective to B)

$B \downarrow \quad P_\perp \downarrow \quad P_\perp < P_\parallel$



$B \uparrow \quad P_\perp \uparrow \quad P_\perp > P_\parallel$

Microscopic instabilities are triggered by pressure anisotropy

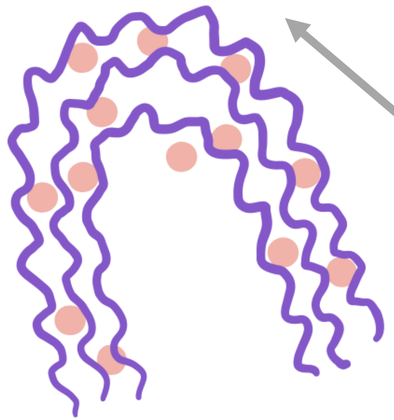
Adiabatic invariance of magnetic moment of a particle: $\mu \equiv m\,P_\perp\,/2B$

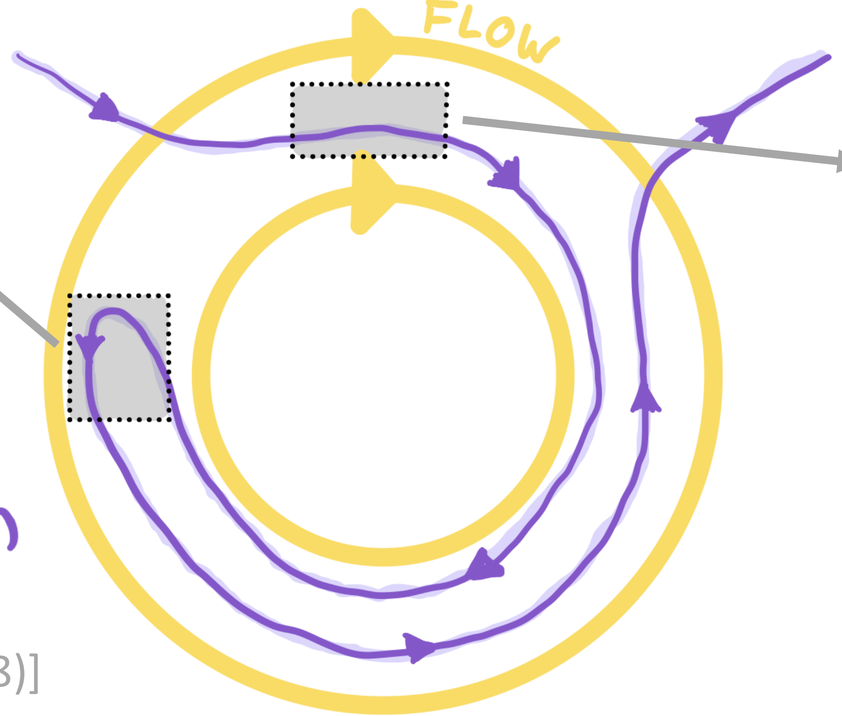
($\perp$ and $\parallel$ with respective to B)

$B\downarrow \quad P_\perp\downarrow \quad P_\perp < P_\parallel$

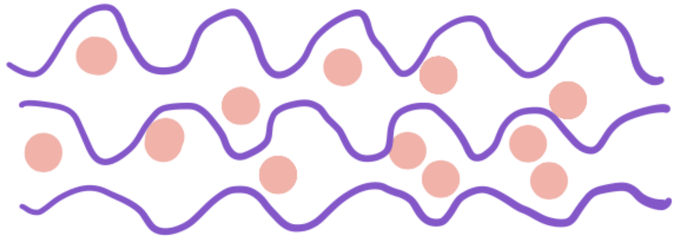
$P_\perp - P_\parallel < \dfrac{-B^2}{4\pi}$

Firehose instability





$B\uparrow \quad P_\perp\uparrow \quad P_\perp > P_\parallel$



$P_\perp - P_\parallel > \dfrac{B^2}{8\pi}$

Mirror instability

[e.g., Rosenbluth+(1956); Parker(1958)]

[e.g., Shapiro+(1964); Barnes(1966)]

Low viscosity is achieved by particles scattering off micro. B fields

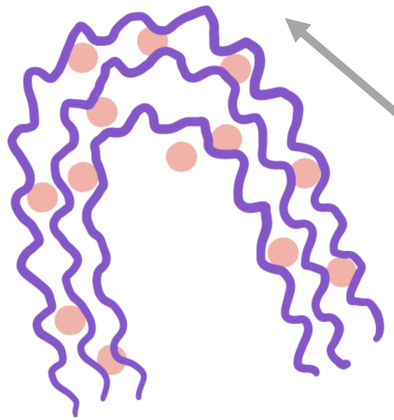
Adiabatic invariance of magnetic moment of a particle: $\mu \equiv m\, P_\perp / 2B$

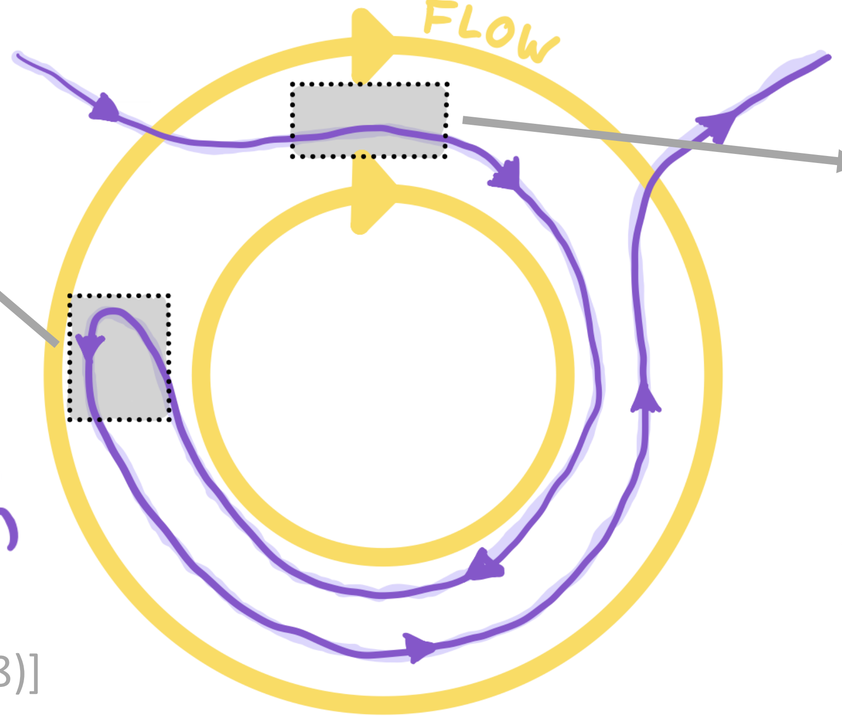
($\perp$ and $\parallel$ with respective to B)

$B \downarrow \quad P_\perp \downarrow \quad P_\perp < P_\parallel$

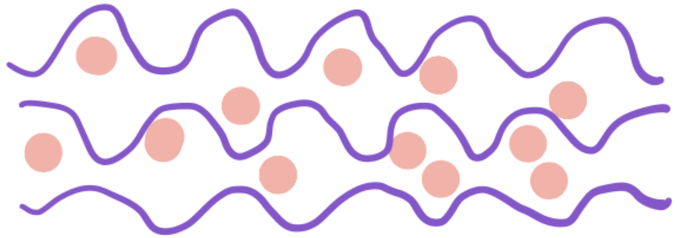
$P_\perp - P_\parallel < \dfrac{-B^2}{4\pi}$

Firehose instability





$B \uparrow \quad P_\perp \uparrow \quad P_\perp > P_\parallel$



$P_\perp - P_\parallel > \dfrac{B^2}{8\pi}$ Mirror instability

[e.g., Rosenbluth+(1956); Parker(1958)]

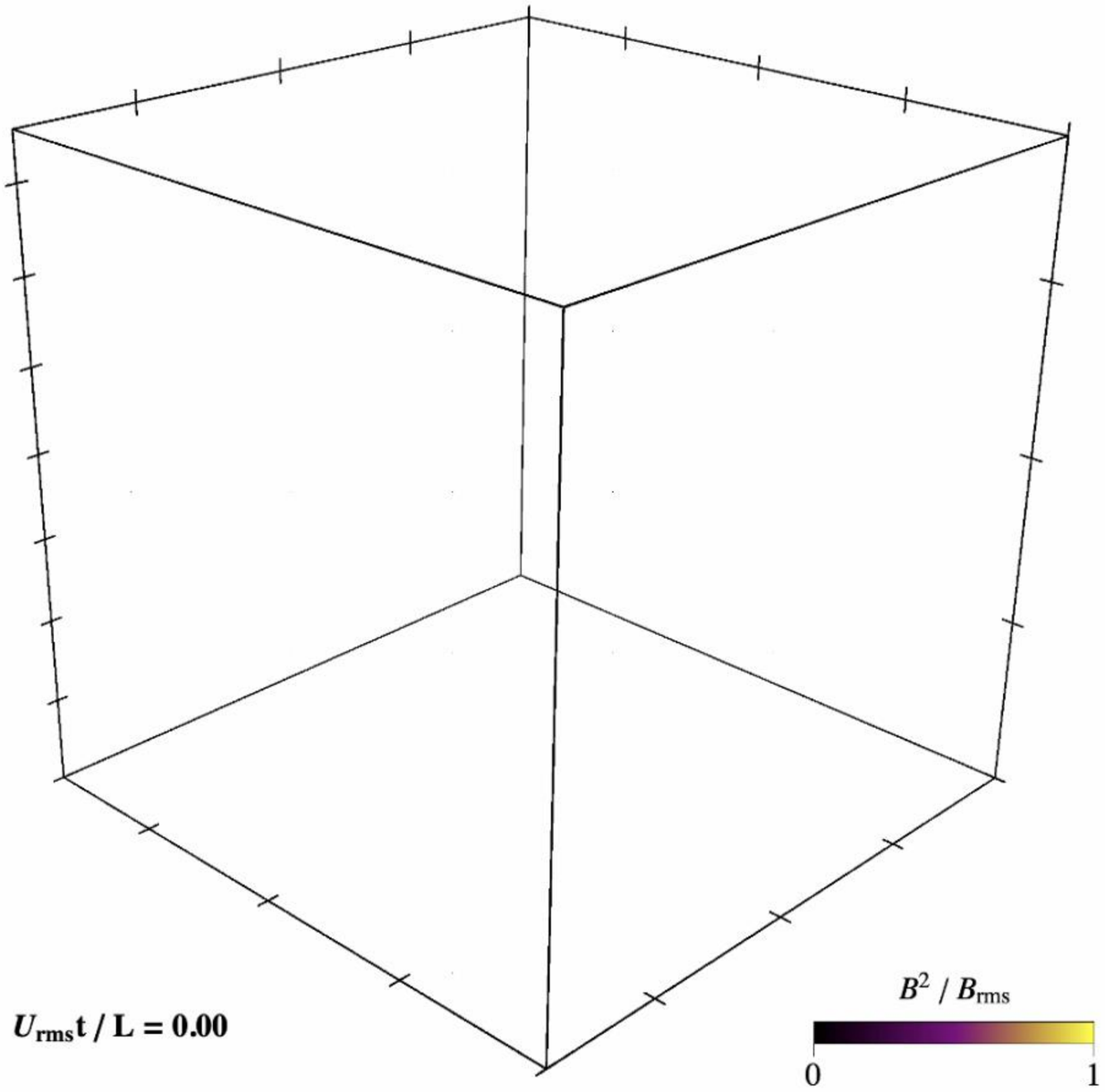
[e.g., Shapiro+(1964); Barnes(1966)]

The scattering of particles off mirror, firehose fluctuations makes the plasma effectively "collisional", with an effectively short λ_{mfp}

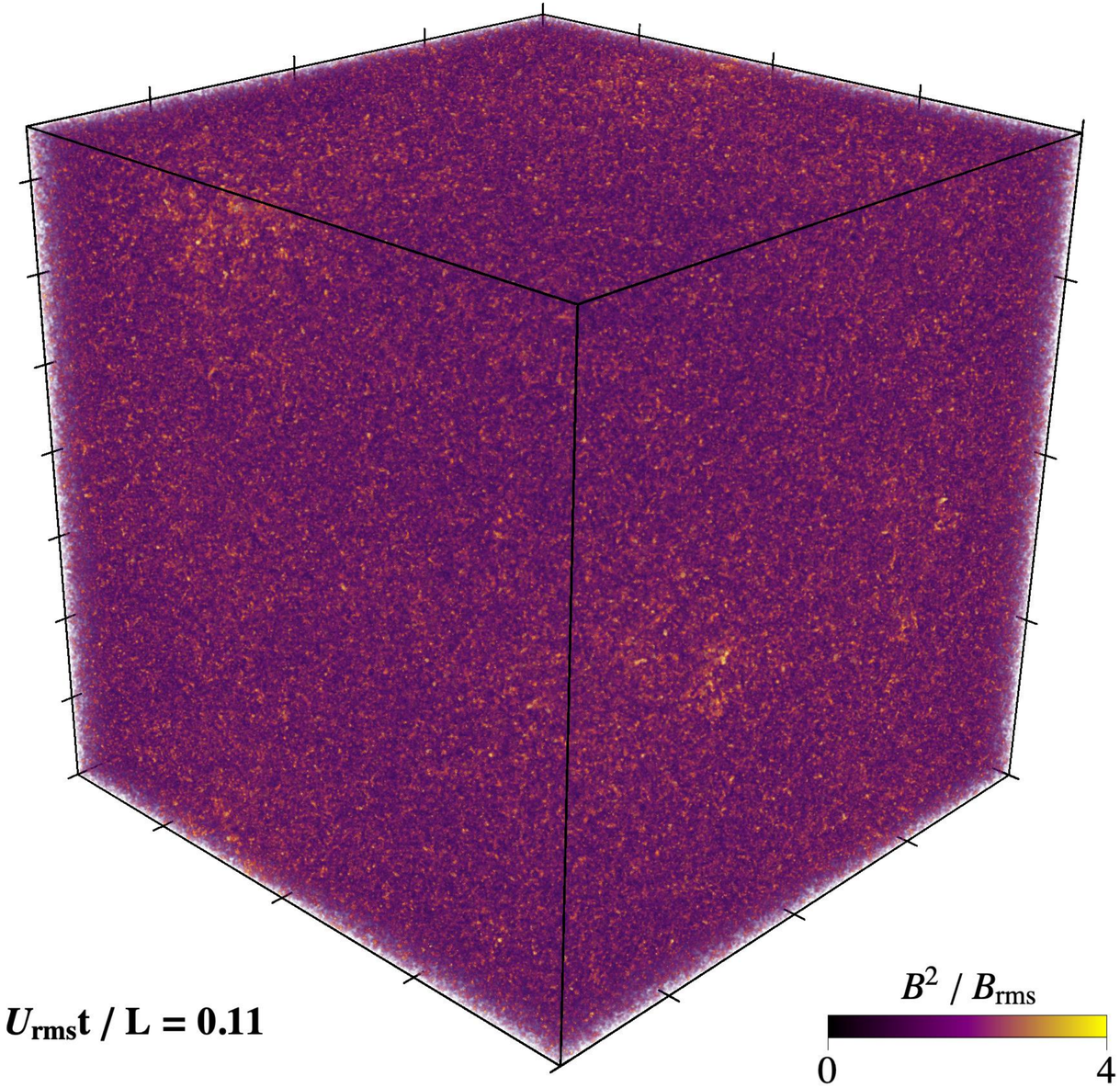


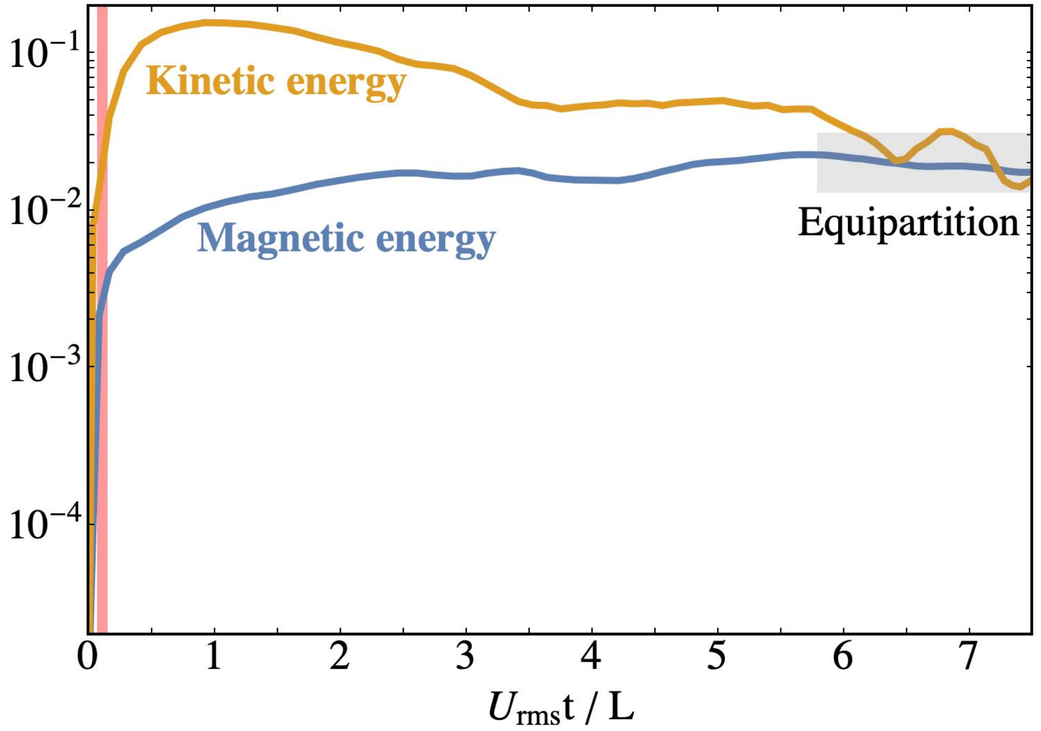
Small effective viscosity ν

Demonstrated by Rincon+(2016) and St-Onge&Kunz (2019) in hybrid kinetic studies

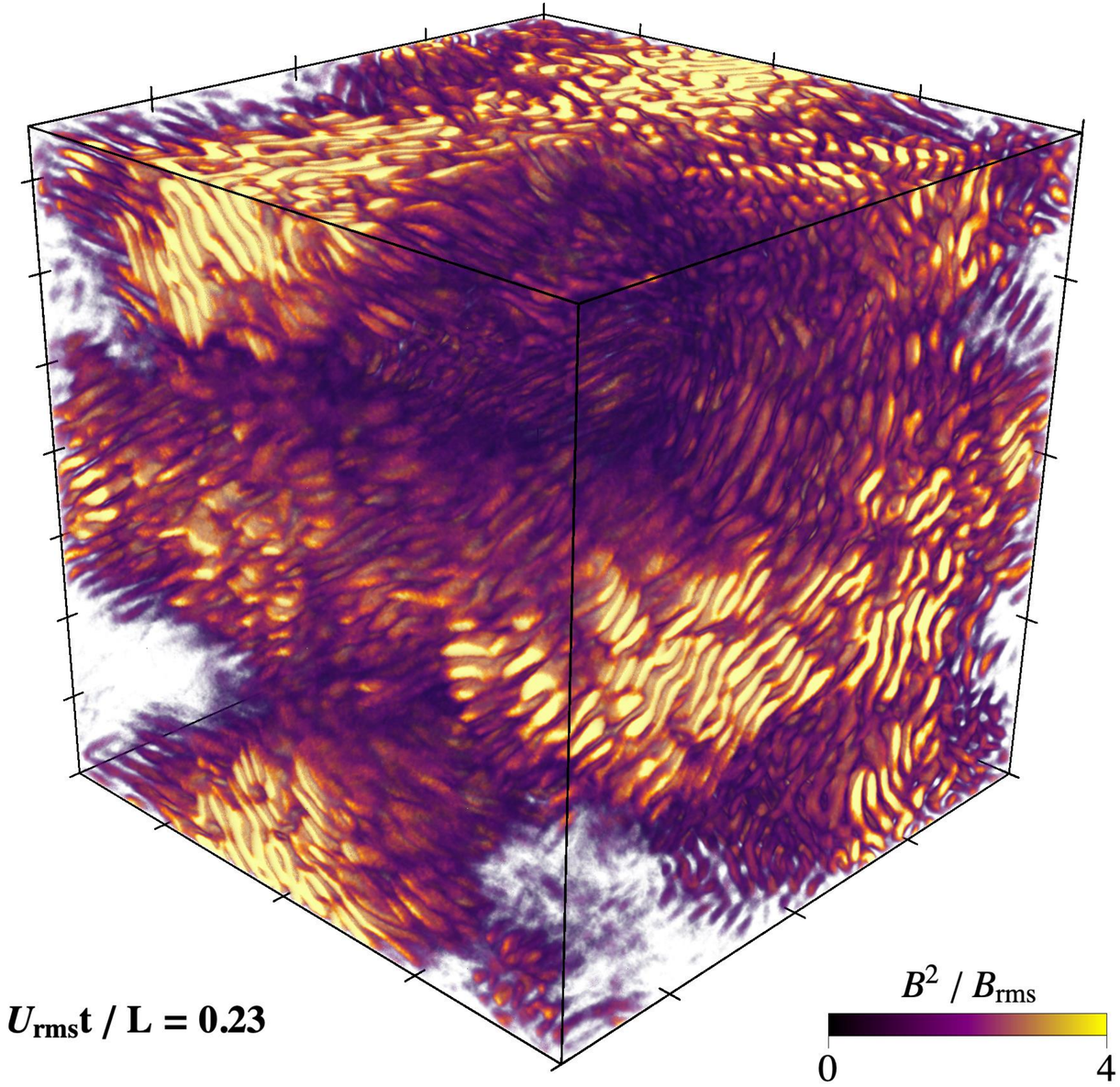


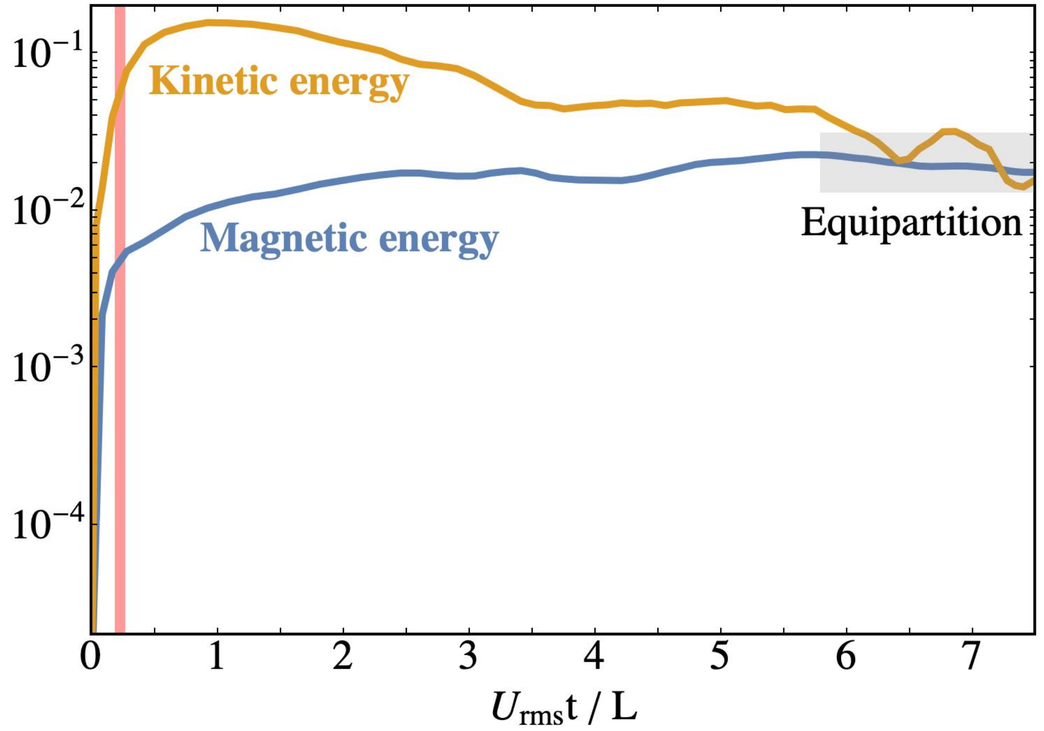
[Zhou+(2024) *ApJ*]



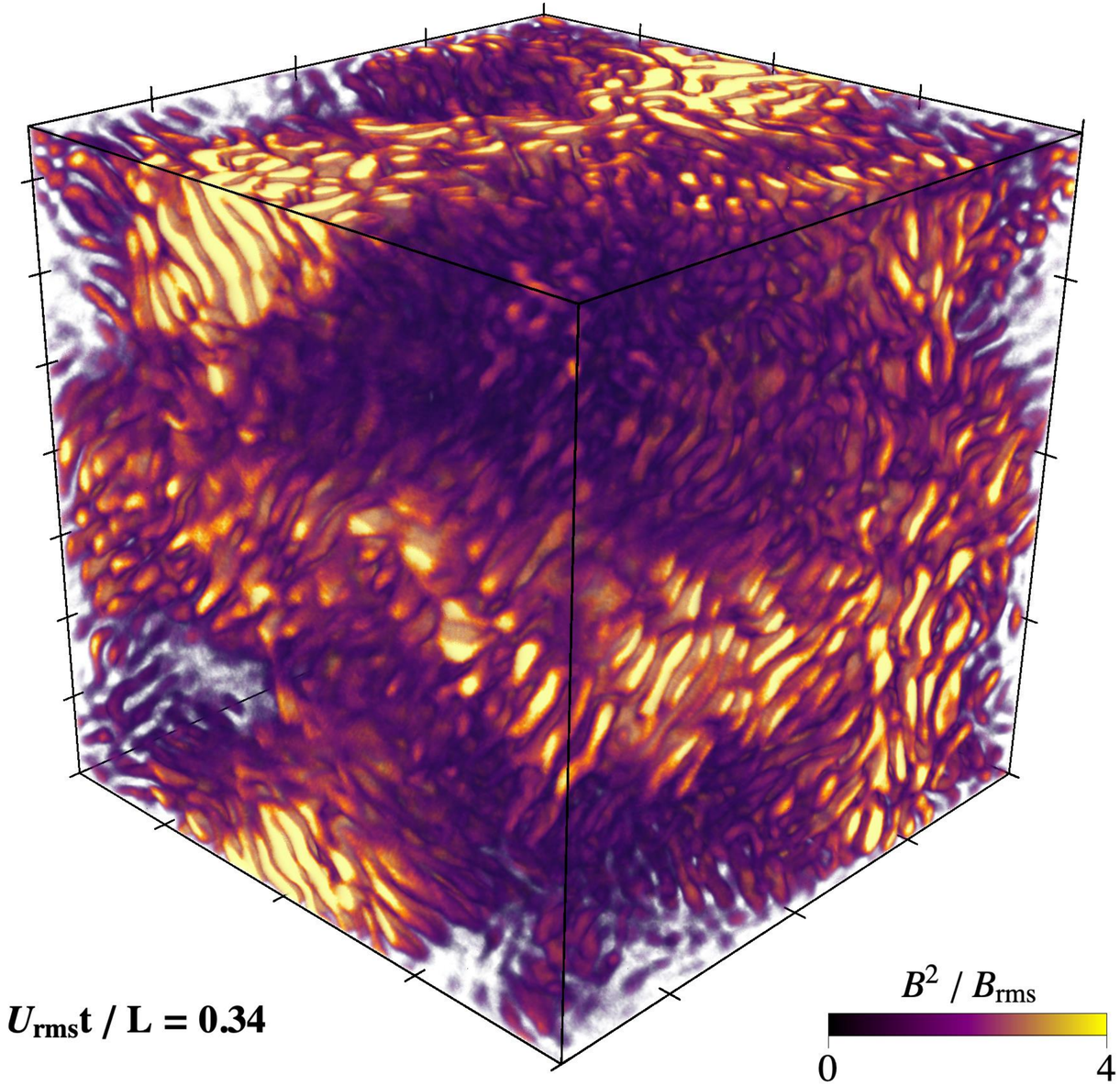


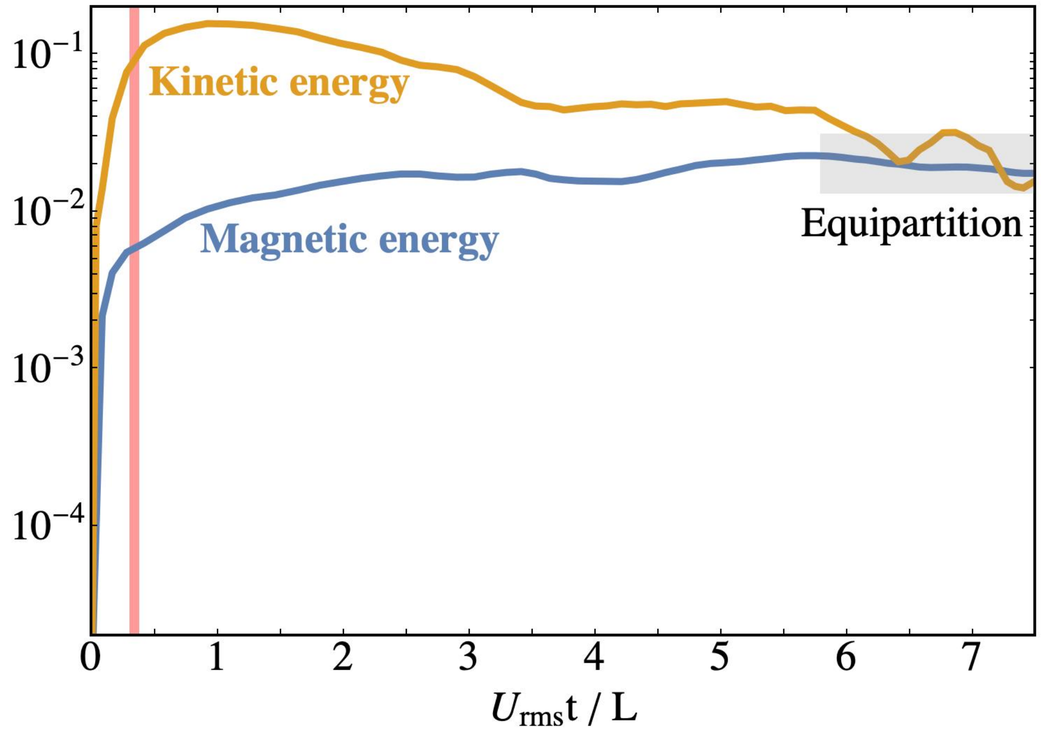
[Zhou+(2024) *ApJ*]





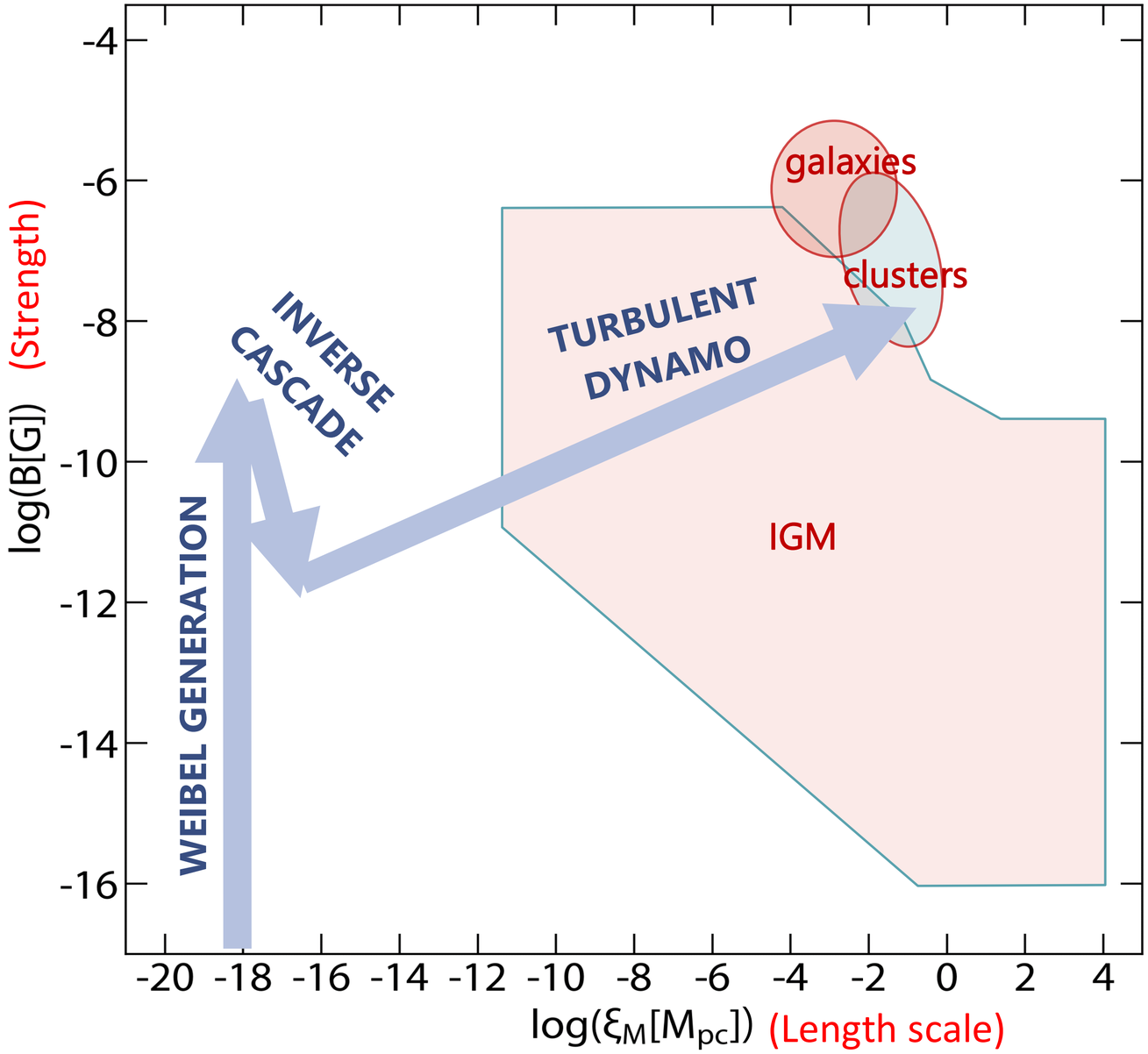
[Zhou+(2024) *ApJ*]





[Zhou+(2024) *ApJ*]

Summary



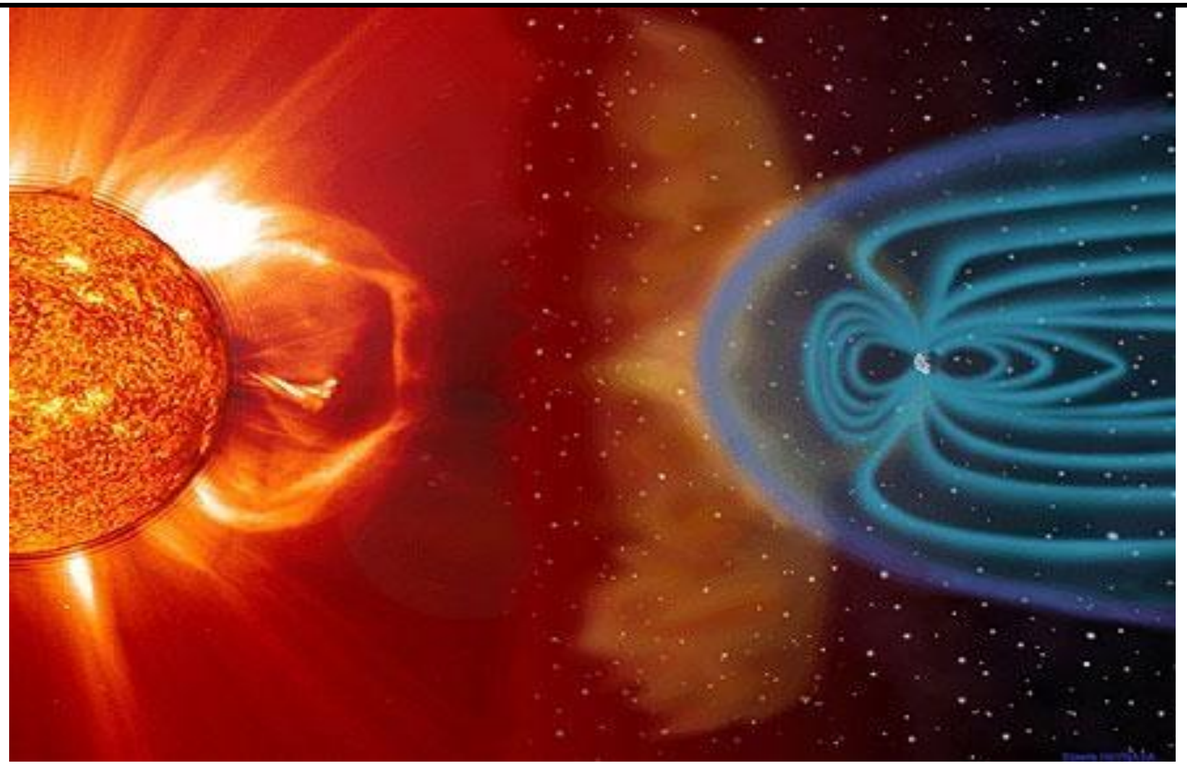
We demonstrate a self-consistent paradigm to generate cosmic magnetic fields by including the effects of nonequilibrium micro-physics of collisionless plasmas in macroscopic astrophysical processes.

Cosmic plasmas are *spontaneously magnetized* through robust kinetic instability.

Generated seed magnetic fields go through a self-similar decay before coupled with the flow.

Ab initio production of equipartition magnetic fields through large-scale astrophysical flows.

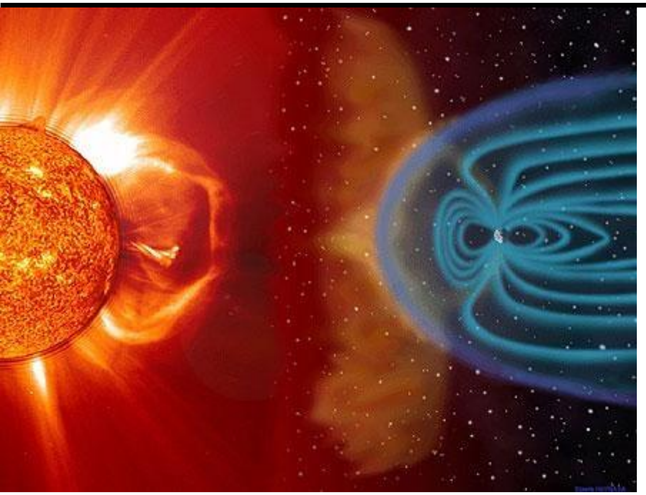
Understanding the nature turbulence is crucial

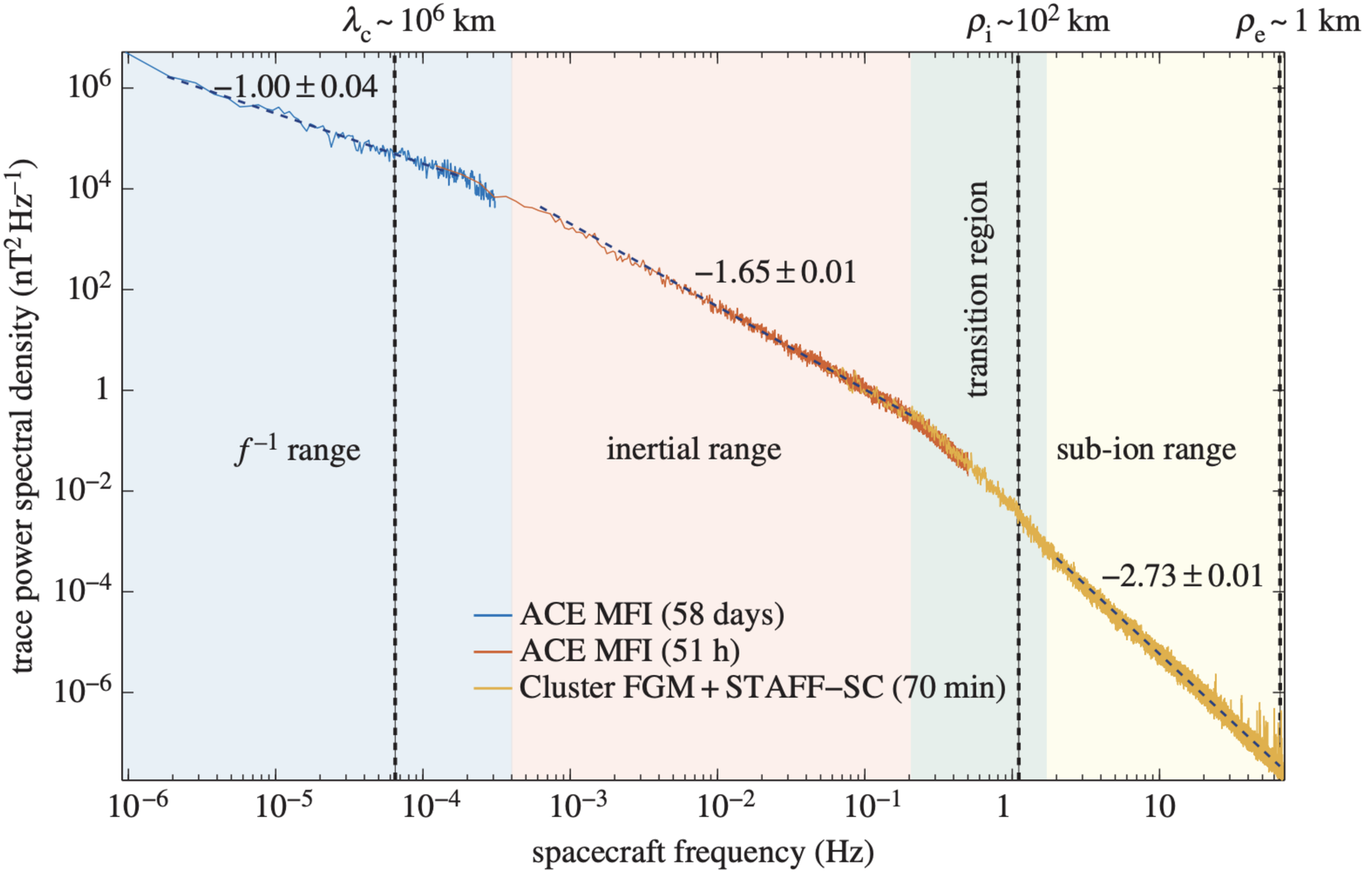


The dissipation of turbulence causes the heating of plasma in space and astrophysical systems.

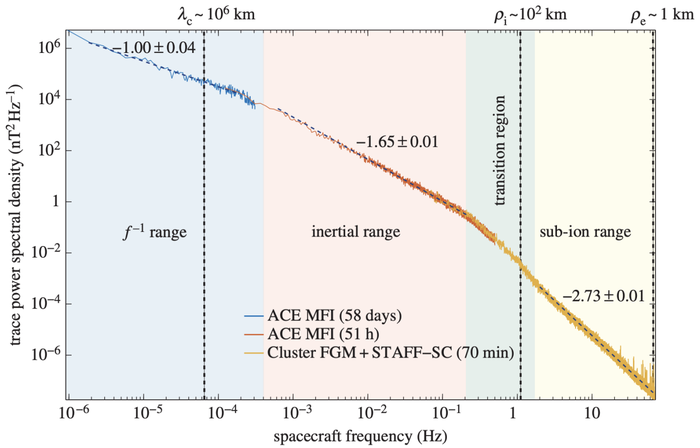
The plasma *heating at microscales impacts the macroscale evolution* of the system.

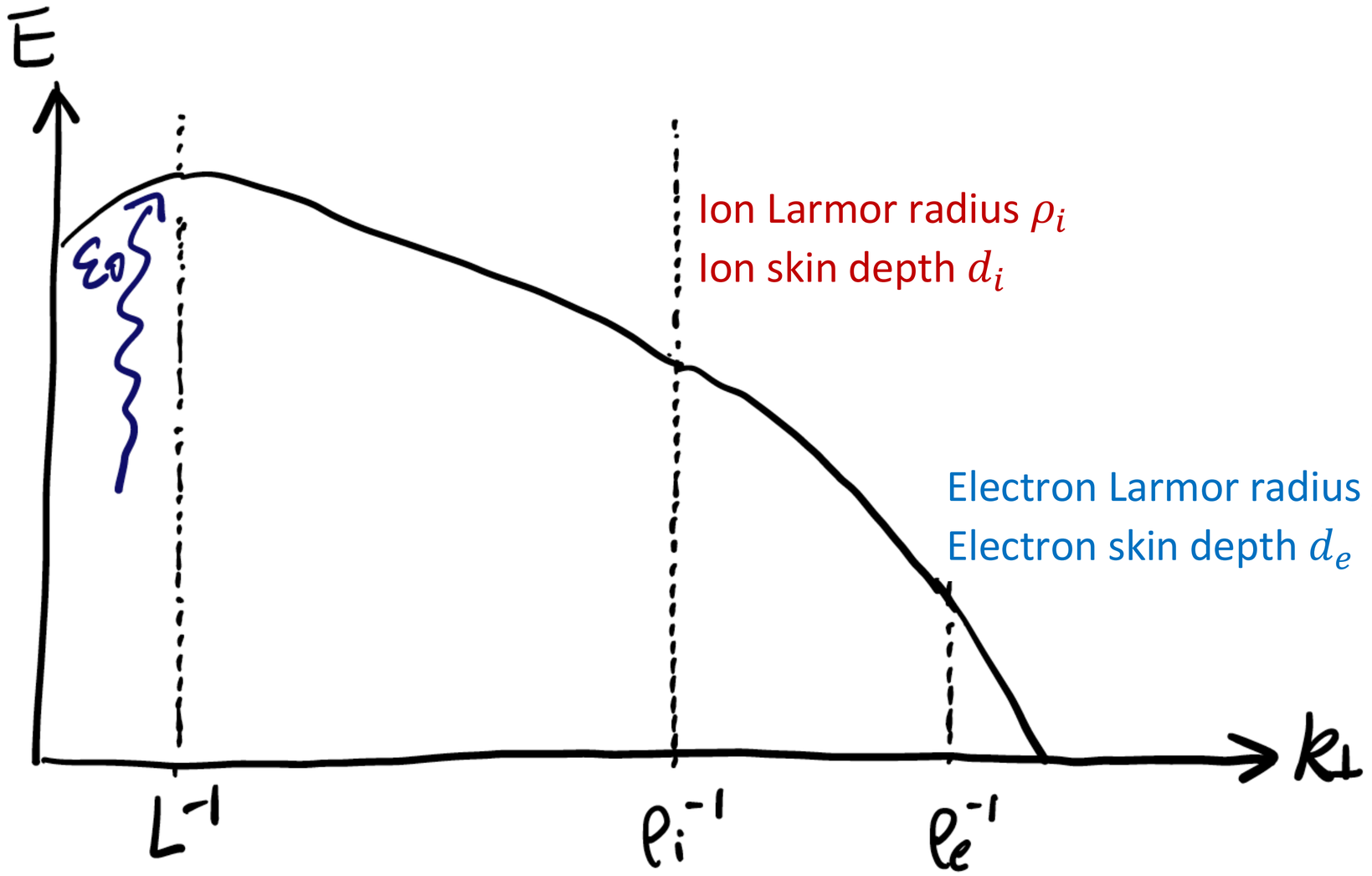
Turbulent spectrum in the Solar wind





Turbulent spectrum in a theorist's mind





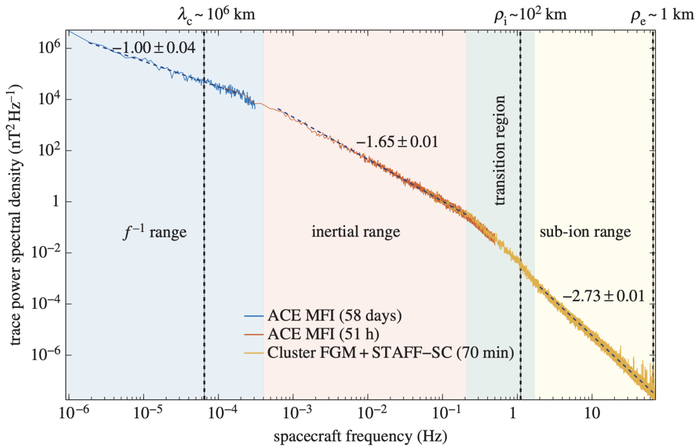
Ion Larmor radius ρ_i

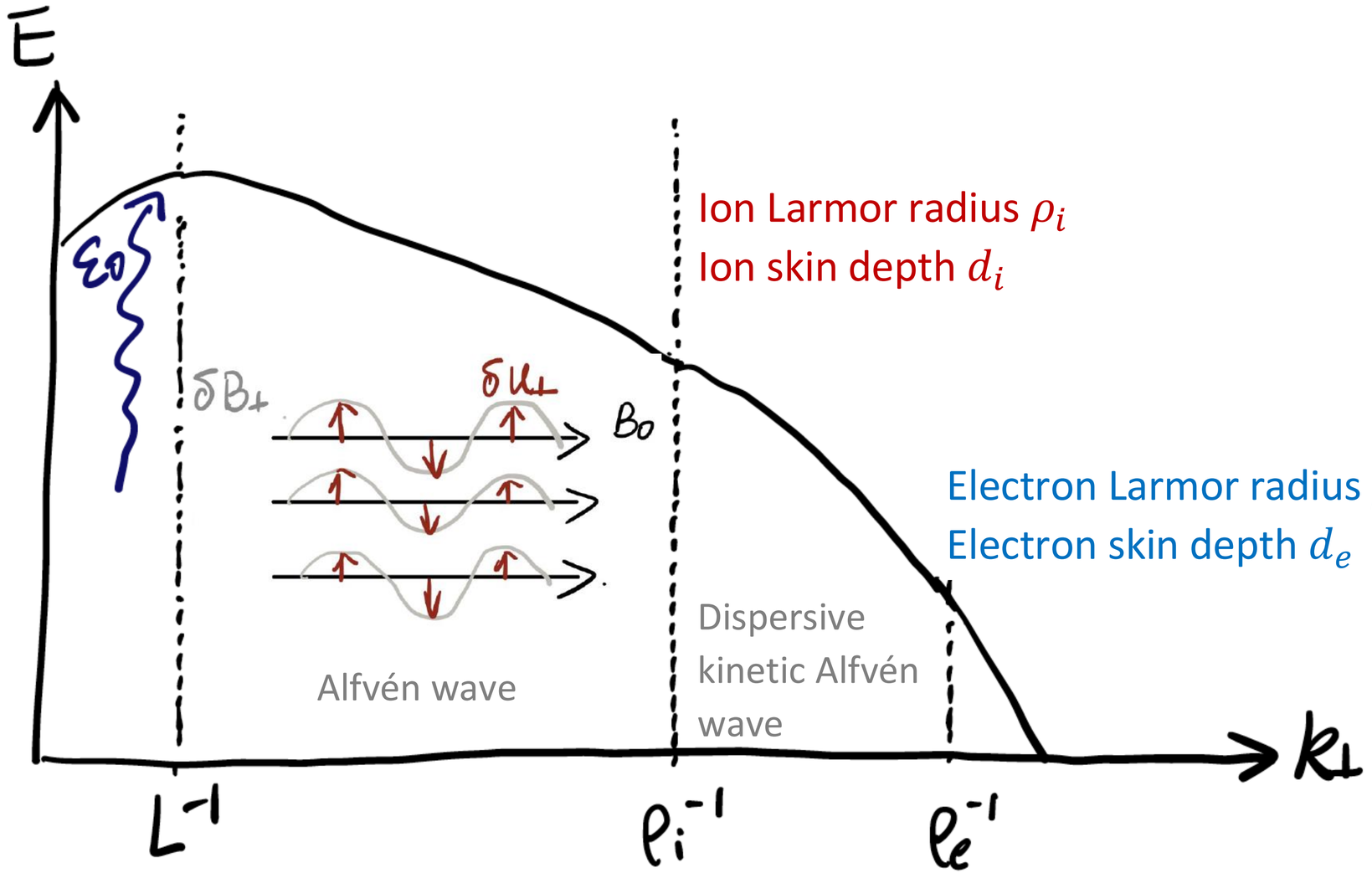
Ion skin depth d_i

Electron Larmor radius ρ_e

Electron skin depth d_e

Alfvén wave (AW) transition to kinetic Alfvén wave (KAW) crossing $\boldsymbol{\rho_i}$





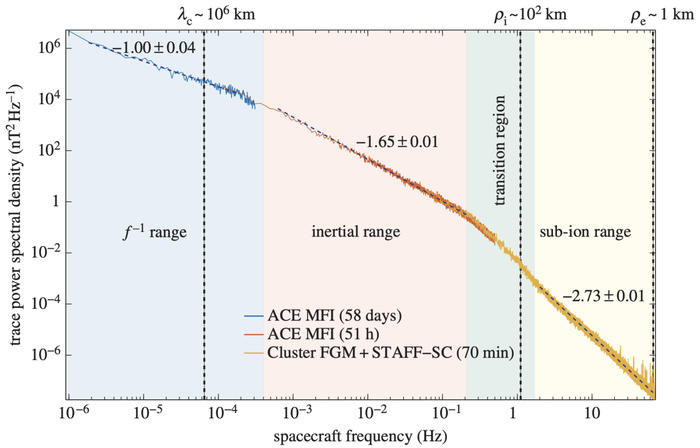
Ion Larmor radius ρ_i

Ion skin depth d_i

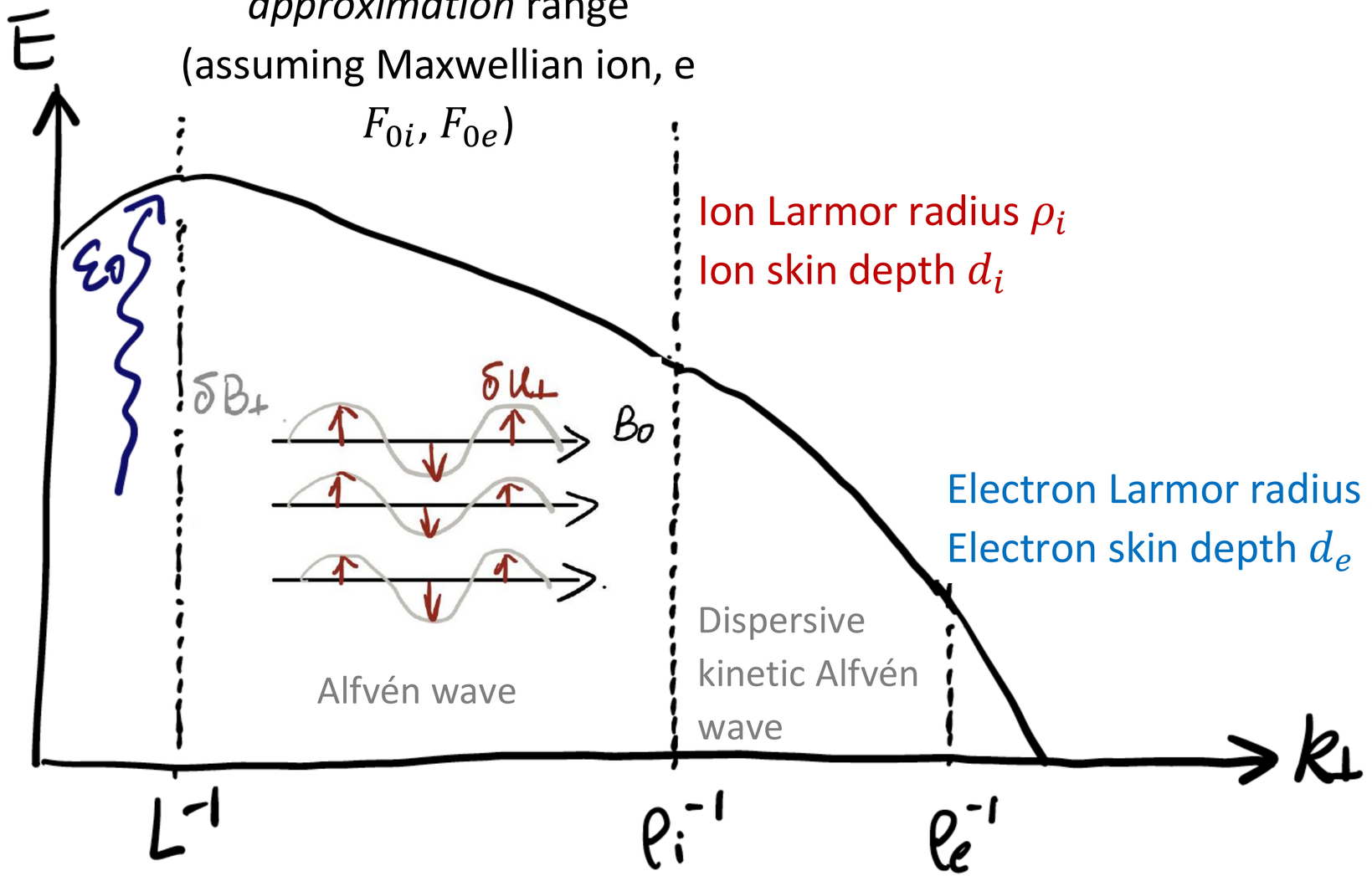
Electron Larmor radius ρ_e

Electron skin depth d_e

Inertial range (MHD approximation)



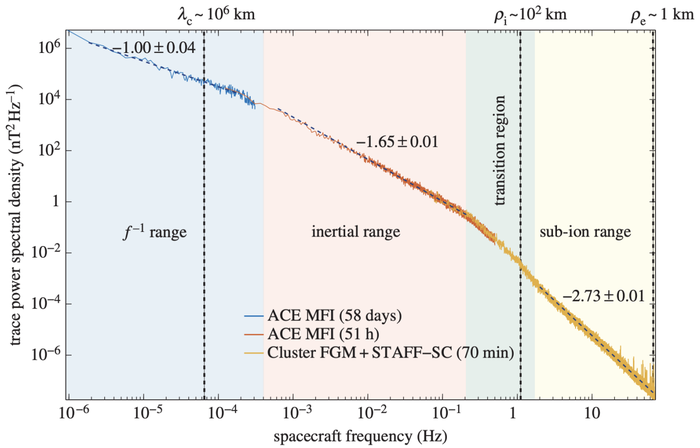
Magnetohydrodynamics (MHD)
approximation range
(assuming Maxwellian ion, e
F_{0i}, F_{0e})



Ion Larmor radius ρ_i
Ion skin depth d_i

Electron Larmor radius ρ_e
Electron skin depth d_e

Kinetic range (electron and ion kinetic/nonequilibrium physics matters)



Magnetohydrodynamics (MHD) *approximation* range (assuming Maxwellian ion, e F_{0i}, F_{0e})

Kinetic range

not necessarily small

$F_i = F_{0i} + \delta f_i$

Ion Larmor radius ρ_i

Ion skin depth d_i

$F_e = F_{0e} + \delta f_e$

Electron Larmor radius ρ_e

Electron skin depth d_e

E

E_0

$\delta B_\perp$

$\delta u_\perp$

B_0

Alfvén wave

Dispersive kinetic Alfvén wave

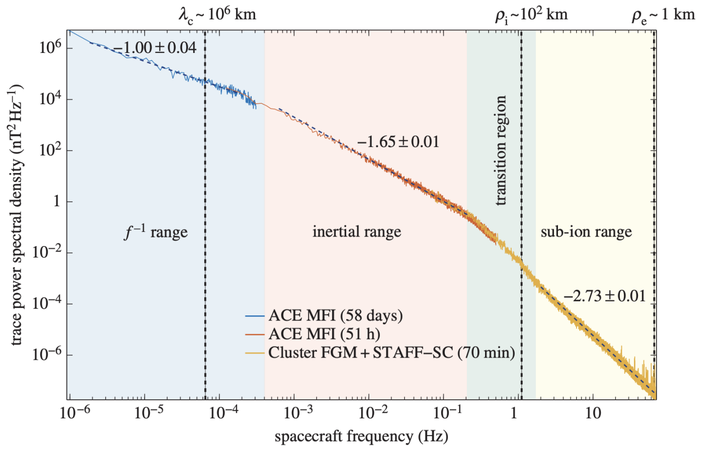
$k_\perp$

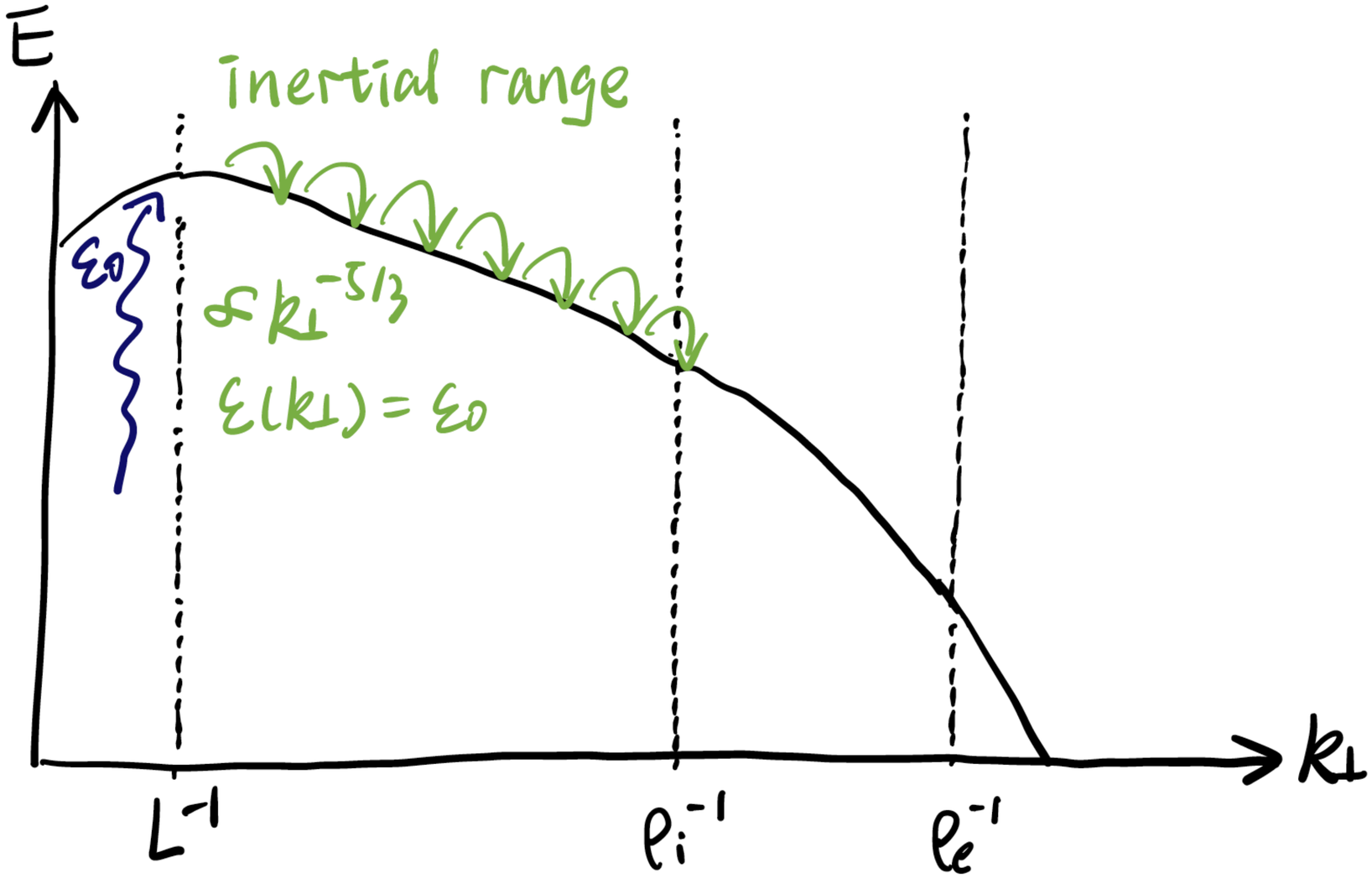
L^{-1}

ρ_i^{-1}

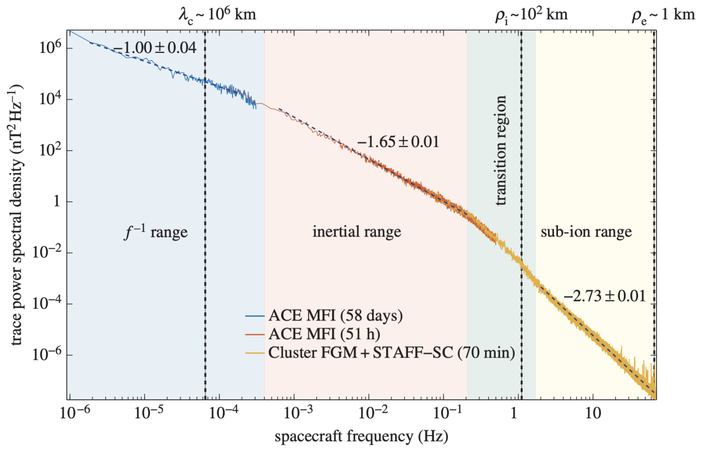
ρ_e^{-1}

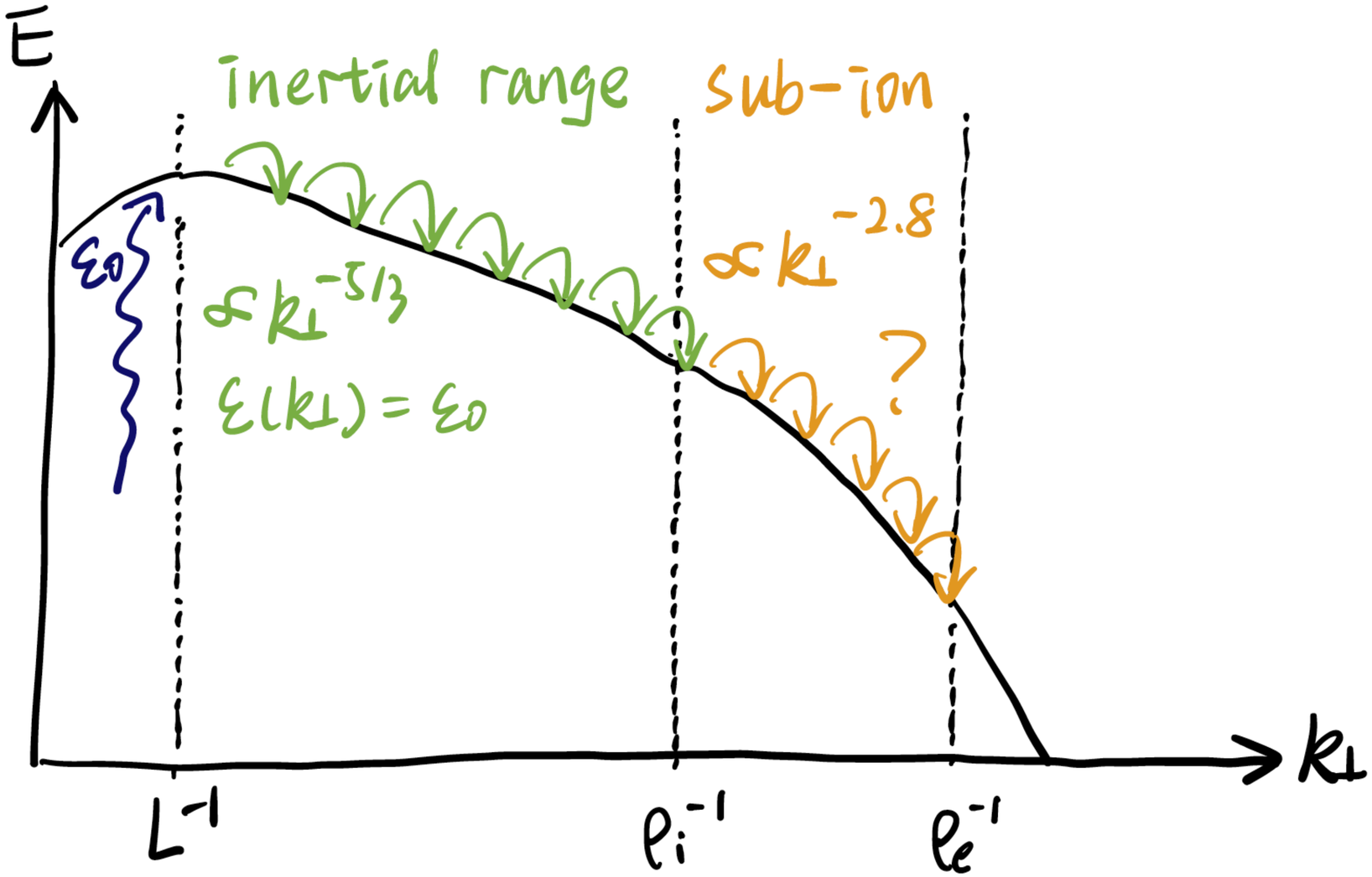
Kolmogorov-like cascade; constant energy flux



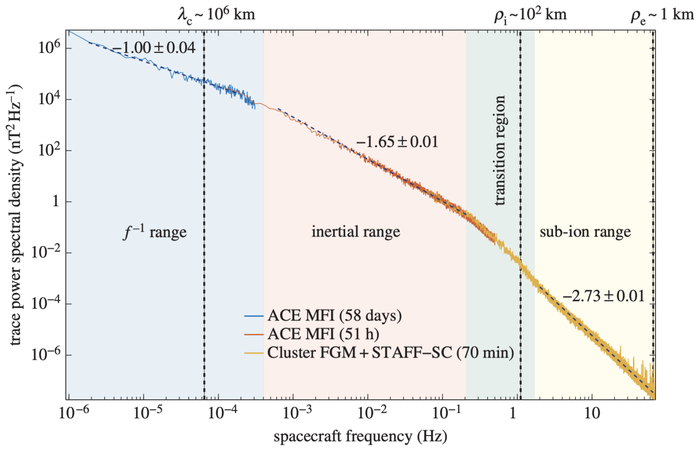


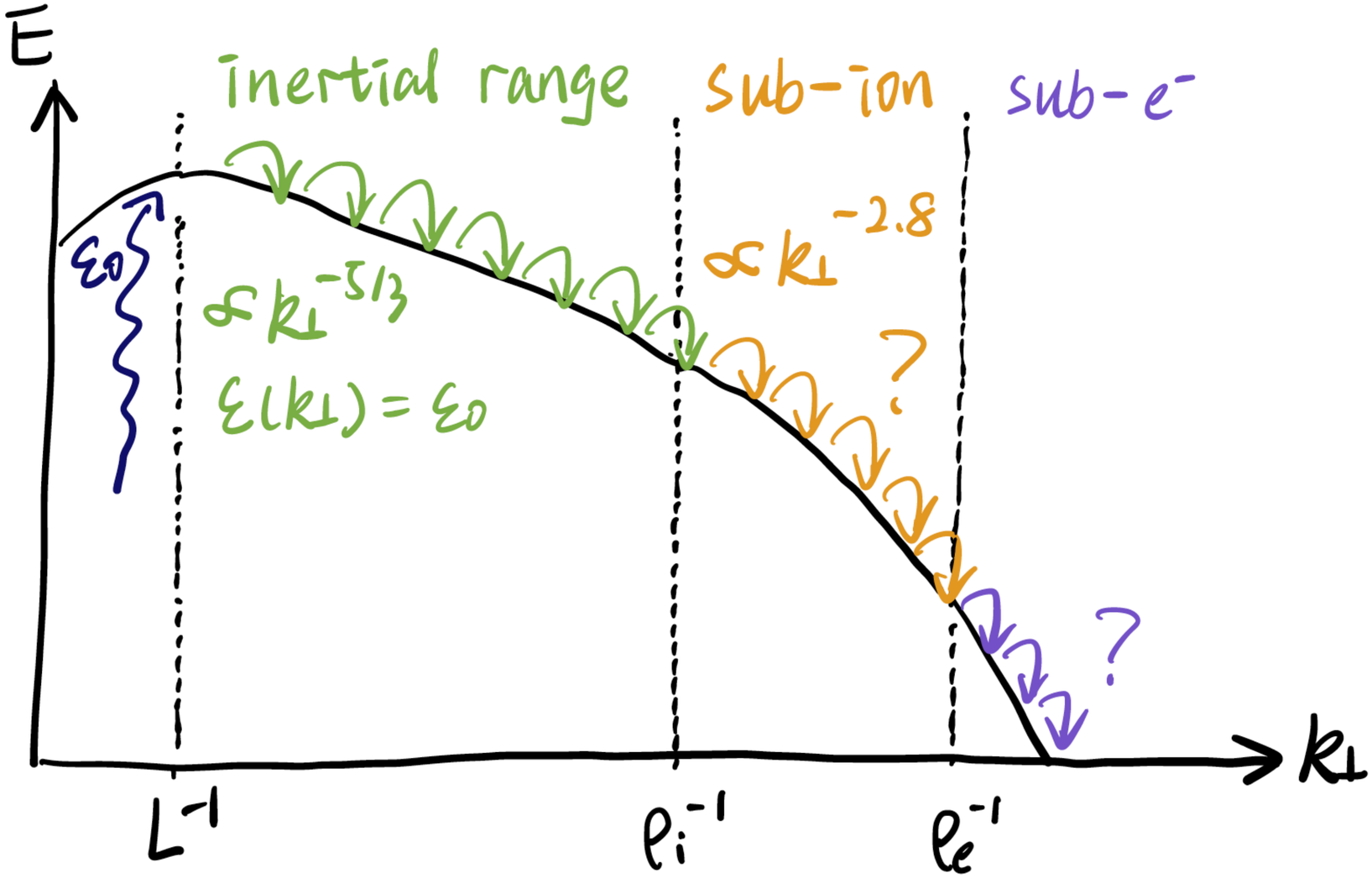
Steepened spectrum; nature of nonlinear cascade changes



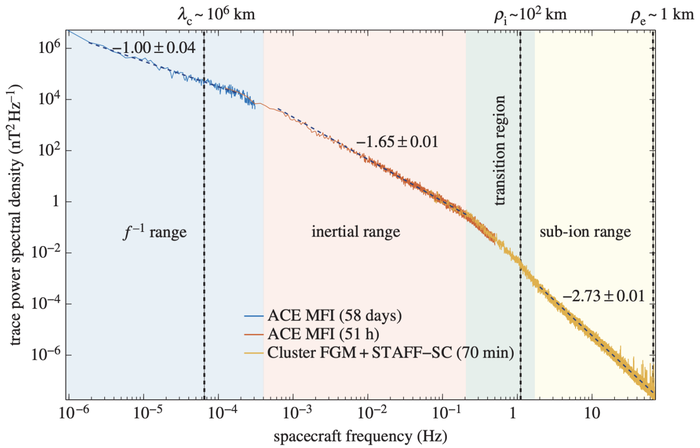


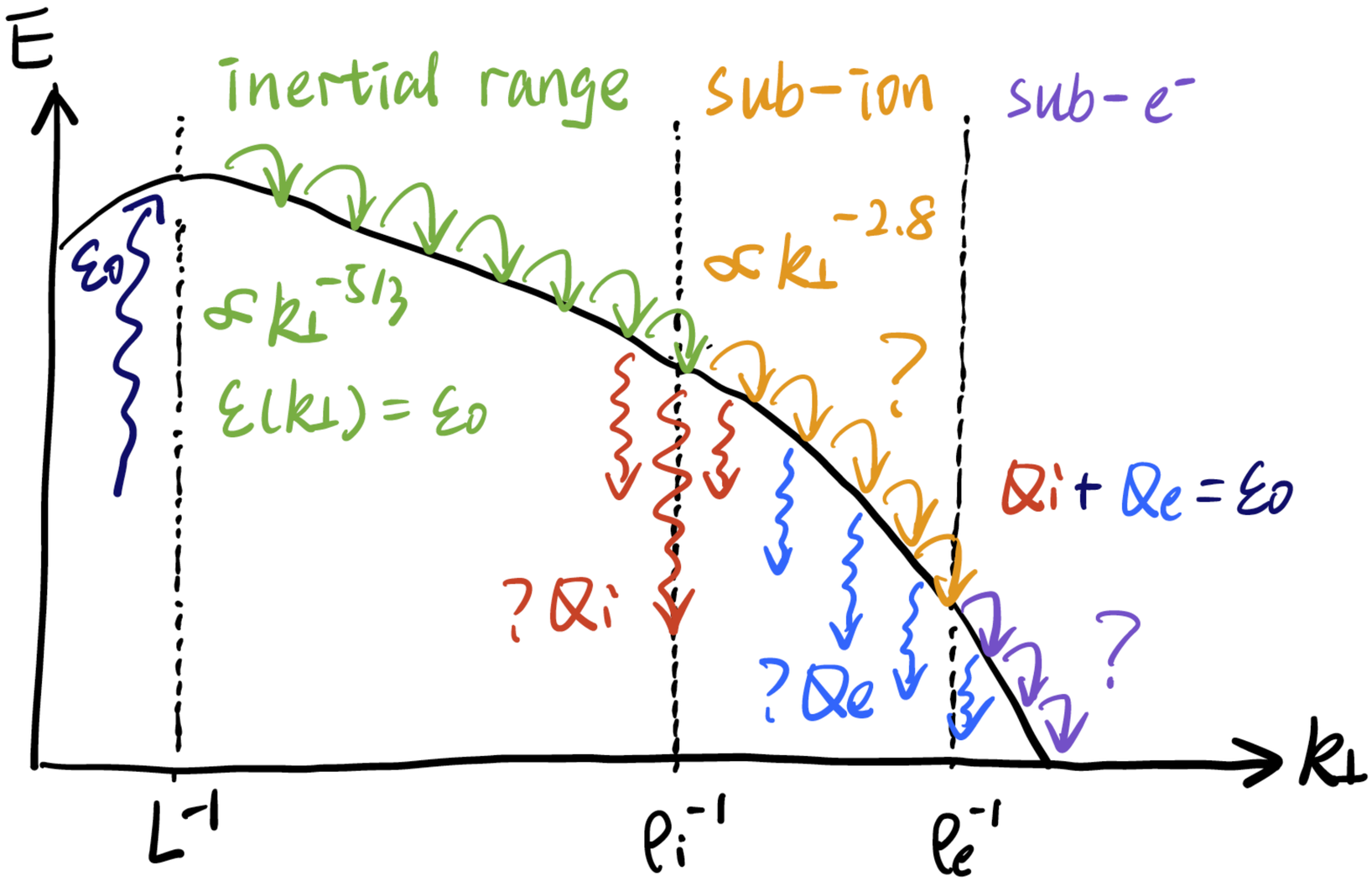
Even steeper spectrum below electron scale



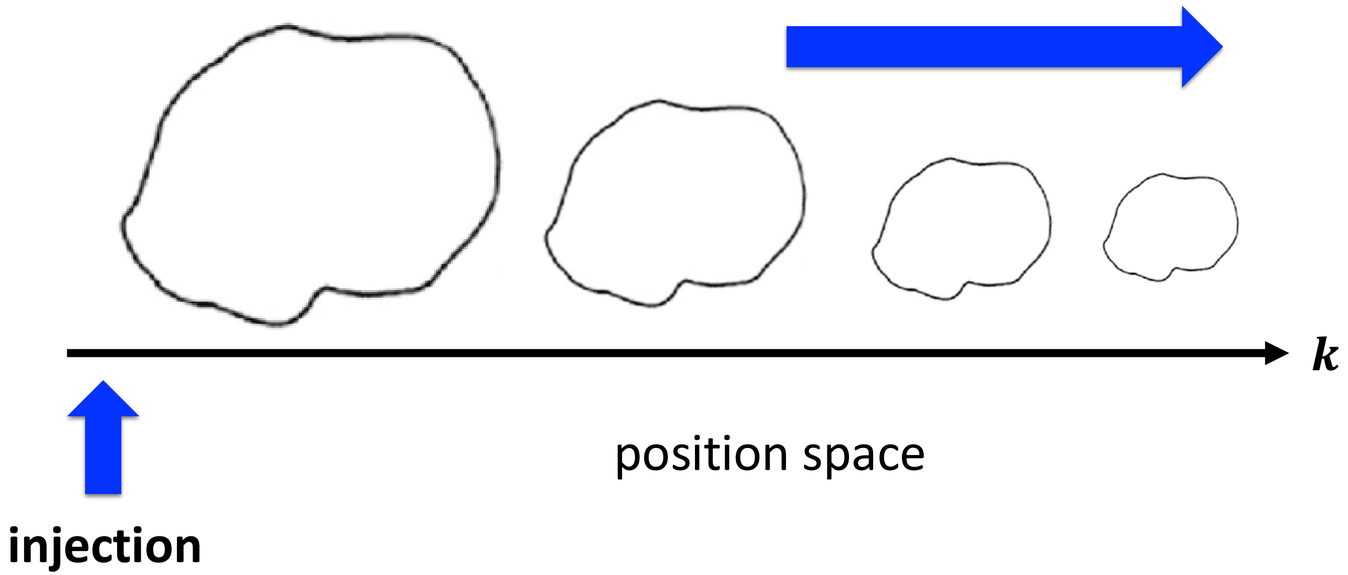


Electromagnetic energy leakage into electron and ion internal energy





Collisional plasma --- “fluid channel” of thermalization

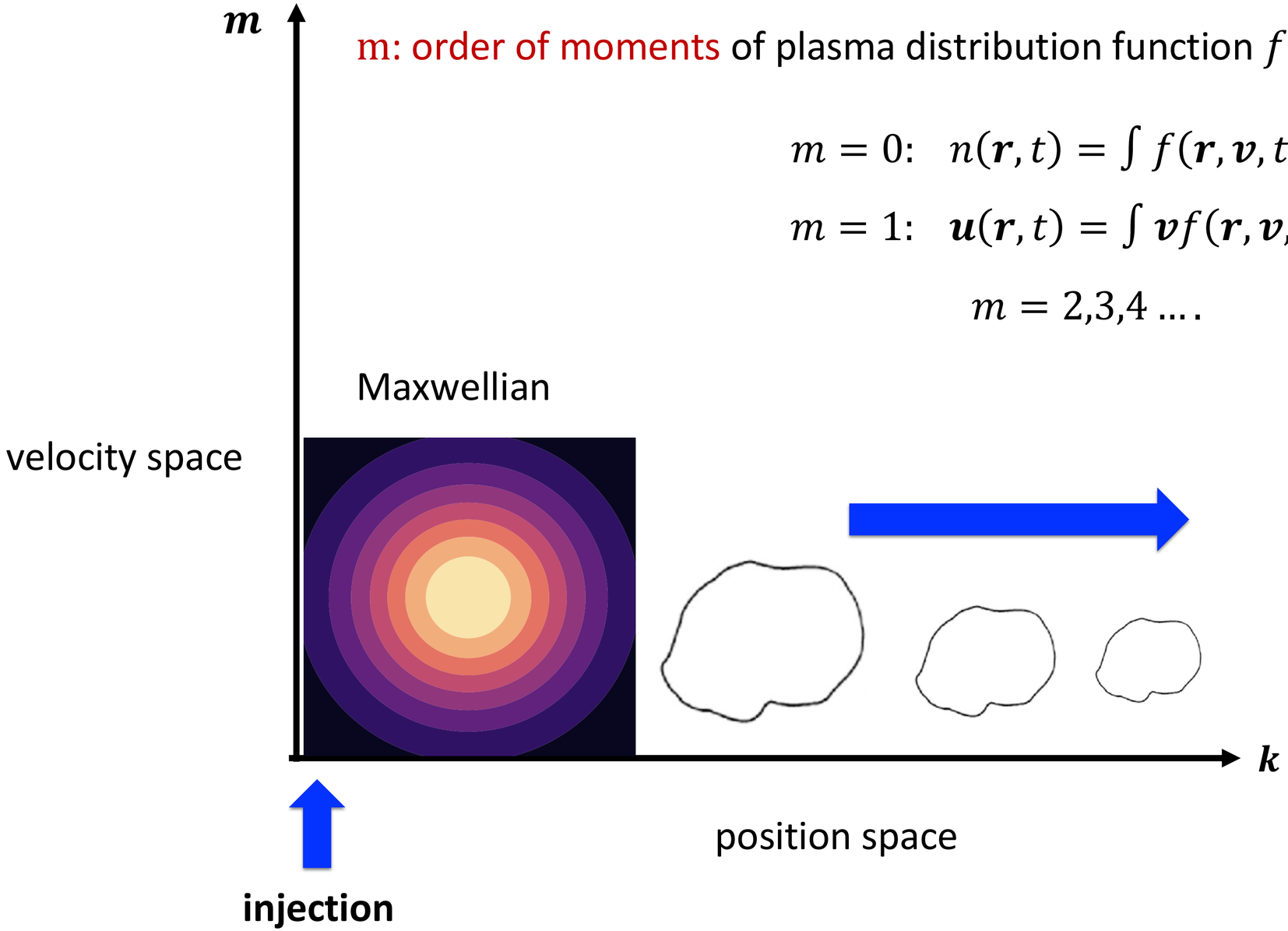


Nonlinear advection

“fluid-channel” of dissipation

at small spatial scales

Non-Maxwellian plasma has nontrivial dynamics in velocity space



m: order of moments of plasma distribution function $f(\boldsymbol{r}, \boldsymbol{v}, t)$

$$m = 0: \quad n(\boldsymbol{r}, t) = \int f(\boldsymbol{r}, \boldsymbol{v}, t) d\boldsymbol{v}$$

$$m = 1: \quad \boldsymbol{u}(\boldsymbol{r}, t) = \int \boldsymbol{v} f(\boldsymbol{r}, \boldsymbol{v}, t) d\boldsymbol{v}$$

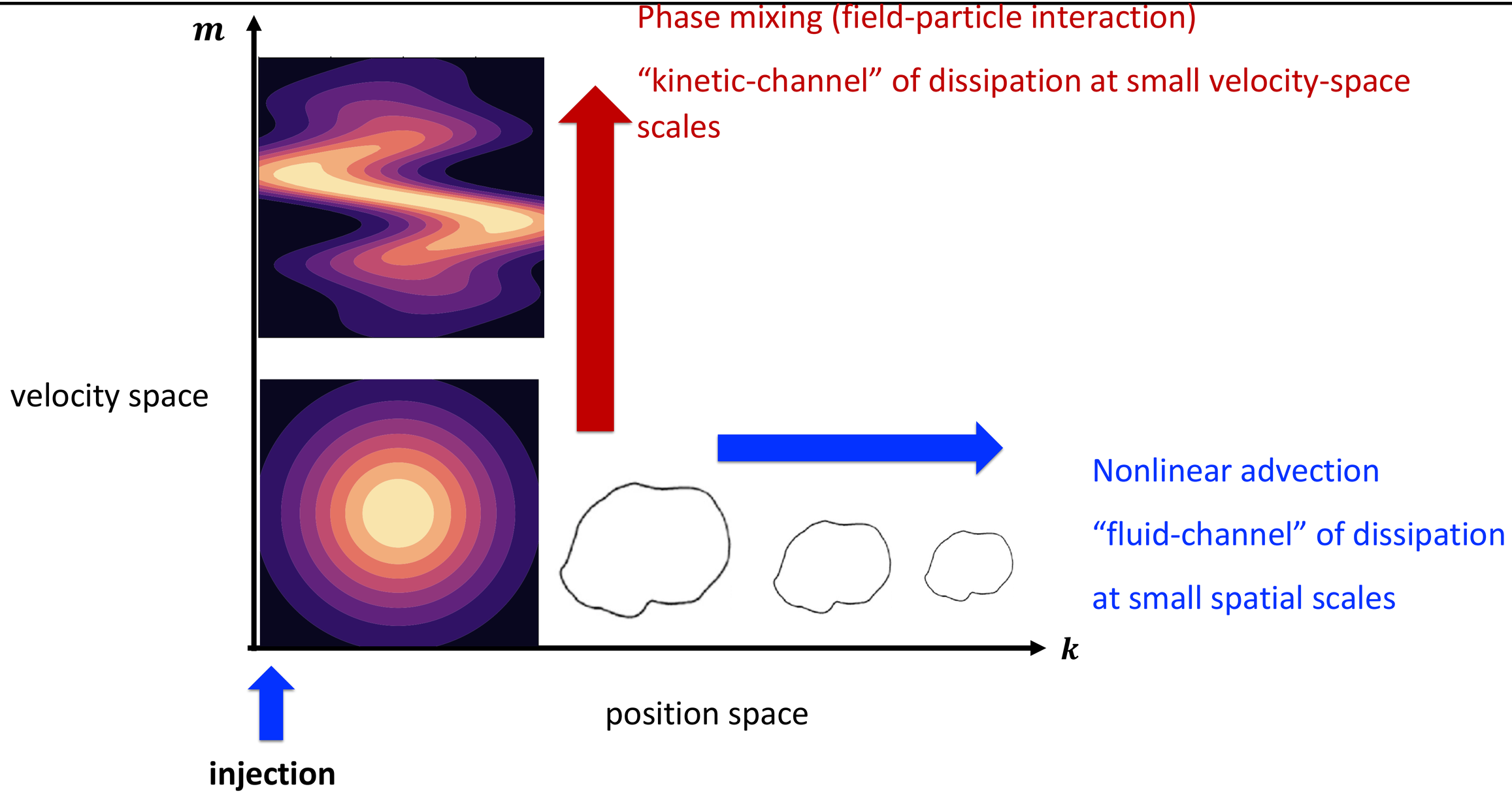
$$m = 2,3,4 \ldots.$$

Nonlinear advection

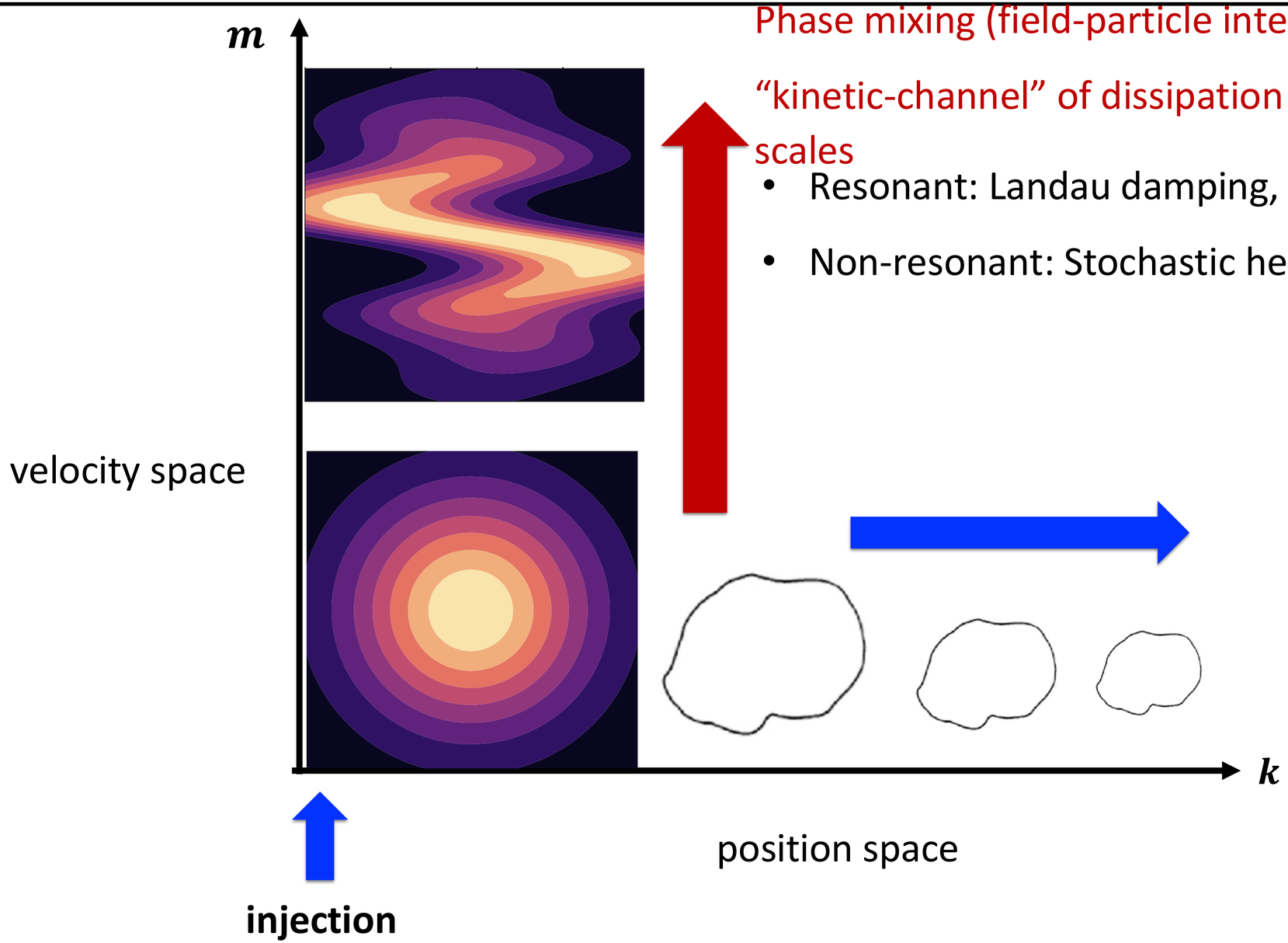
"fluid-channel" of dissipation

at small spatial scales

“kinetic-channel” of dissipation exists for weakly collisional plasma



The whole phase-space dynamics is crucial for particle heating



Phase mixing (field-particle interaction)

"kinetic-channel" of dissipation at small velocity-space scales

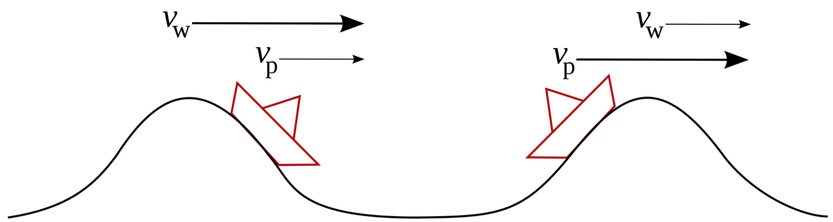
- Resonant: Landau damping, cyclotron damping, etc.
- Non-resonant: Stochastic heating, magnetic pumping, etc

Nonlinear advection

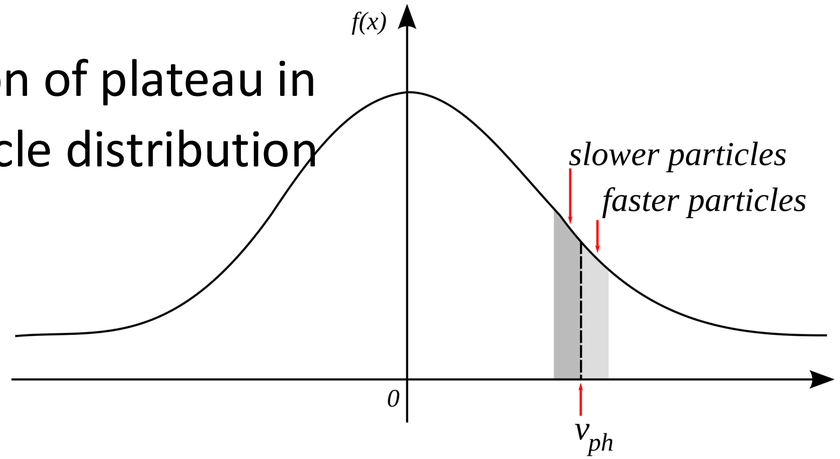
"fluid-channel" of dissipation at small spatial scales

Kinetic processes for energizing particles

Landau damping of kinetic Alfvén wave (KAW)



Formation of plateau in the particle distribution

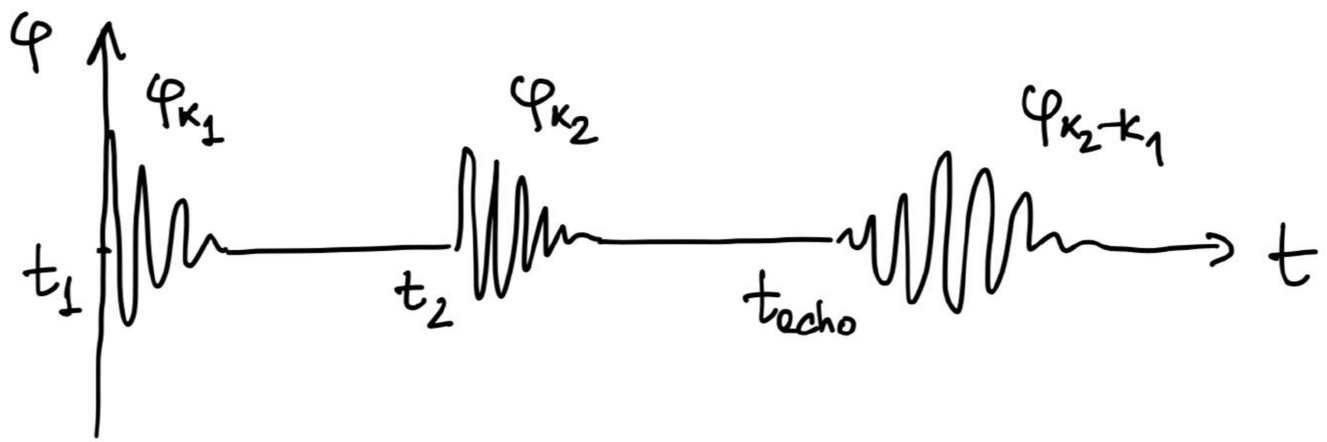


Phase mixing (field-particle interaction)

"kinetic-channel" of dissipation at small velocity-space scales

- Resonant: Landau damping, cyclotron damping, etc.
- Non-resonant: Stochastic heating, magnetic pumping, etc

Plasma echo



$$\delta f_{k_1} = a_1 e^{-ik_1 v(t-t_1)},$$

$$\delta f_{k_2} = a_2 e^{-ik_2 v(t-t_2)}.$$

FIGURE 42. Two subsequent pulses, then an echo at time (11.4).

$$\delta f_{k_2-k_1} \propto \delta f_{k_2} \delta f_{k_1}^* = a_1^* a_2 e^{-ik_2 v(t-t_2)+ik_1 v(t-t_1)} = a_1^* a_2 e^{-i[(k_2-k_1)t-(k_2 t_2-k_1 t_1)]v}.$$

$$C_{\boldsymbol{k}}(\boldsymbol{v}, \boldsymbol{v}') = \langle \delta f_{\boldsymbol{k}}(\boldsymbol{v}) \delta f_{\boldsymbol{k}}^*(\boldsymbol{v}') \rangle.$$

$$\boxed{t_{\text{echo}} = \frac{k_2 t_2 - k_1 t_1}{k_2 - k_1}},$$

$$\varphi_{k_2-k_1} \propto \int \mathrm{d}v \, \delta f_{k_2-k_1}$$

This has been observed in experiment!!!

https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.20.95

Kinetic Reduced Electron Heating Model (KREHM) framework

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β: $\beta_e \sim m_e/m_i$) → KREHM

Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

Continuity $$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law $$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}}\right)$$

$\vdots$

m: order of moments

δf_e

$\downarrow$

$m=0$ $\quad \delta n_e$

$\downarrow$

$m=1$ $\quad u_{\parallel e}$ $(A_\parallel)$

$\downarrow$

$m=2$ $\quad \delta T_{\parallel e}$

$\downarrow$

$\vdots$

For simplicity: assume isothermalality (no leakage to velocity space)

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β, $\beta_e \sim m_e/m_i$) → KREHM

Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

m: order of moments

Continuity $$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law $$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}}\right)$$

δf_e

$m = 0$ δn_e

$m = 1$ $u_{\parallel e}$ $(A_\parallel)$

$m = 2$ $\delta T_{\parallel e}$

⋮

For simplicity: assume isothermal electrons (no leakage to velocity space)

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β, $\beta_e \sim m_e/m_i$) → KREHM

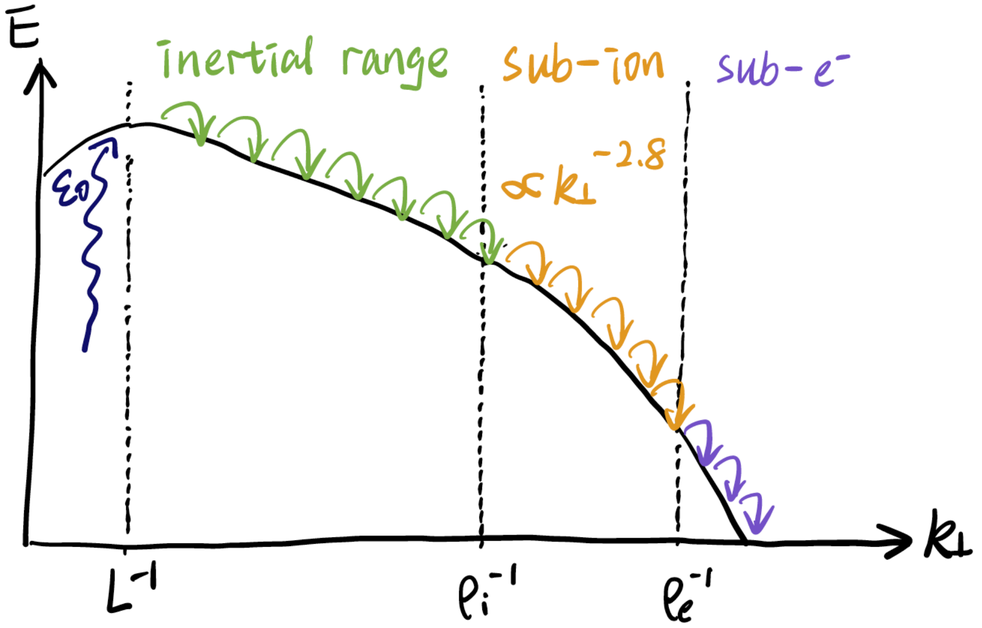
Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

Continuity $$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law $$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_e}{T_{0e}}\right)$$



Generalized Helicity as a quadratic invariant:

$$H \equiv \int \frac{(1-\hat{\Gamma}_0)\varphi}{\rho_i^2}(1-d_e^2\nabla_\perp^2)A_z \mathrm{d}V$$

$H \propto \int \delta n_e A_z \mathrm{d}V$ in the range $\rho_i \gg k_\perp^{-1} \gg d_e$

$H \propto \int \delta \boldsymbol{u}_\perp \cdot \delta \boldsymbol{B}_\perp \mathrm{d}V$ at MHD scales ($k_\perp \rho_i \ll 1$)

Remove the isothermal assumption to study electron heating

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β, $\beta_e \sim m_e/m_i$) → KREHM

Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

m: order of moments

Continuity $$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law $$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}}\right)$$

$\vdots$

δf_e

$m = 0$ δn_e

$m = 1$ $u_{\parallel e}$ $(A_\parallel)$

$m = 2$ $\delta T_{\parallel e}$

$\vdots$

Remove the isothermal assumption to study electron heating

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β: $\beta_e \sim m_e/m_i$) → KREHM

Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

Continuity $$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law $$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}}\right)$$

Kinetic equation $$\frac{dg_m}{dt} = -v_{\mathrm{the}}\hat{b}\cdot\nabla\left(\sqrt{\frac{m+1}{2}}g_{m+1} + \sqrt{\frac{m}{2}}g_{m-1} - \delta_{m,1}g_2\right) - \sqrt{2}\delta_{m,2}\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

m: order of moments

	δf_e
$m=0$	δn_e
$m=1$	$u_{\parallel e}$ $(A_\parallel)$
$m=2$	$\delta T_{\parallel e}$
	$\vdots$

$m \geq 2$: g_m

Remove the isothermal assumption to study electron heating

Fully kinetic → (Ring average) → Gyro-kinetics → (low-β, $\beta_e \sim m_e/m_i$) → KREHM

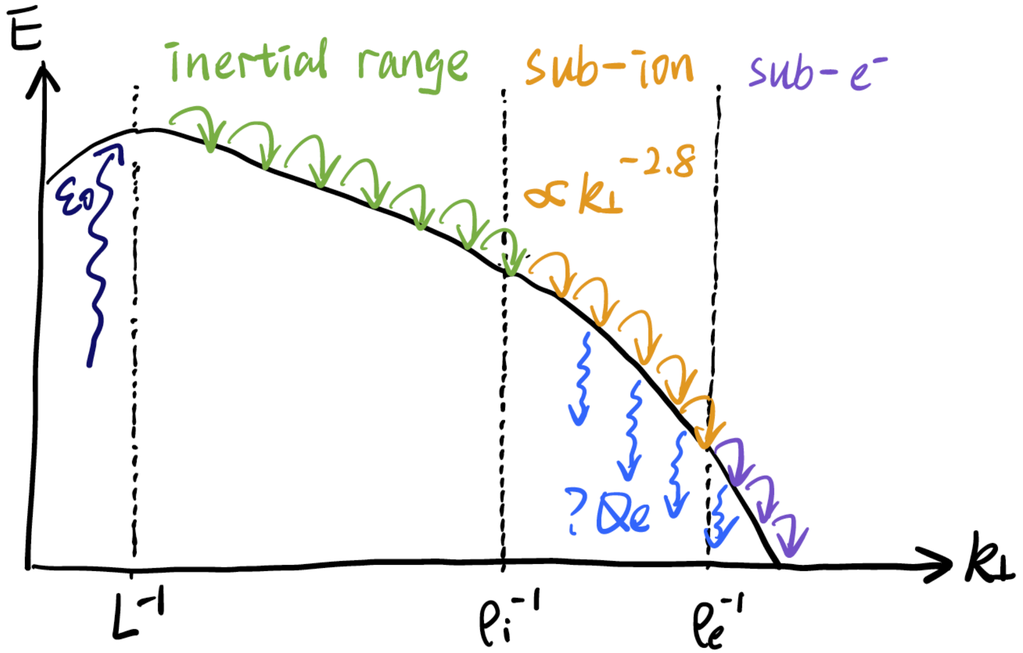
Simplest yet rigorous eqns containing fluid & kinetic physics [Zocco+ 2011]

Ions become isothermal and electrostatic, only introducing ρ_i scale.

Electrons are described by δf_e (deviation from F_{0e})

Continuity
$$\frac{1}{n_{0e}}\frac{d\delta n_e}{dt} = -\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

Ohm's law
$$\frac{d}{dt}(A_\parallel - d_e^2\nabla_\perp^2 A_\parallel) = -c\frac{\partial\varphi}{\partial z} + \frac{cT_{e0}}{e}\hat{b}\cdot\nabla\left(\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}}\right)$$



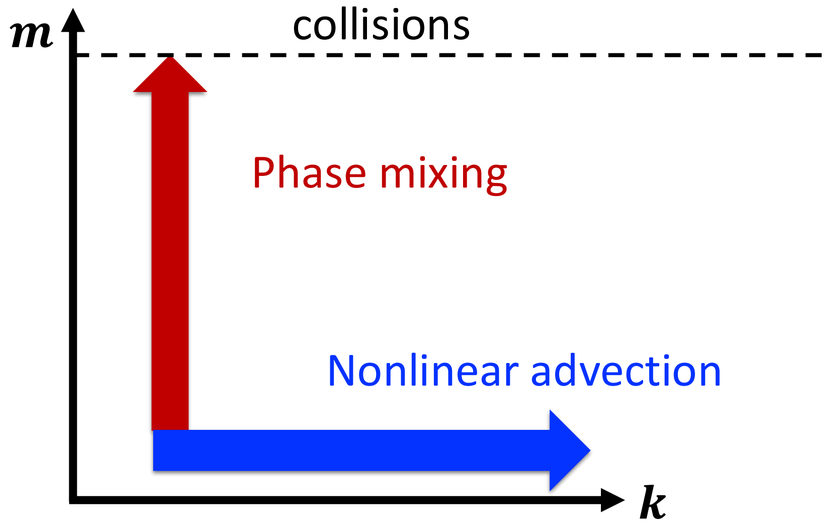
Kinetic equation
$$\frac{dg_m}{dt} = -v_{\rm the}\hat{b}\cdot\nabla\left(\sqrt{\frac{m+1}{2}}g_{m+1} + \sqrt{\frac{m}{2}}g_{m-1} - \delta_{m,1}g_2\right) - \sqrt{2}\delta_{m,2}\hat{b}\cdot\nabla\frac{e}{cm_e}d_e^2\nabla_\perp^2 A_\parallel$$

With kinetic electrons --- Phase space cascade and electron heating

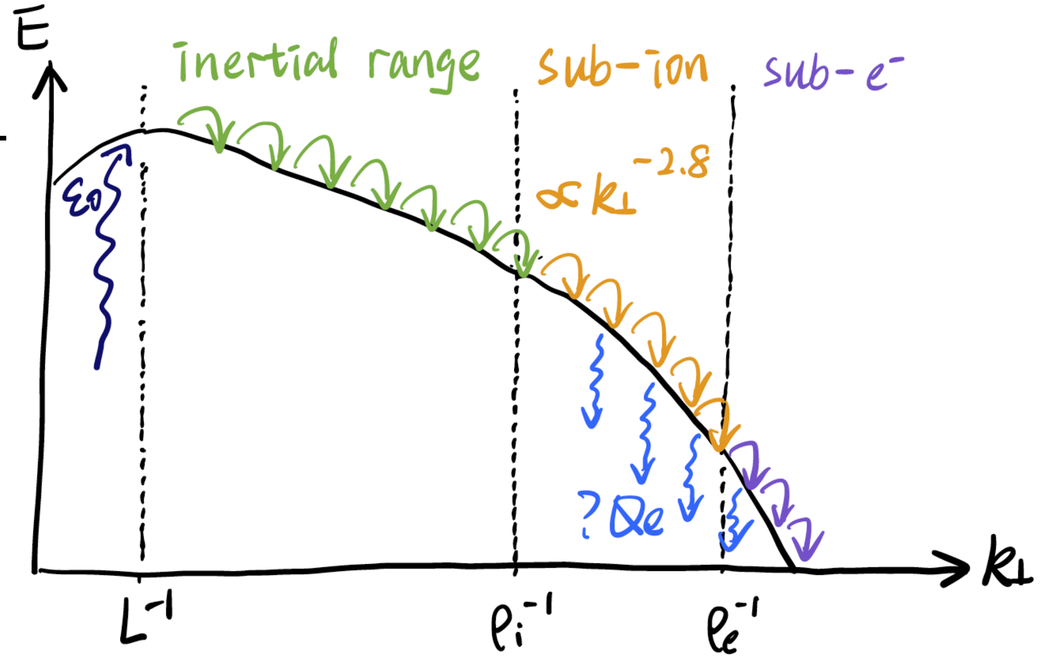
Nonlinear advection

Source of free energy in velocity space $\hat{b} \cdot \nabla J_{||}$

$$\frac{dg_m}{dt} = \underbrace{-v_{\rm the}\hat{\mathbf{b}} \cdot \nabla \left(\sqrt{\frac{m+1}{2}} g_{m+1} + \sqrt{\frac{m}{2}} g_{m-1} - \delta_{m,1} g_2 \right)}_{\text{Phase mixing}} - \sqrt{2}\delta_{m,2}\hat{\mathbf{b}} \cdot \nabla J_{||}$$



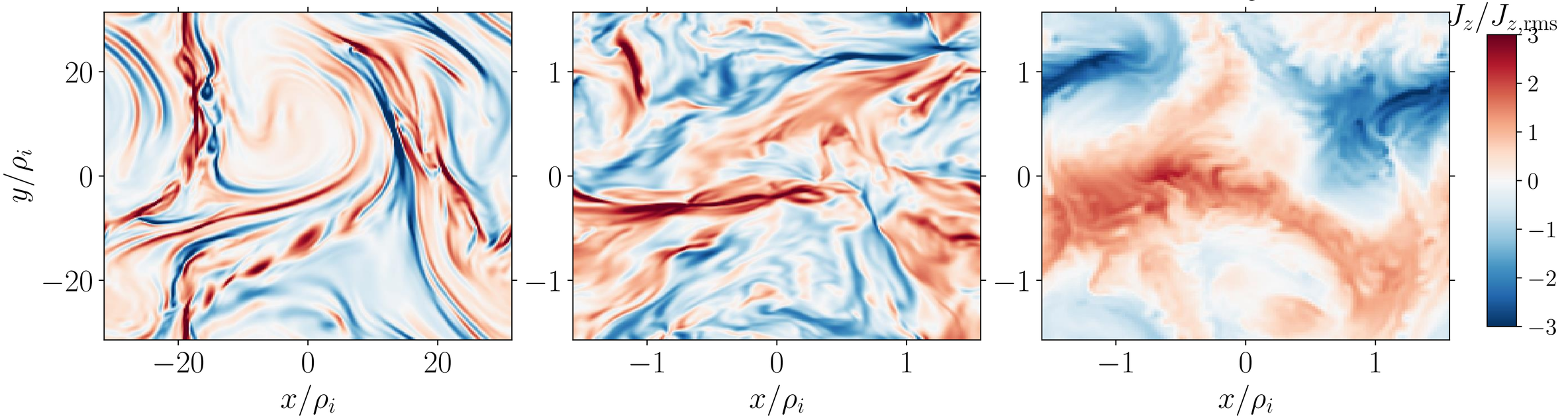
Simulations with kinetic electrons



Inertial transition to ρ_i

sub-ρ_i, transition to d_e

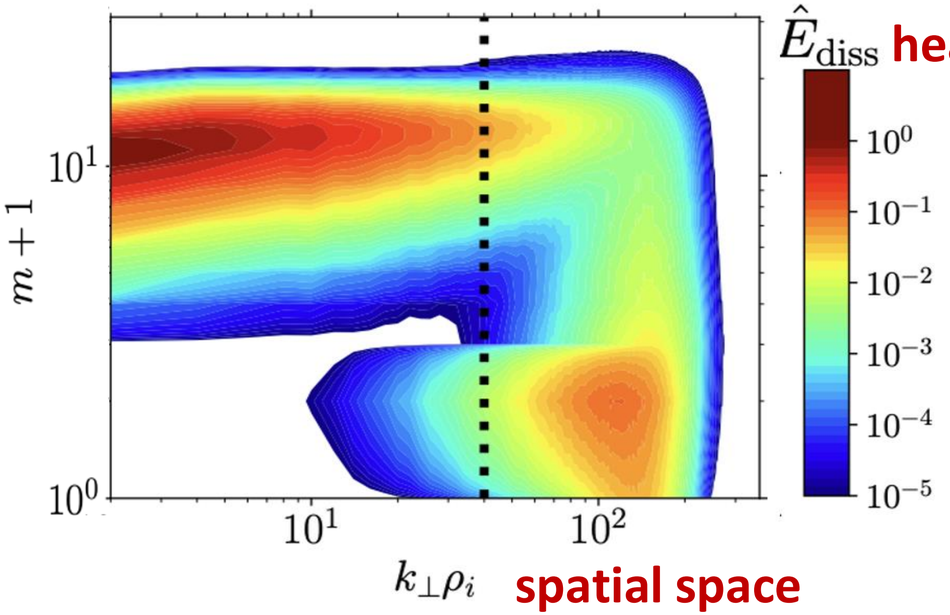
sub-d_e



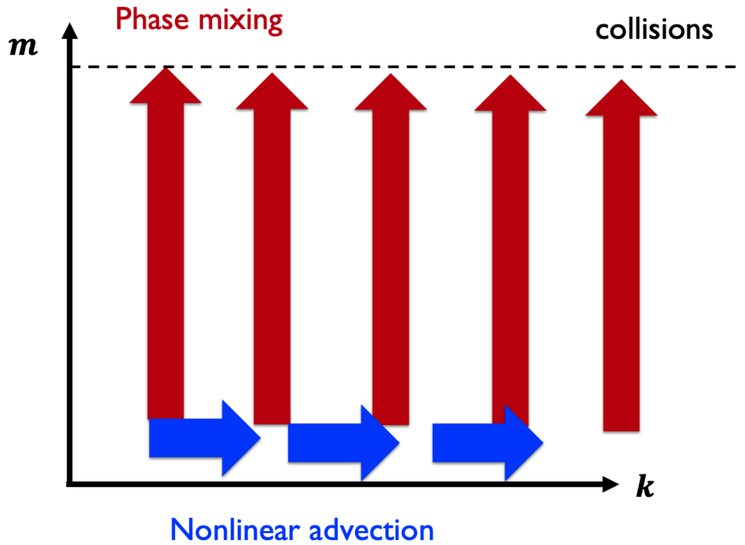
Landau damped EM energy matches electron heating

≈ electron heating

velocity space

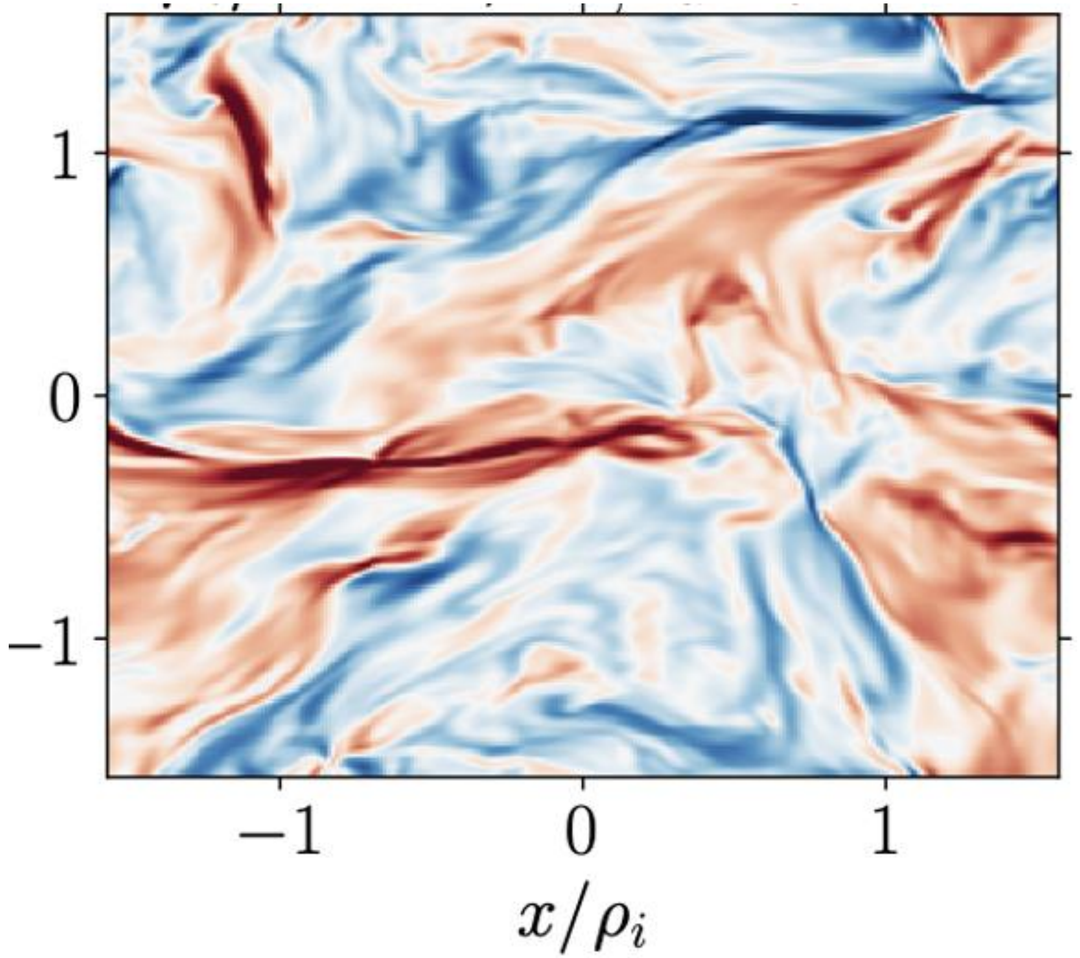


spatial space



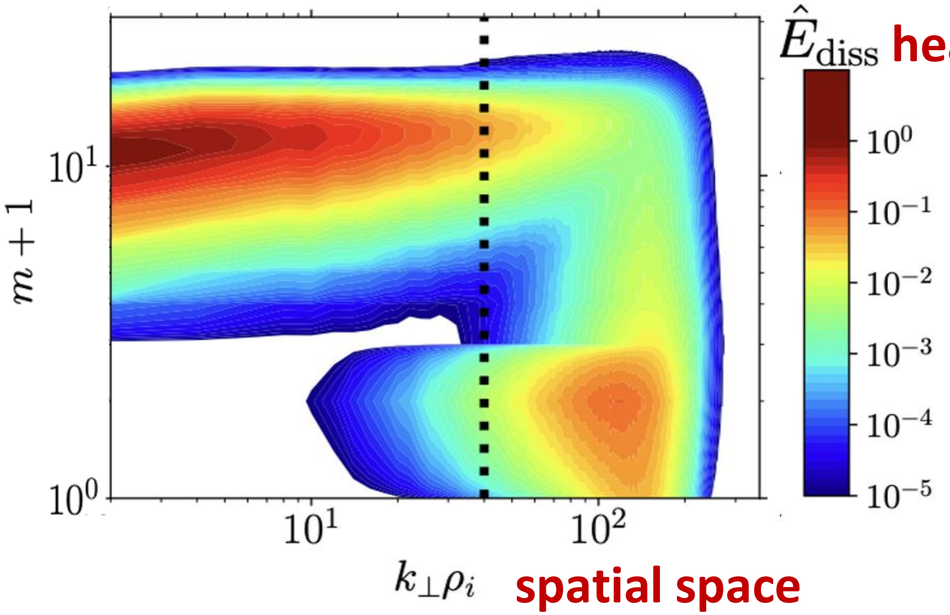
EM fluctuations are Landau-damped at each scale independently → phase mixing domination

sub-ρ_i, transition to d_e



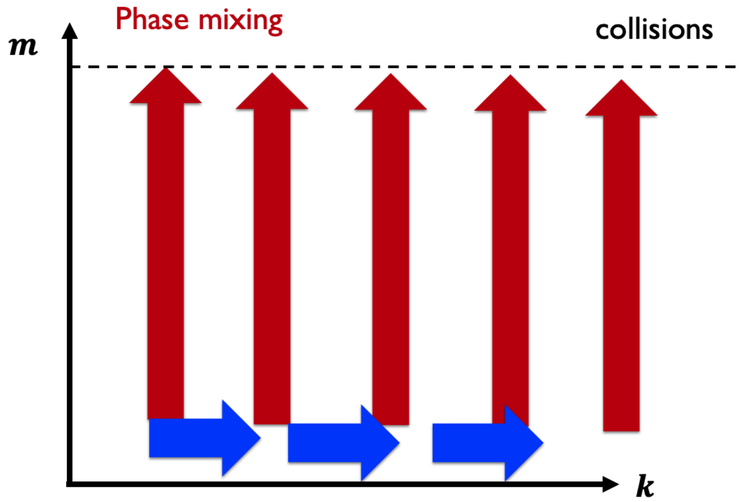
Towards a prescription for electron heating

velocity space



≈ electron heating

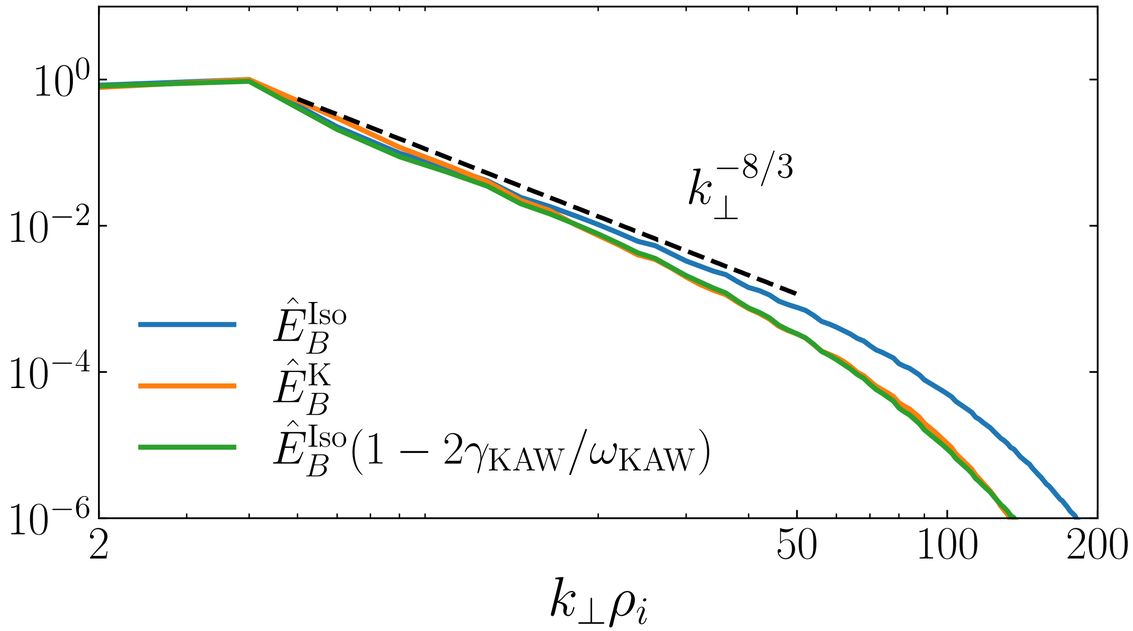
spatial space



EM fluctuations are Landau-damped at each scale independently → phase mixing domination

From the linear dispersion relation of KAWs: damping rate γ_{KAW} and frequency ω_{KAW}

$$\hat{E}_B^{\rm K}(k_\perp) \approx \hat{E}_B^{\rm iso}(k_\perp)(1 - 2\gamma_{\rm KAW}/\omega_{\rm KAW}).$$



[See also Howes+2008,2011; TenBarge+ 2013]

A zeroth-order solution in the velocity space

$$\frac{dg_m}{dt} = -v_{\rm the}\hat{\mathbf{b}}\cdot\nabla\left(\sqrt{\frac{m+1}{2}}g_{m+1} + \sqrt{\frac{m}{2}}g_{m-1} - \delta_{m,1}g_2\right) - \sqrt{2}\delta_{m,2}\hat{\mathbf{b}}\cdot\nabla J_{\|}$$

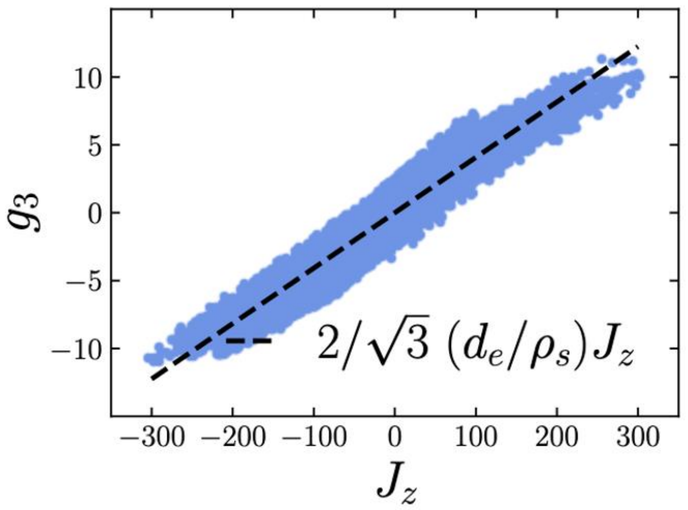
Nonlinear advection

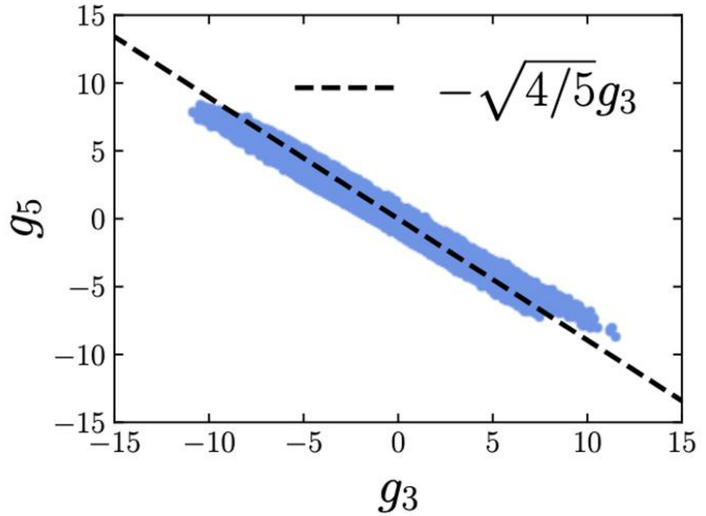
phase mixing

In the phase-mixing dominated regime, *l.h.s. << r.h.s.*

To the lowest order:

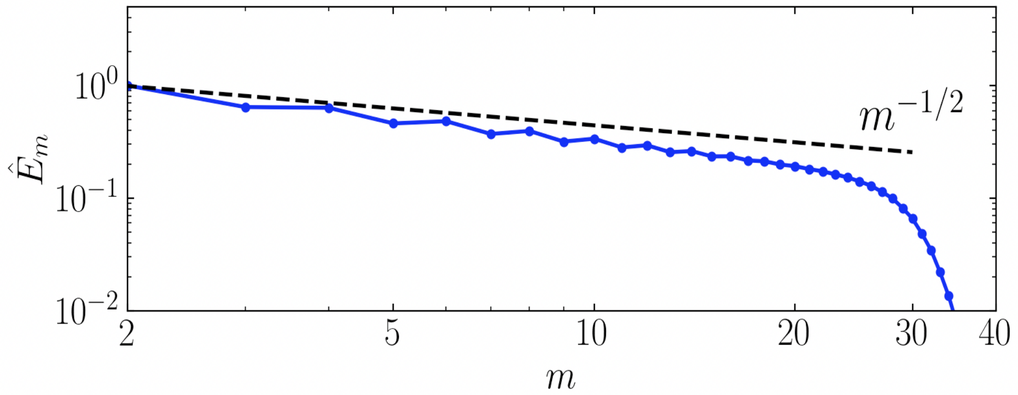
$$g_{m+1} = -\sqrt{m/(m+1)}\,g_{m-1}$$
$$g_3 = -\sqrt{2/3}\,(\rho_s/d_e)J_{\|}$$





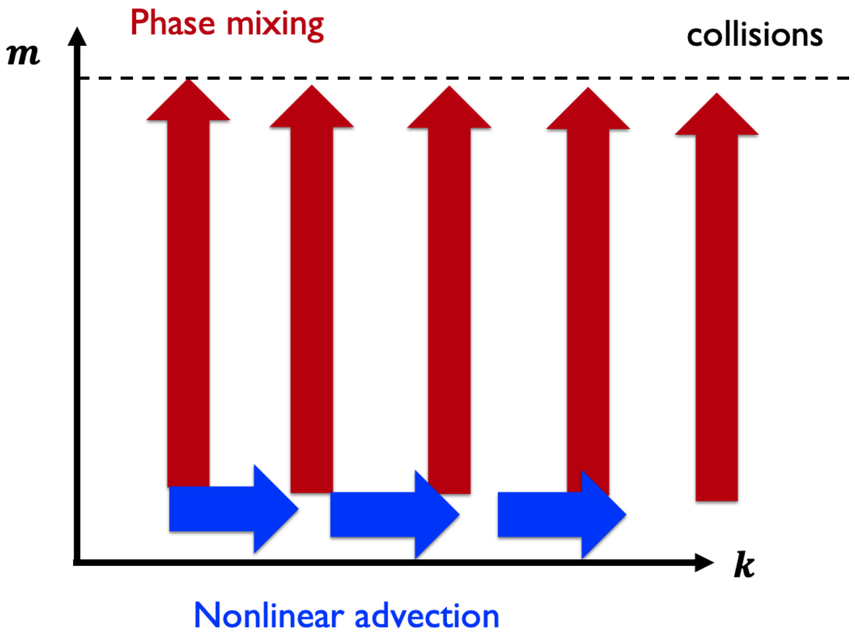
The Hermite spectrum of g_e (velocity-space spectrum):

$$E_m \propto m^{-1/2}$$

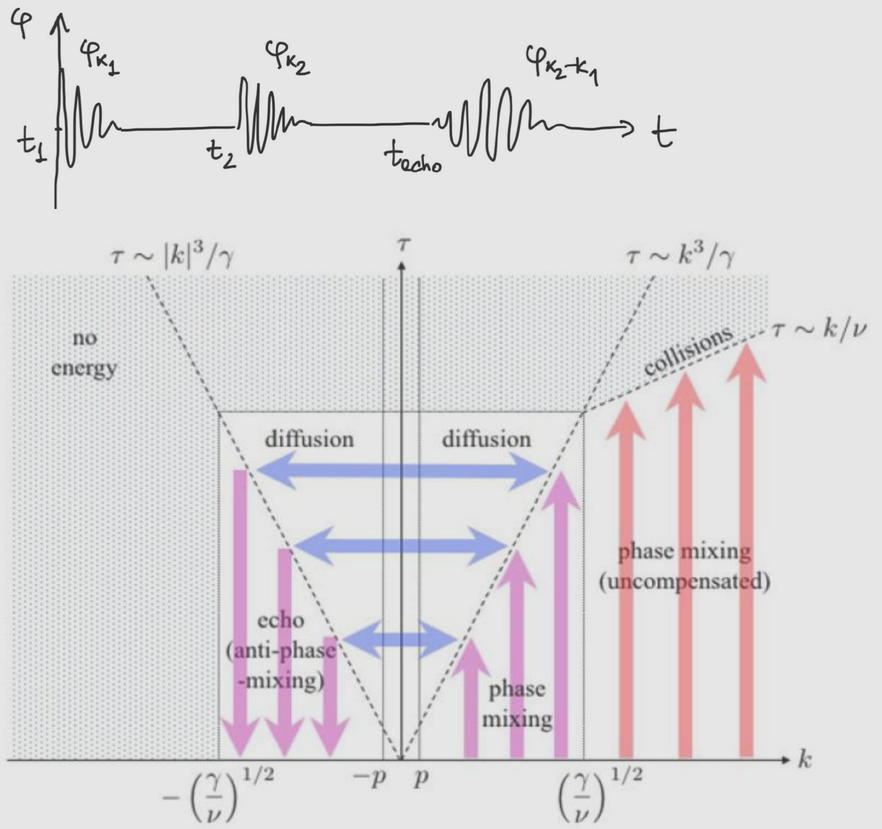


The measured phase-space trajectory is not trivial

Why do we see this?

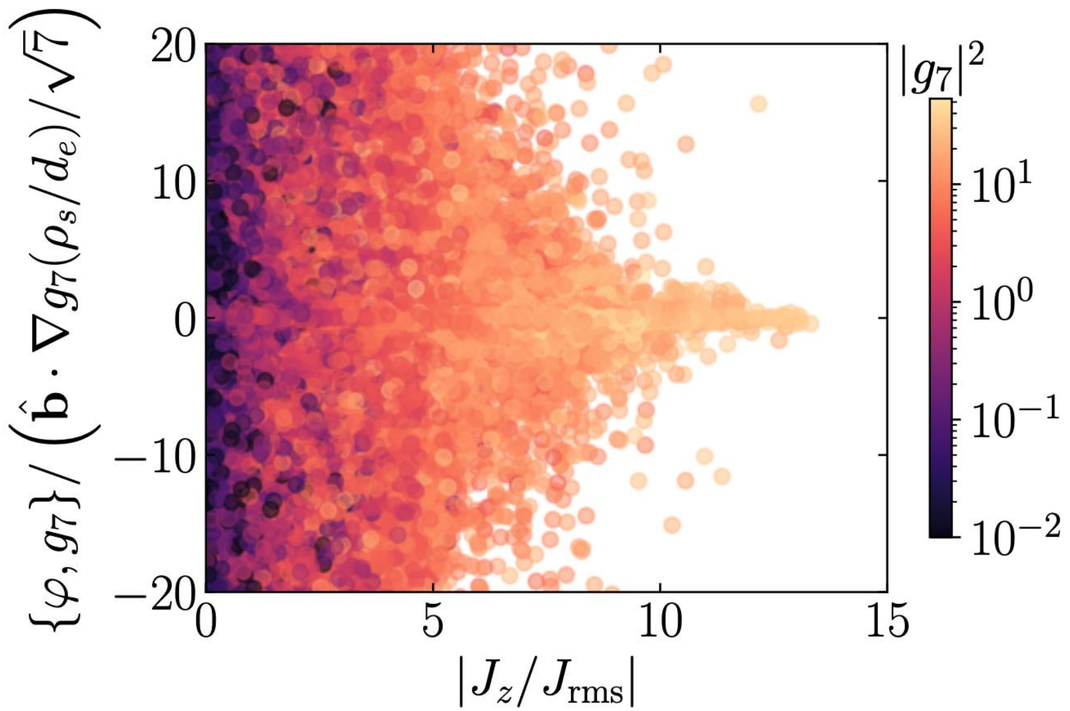


Caveat: Plasma echo/phase-unmixing phenomena



[Adkins&Schekochihin(2018)]

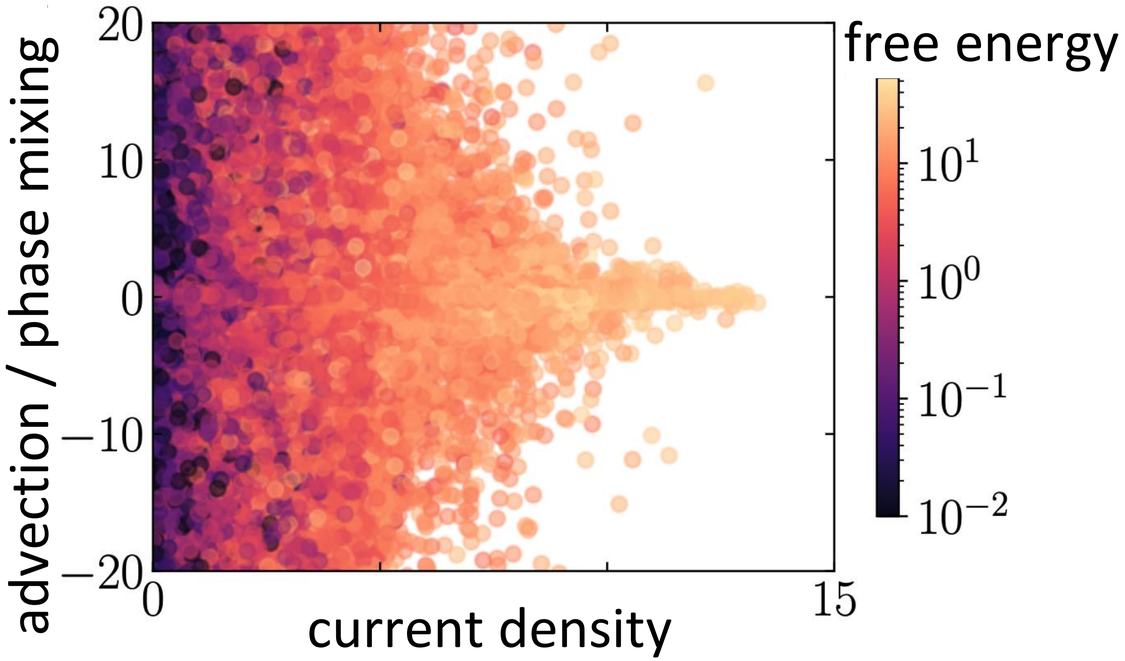
Local weakening of nonlinearity around current sheets enables strong phase mixing



Free energy concentrates around current sheets

At current sheets, nonlinear advection weak, phase mixing strong

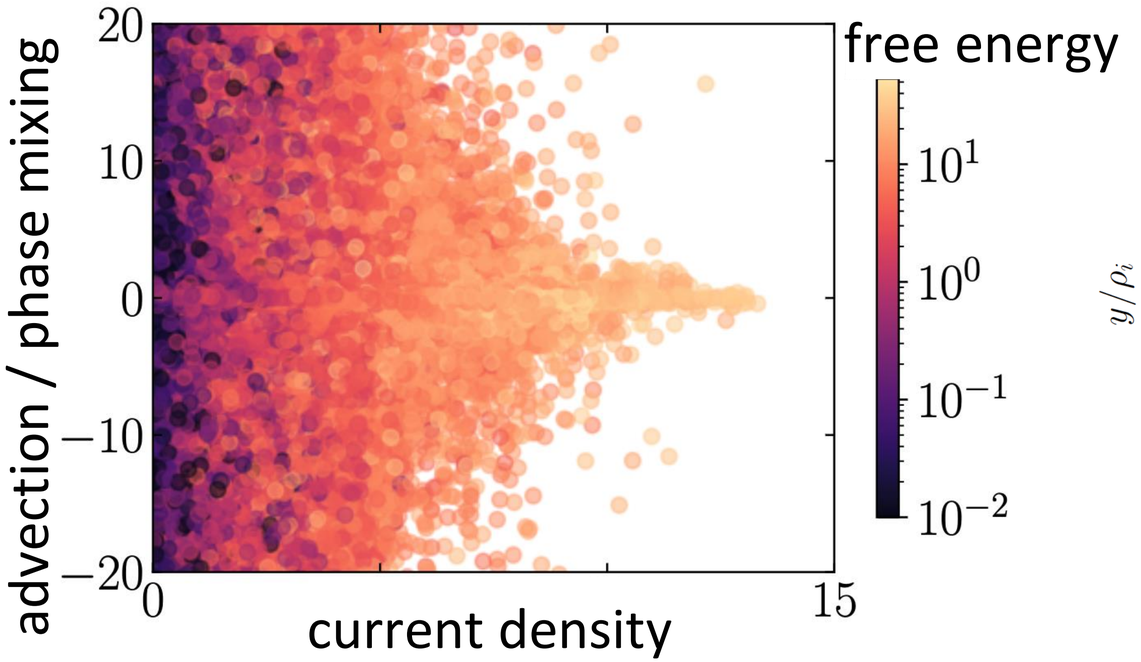
Local weakening of nonlinearity around current sheets enables strong phase mixing

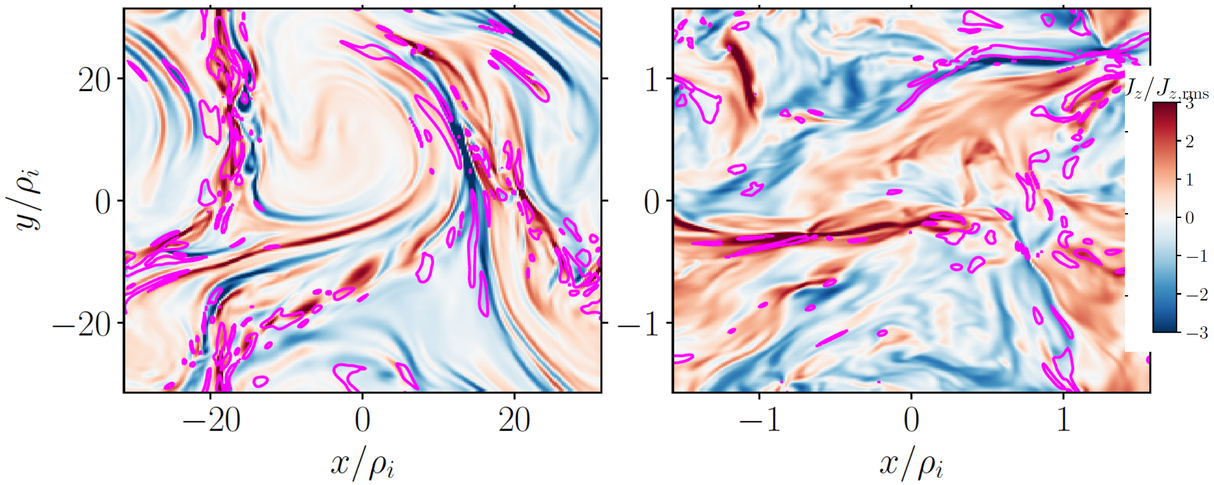


Free energy concentrates around current sheets

At current sheets, nonlinear advection weak,
phase mixing strong

Heating occurs around current sheets





Free energy concentrates around current sheets

At current sheets, nonlinear advection weak, phase mixing strong

Magenta contours indicates regions with strong electron heating

Heating overlaps with current sheets

Where should we go from here?

Multi-range turbulence + phase-space dynamics

